

**Presence at a distance: Empirical investigations toward
understanding, modeling, and enhancing social presence in
online distance learning environments**

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SUMMARY

In a technology-rich world, human relations are increasingly mediated through the interfaces of our digital devices. This is also true for certain arenas of the educational sector. More likely than not, a student enrolling in any higher education program today will be confronted with one or more pedagogical situations in which human interaction is not immediate but experienced through technology. While some students may take a class or two online in an otherwise face-to-face program, others may traverse a full learning trajectory without ever physically meeting their instructors or fellow students, as is the case in distance education. While increasingly sophisticated communication technologies may coax our senses, the fact remains: if our communication partner is not really there but physically absent, this will likely have psychological ramifications. Although the peculiarities of this increasingly prevalent mode of communication have long been the subject of investigation, with abounding speculation as to the wide-ranging consequences on communication processes and, in extension, learning processes relying on communication, the actual effects of mediation and their relevance for online distance learning are still not well understood.

The notion of social presence is one of the most prominent attempts at understanding the unique psychological properties of mediated communication. It refers to the degree to which we fall for the illusion of non-mediation, or in other words, our failure to perceptually integrate what we know in the abstract: That we are talking to a screen. Notably, even in communication media with modest face-to-face fidelity, for example text-based asynchronous messaging, it is possible to experience relatively high degrees of social presence. That is, we feel that our communication partner is “real” and “there”. Due to this paradox, we cannot deduce a simple and monocausal relation between technological sophistication of the communication medium and the experience of social presence. In fact, it appears that there are multiple processes and determinants at work accounting for this phenomenon, which, however, are still largely uncharted. As evocative and relevant this concept may appear—the sheer volume of studies attest to its popularity—, its practical benefits in advancing our knowledge on the psychology of mediated communication in online distance learning have been relatively small, a fact that may be traced back to a history of inconsistent definitions and the resulting lack of cumulative research progress. Section A “Introduction and Theoretical Foundations” lays out what we currently do and don’t know about the psychology of mediated communication, how this relates to online distance learning and some of its classic issues, as well as the state of research in this educational arena, with a specific focus on the online environments in which mediated learning experiences take place. Following this, a thorough review of the concept of social presence identifies central challenges of the literature and a model to guide the following steps is proposed.

Section B “Empirical Studies” lays out the central research questions of this thesis and presents five chapters to answer them. As a whole, the quantitative studies within these chapters are geared toward a better understanding of the phenomenon of social presence, a more comprehensive theoretical modeling around it, as well as practical application of this knowledge toward enhancing online distance learning experiences. To this end, data from a total of 1023 students at FernUniversität in Hagen was collected and analyzed. Zooming in on each individual study and its contribution, the following steps lead to this thesis’ goals: Building on important previous work, the first investigation, a scale validation study, aimed to put a stake in the ground with respect to the definition and measurement of social presence for the remainder of this thesis and, hopefully, the field more generally. In a second step, the explanatory scope was widened to account for processes and variables, that, together, may provide a more comprehensive picture of social presence and related socio-emotional variables in online distance learning. Structural equation modeling yielded an empirically supported and theoretically sound model with practical implications. In the third study, drawing from the tenets of the previously established model, a central proposition to enhance social presence in online distance learning environments was tested experimentally, further supporting the model through causal evidence while also yielding novel design recommendations. For the remaining two studies, the scope of theorizing was again widened. The fourth study investigated an emerging psychological phenomenon observed in social networking sites, ambient awareness. In a first sample, its prevalence in online distance learning environments was explored and, in a second sample, a theoretically plausible connection to social presence was assessed. Lastly, due to the historic roots of social presence, most empirical investigations have neglected the potential role of individual differences. For this reason, the fifth chapter investigated the merits of a trait-level view of social presence by examining its relationship to personality.

The integration and cumulation of newly generated knowledge is especially important, as this synthesizing work has been found to be lacking in much of the literature on social presence. For this reason, a comprehensive discussion is presented in section C “Discussion and Future Directions”, in which the contributions of this thesis toward the challenges of social presence are laid out. Despite some questions remaining unanswered, the interpretation of findings allows us to conclude that significant progress has been made toward a better identification, explanation, and application of social presence. As a visual summary, the final research model is presented in its modified and extended form, detailing all hypotheses examined in this thesis. Further condensing the findings of this thesis, a set of questions informing a tentative theory of social presence are answered succinctly. Through these summaries on different levels of elaboration, we learn that social presence is a complex psychological phenomenon that is multi-causally determined and nuanced in terms of its benefits for online distance learning. Taking into account

some limitations within the works presented in this thesis, potential steps for improving future research efforts are discussed in light of recent developments in the field of online distance learning. Critically, however, in order to further build our knowledge of social presence in online distance learning, two major road-blocks on the path to a new phase of social presence research need be considered specifically. The first relates to the dubious connection to learning achievement. The second pertains to a lack of integration of social presence into our broader knowledge base of human psychology. Two frameworks that specifically address these problems are proposed, with consequential implications for future research. The thesis ends with a brief reflection.

ZUSAMMENFASSUNG

In einer zunehmend technisierten Welt wird zwischenmenschliche Interaktion immer häufiger durch die Schnittstellen unserer digitalen Endgeräte vermittelt. Dies gilt ebenso für Bereiche des Bildungssektors. Die meisten der angehenden Studierenden, die sich heute in einer Hochschule einschreiben, werden unweigerlich im Laufe ihres Studiums mit pädagogischen Szenarien konfrontiert werden, in denen menschliche Interaktion mit Hilfe von Technologie vermittelt wird. Dies mag in manchen Fällen lediglich bedeuten, dass Studierende einzelne Einheiten oder Abschnitte eines Präsenzstudiums online absolvieren. Andere durchlaufen möglicherweise ein ganzes Studium, ohne ihren Mitstudierenden oder Lehrkräften jemals physisch zu begegnen, wie es in einem Fernstudium der Fall sein kann. Mag es besonders bei raffinierten und technisch ausgereiften Kommunikationsmedien nicht immer auffallen, bleibt dennoch eine wichtige Tatsache bestehen: Wenn unser Gegenüber nicht tatsächlich da, also physisch abwesend statt präsent ist, wirkt sich das auf unsere zwischenmenschliche Wahrnehmung aus. Obwohl die psychologischen Besonderheiten dieser zunehmend verbreiteten Art der Kommunikation seit langem Gegenstand von Untersuchungen sind und wichtige Effekte auf Kommunikations-, sowie auf Lernprozesse vermutet werden, sind die tatsächlichen psychologischen Auswirkungen, insbesondere mit Blick auf das Lernen im Online- und Fernstudium, nach wie vor nicht ausreichend erforscht.

Der Begriff *Social Presence* stellt eine der prominentesten Bemühungen dar, sich der besonderen psychologischen Herausforderungen computervermittelter Kommunikation konzeptionell zu nähern. Er bezieht sich auf das Ausmaß, zu dem wir bei diesen Formen der Kommunikation ein Gefühl der Unmittelbarkeit erleben. Somit impliziert der Begriff *Social Presence* gewissermaßen eine Illusion oder anders gesagt, das Versäumnis in unserer Wahrnehmung, der Tatsache kontinuierlich gewahr zu sein, dass wir eigentlich mit einem Bildschirm kommunizieren. Interessanterweise ist es auch bei Kommunikationsmedien, die mit dem Naturell von face-to-face Gesprächen relativ wenig Gemeinsamkeiten aufweisen, zum Beispiel asynchrone textbasierte Chats, ebenfalls möglich, ein hohes Maß an *Social Presence* zu erfahren, also dem Gefühl zu erliegen, dass unser Kommunikationspartner „echt“ und „da“ ist. Diese paradoxe Beobachtung ist der Grund, dass wir keinen monokausalen Zusammenhang zwischen der technologischen Raffinesse des Kommunikationsmediums und der Wahrnehmung von *Social Presence* ableiten können. Tatsächlich scheint es stattdessen eine Vielzahl von Faktoren zu geben, die für dieses psychologische Phänomen verantwortlich sind, von denen jedoch viele nicht ausreichend erforscht sind. So ansprechend und relevant dieses Konzept zu sein scheint –dafür sprechen unzählige Studien, die sich mit dem Konstrukt beschäftigen–, hat seine Anwendung und Erforschung bislang vergleichsweise wenig zur Weiterentwicklung unseres Verständnisses über die Psychologie computervermittelter Kommunikation beigetragen; eine

Tatsache, die möglicherweise auf die lange Geschichte uneinheitlicher Auslegungen dieses Konzepts und den daraus resultierenden Schwierigkeiten, Wissensbestände zu kumulieren, zurückzuführen sein könnte. Abschnitt A „Introduction and Theoretical Foundations“ beschreibt den aktuellen Forschungsstand bezüglich der Psychologie computervermittelter Kommunikation und stellt einen Bezug zur Fernlehre und dem Online-Lernen her, insbesondere mit Blick auf die vielzitierten Herausforderungen, die mit diesen Lernkontexten einhergehen. Ein besonderer Fokus liegt dabei auf den Online-Umgebungen, in denen die Lernaktivitäten dieser Szenarien üblicherweise stattfinden. Im Anschluss daran erfolgt eine Besprechung von Social Presence, in der zentrale Herausforderungen in der Erforschung dieses Konzepts diskutiert werden. Dies mündet in die Vorstellung eines theoretischen Modells, das für die folgenden empirischen Arbeiten handlungsleitend ist.

In Abschnitt B „Empirical Studies“ werden die zentralen Forschungsfragen dieser Arbeit dargelegt sowie fünf Studien zu ihrer Beantwortung präsentiert. Zusammengefasst zielen diese empirischen Untersuchungen auf ein besseres grundlegendes Verständnis des Phänomens Social Presence, auf eine theoretisch plausible und empirisch gestützte Modellierung, sowie auf die praktische Anwendung dieses Wissens zur Verbesserung der Online- und Fernlehre ab. Insgesamt wurden zu diesem Zweck Daten von 1023 Studierenden der FernUniversität in Hagen erhoben und analysiert. In der ersten Untersuchung, einer Skala-Validierungsstudie, wird aufbauend auf der sich häufenden Kritik an der Social Presence Forschung versucht, wichtige Grundlagen bezüglich der Definition und Messung des Konstrukts zu schaffen. In der zweiten Studie wird dann der Betrachtungshorizont mit Hilfe eines Strukturgleichungsmodells erweitert, um zu einem umfassenderen Bild von Social Presence und verwandten sozio-emotionalen Prozessen und Variablen zu gelangen. Dies resultiert in einem theoretisch fundierten und empirisch gestützten Modell mit praxisrelevanten Implikationen. Jene Implikationen zur Förderung und Begünstigung von Social Presence in Online-Lernumgebungen werden in der dritten Studie mit Hilfe eines Experiments empirisch überprüft, wodurch weiterhin Gestaltungsempfehlungen für derartige Lernumgebungen abgeleitet werden können. Für die verbleibenden zwei Studien wird der theoretische Blick erweitert, sodass sich das vierte Kapitel mit einem bislang nur aus sozialen Netzwerken bekannten psychologischen Phänomen beschäftigt, Ambient Awareness. Mit Hilfe zweier Stichproben wird zunächst seine Prävalenz in Online-Lernumgebungen untersucht, um dann eine theoretisch naheliegende Verbindung zu Social Presence zu überprüfen. Zuletzt wird in der fünften Studie ein neuartiger Blickwinkel auf das Konstrukt eingenommen. Wurde in der Vergangenheit Social Presence aufgrund seiner sozialpsychologischen Wurzeln meist als sozial und kontextuell determiniert verstanden, wird hier die potenzielle Rolle individueller Unterschiede, genauer gesagt der Persönlichkeit der Studierenden, bei dem Erleben von Social Presence in den Fokus gerückt.

Der Integration und Kumulierung der neu gewonnenen Erkenntnisse kann deshalb eine besondere Bedeutung zugeschrieben werden, da diese Synthesearbeit in der Social Presence Literatur als defizitär erachtet werden kann. Aus diesem Grund wird in Abschnitt C „Discussion and Future Directions“ eine umfassende Diskussion und Einordnung der Ergebnisse, insbesondere mit Blick auf die Schwächen der Social Presence Literatur vorgenommen. Trotz verbleibender offener Fragen, können die Ergebnisse der empirischen Arbeit so interpretiert werden, dass deutliche Fortschritte bei der Identifizierung, der theoretischen Erklärung sowie der praxisnahen Anwendbarkeit von Social Presence erzielt wurden. Als visuelles Resümee wird das endgültige Forschungsmodell in seiner durch die Studien modifizierten und erweiterten Form präsentiert, in dem alle in dieser Arbeit untersuchten Hypothesen enthalten sind. Um die zentralen Erkenntnisse weiter zu verdichten, wird eine Reihe von Fragen, die zusammengekommen die Grundlage für eine Theorie von Social Presence darstellen, fokussiert beantwortet. Durch diese Zusammenfassung auf verschiedenen Abstraktionsebenen entsteht ein Bild von Social Presence als ein komplexes psychologisches Phänomen, das mit Blick auf seine Relevanz für das Online- und Fernstudium multikausal bestimmt und nuanciert zu sein scheint. Mit Blick auf Limitationen dieser Arbeit sowie aktueller mediendidaktischer Entwicklungen werden mögliche Schritte zur Verbesserung künftiger Forschungsbemühungen vorgeschlagen. Darüber hinaus werden zwei zentrale, weiterhin bestehende Hindernisse auf dem Weg zu einer ambitionierten und fundierten Social Presence Forschung aufgezeigt. Ersteres bezieht sich auf die bislang inkonsistente Erforschung des Zusammenhangs zwischen Lernerfolg und Social Presence. Letzteres bezieht sich auf die fehlende Einbettung des Phänomens Social Presence in die menschliche Psychologie. Als ersten Schritt zu Überwindung dieser Hindernisse werden zwei theoretische Frameworks zur Gestaltung künftiger Forschung vorgeschlagen. Die Arbeit endet mit einer kurzen Reflexion.

Empirical chapters in this thesis

- Weidlich, J., Kreijns, K., Rajagopal, K. & Bastiaens, T. (2018). What Social Presence is, what it isn't, and how to measure it: A work in progress. In *Proceedings of EdMedia: World Conference on Educational Media and Technology* (pp. 2135-2143). Amsterdam, Netherlands: AACE
- Weidlich, J., & Bastiaens, T. J. (2017). Explaining social presence and the quality of online learning with the SIPS model. *Computers in Human Behavior*, 72, 479–487.
- Weidlich, J., & Bastiaens, T. J. (2019). Designing sociable online learning environments and enhancing social presence: An affordance enrichment approach. *Computers & Education*, 142, 103622.
- Weidlich, J., & Bastiaens, T. J. (2019). Pathway to presence: an investigation of ambient awareness in online learning environments. *Interactive Learning Environments*, 1-14.
- Weidlich, J., & Bastiaens, T. J. *A trait-level view of social presence: Exploring personality in online distance learning* [Unpublished: Manuscript in preparation]

As is common for a cumulative science, the foundation of the work presented here was laid long before the first steps of this thesis were taken. In this case, the foundation was laid by Karel Kreijns, who in his dissertation significantly and permanently advanced our understanding of social presence, the psychological construct at the heart of this thesis. As a result, his contributions toward this thesis reach beyond what is indicated by authorship.

However, with respect to the manifest steps contributing to the publication of manuscripts, the following applies: Joshua Weidlich came up with the theme and aim of this thesis, was primarily responsible for developing research questions, planning research designs, conducting data collection and analysis, as well as preparing manuscripts for publication. The first study differs in that the idea was jointly developed by Joshua Weidlich and Karel Kreijns and the following steps were divided such that Joshua Weidlich collected the data, lead the manuscript writing, while Karel Kreijns conducted data analysis. Theo J. Bastiaens mentored and supervised all five projects, gave substantive feedback at different stages of projects, and was involved in the manuscript preparation process.

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SECTION A

INTRODUCTION AND THEORETICAL FOUNDATIONS

1 GENERAL INTRODUCTION

'But that's exactly it. Seeing isn't involved.' She reddened a trifle and looked down. 'I hope you don't think I'd ever do anything like that. I mean, just step out of the drier, if anyone were seeing me. It was just viewing.'

'Same thing, isn't it?' said Baley.

'Not at all the same thing. You're viewing me right now. You can't touch me, can you, or smell me, or anything like that. You could if you were seeing me. Right now, I'm two hundred miles away from you at least. So how can it be the same thing?'

Baley grew interested. 'But I see you with my eyes.'

'No, you don't see me. You see my image. You're viewing me.'

'And that makes a difference?'

'All the difference there is.'

The Naked Sun (1957), Isaac Asimov

In *The Naked Sun*, second of the famous *Robots* series by science fiction writer Isaac Asimov, our protagonist Elijah Baley finds himself investigating a murder on the planet of Solaria. Solarians are embedded in culture and tradition foreign to the detective; the planet sparsely inhabited with a rigidly controlled human population while a large number of robots maintain infrastructure, agriculture, and technology. Notably, all its inhabitants are taught from birth to avoid personal contact, and, as a result, face-to-face conversation (what is called *seeing*) is extremely rare. Instead, communication is conducted via two-way teleconference technology, a holography system. This is called *viewing*. Our protagonist is ignorant of the fact that Solarians perceive these two modes of communication to be strikingly different and, thus, some perplexing communication episodes ensue.

On the face of it, humans and Solarians do not share many characteristics. On earth, *seeing* is frequent and natural. We regularly meet our friends at cafes, visit family, and talk over dinner, all while sharing the same physical space. But increasingly, there are parallels to Solaria. Through technological innovations of the last decades, we, too, do much of our communicating in the mode of *viewing*, talking and writing from a distance in a myriad of ways, many through the interfaces of our technologies. The difference between Solarians and us is that they are fully aware of the striking psychological differences between face-to-face and mediated communication. We, however, remain comparatively naïve. Most of us have some anecdote of missed sarcasm or even graver misunderstandings due to the limitations of text-based messaging

but rarely are we tuned in to how much mediated communication *feels* different. When is this the case and what does this mean for the way we communicate?

There is an idea that addresses this and it is called social presence. With the notion of social presence we try to capture the psychological effects of communicating with others through technological interfaces. Long before it was first coined in social psychology, Isaac Asimov birthed the idea of presence in *The Naked Sun*. Prescient of technologies that wouldn't emerge until almost half a century later, he hypothesized that even through high-fidelity teleconferencing technologies, there would be a difference; a difference in how we communicate, a difference in how we present ourselves, a difference in how we perceive our communication partner. Social presence is the attempt at capturing this difference.

The following chapters will show why this investigation is important and timely, framing it in recent developments in distance education and the online technology mediating this educational trajectory. Then, by reviewing the current state of research on online distance education, social presence, and related pedagogical and psychological concepts, gaps in research and central issues will be identified. Building on this, an overarching framework is presented that will guide us through the empirical studies of this thesis and allows us to formulate general research questions of this thesis. After these necessary preparatory steps, five empirical studies will be presented that, together, form the main part of this theses. Each study will make up one chapter and, hopefully, add one piece to the puzzle of social presence. Following this empirical part is a discussion answering the research questions and interpreting the findings in terms of the bigger picture. After this, insights gained along the way will be summarized and discussed. Finally, this updated state of knowledge will be used toward sketching out future directions for social presence research.

2 FRAMING THE RESEARCH

2.1 Distance Education

Mediated interaction is ubiquitous; in many domains of collective human endeavor, the modern workplace as much as private leisure are we involved in interaction and communication without being ‘there’ (Newport, 2014; Colbert, Yee, & George, 2016; Arminen et al., 2016). But perhaps nowhere is the experience as fully mediated as in distance education, where students pass through a whole educational trajectory remotely. In fact, it is possible for a distance education student to move through an entire curriculum and exiting with a degree in hand while never having stepped foot into a university building. In this case, every part of the educational journey is experienced from a distance, through mediation. Thus, a central characteristic of distance education is this physical separation between a student and the education institution (Holmberg, 2005; Zawacki-Richter, 2015). Moreover, besides the physical separation, there is a psychological distance between students and their peers as well as their instructors (Moore, 1993). Even if we merely grant minute differences between mediated and non-mediated communication, in adding up, these differences may profoundly shape the overall experience of students in this mode of education. This characteristic makes distance education uniquely interesting for investigations of social presence and, inversely, makes social presence critically relevant for distance education practices, and, thus, a popular topic of academic investigation. The practical relevance of this is amplified by the fact that distance education is growing and changing.

The growth of distance education is a worldwide phenomenon, with today more than 8.5 million higher education students enrolled in at least one distance course in Australia, Brazil, Canada, Germany, the United Kingdom, and United States (Quayyum & Zawacki-Richter, 2018) and a growth rate of 5.6% from 2015 to 2016 (Seaman et al., 2018). Additionally, the distance education landscape is seeing changes, as many face-to-face education institutions, including private ones, expand their portfolio to distance learning offerings (Litto, 2018). This dual mode activity of campus-based institutions is at the same time a driver of innovation for these institutions but also, in turn, creates diversity and fosters competition in the distance education landscape (Mays et al., 2018; Xiao, 2018). Traditional distance institutions like Open Universities are challenged to improve on their hallmark strengths, openness, flexibility, and accessibility. These developments track the change from distance education being a “relatively exotic form of ‘alternative’ education” (Kearsley, 1998) to a mainstream trajectory of higher education (Zawacki-Richter, 2015).

2.2 Online distance learning

A major catalyst for this growth and change of distance education has been the emerging ubiquity of the internet and the accompanying rise of accessible and affordable information and communication technology. Through this, distance education has moved from correspondence learning and educational multimedia, which it was throughout much of the 20th century, to the online-based learning that it is today such that now online learning is the primary delivery mode of distance education (Holmberg, 1995; Holmberg, 2005). Online learning provides the hallmarks of correspondence learning, maximum flexibility regarding time and place of study, whilst also bridging distance to fellow students and instructors (Anderson, 2004).

While today online learning is intertwined with distance education, they are not synonymous. Not all distance education is conducted through pure online learning and even more so, not all online learning is in the context of distance education. In fact, much of the recent growth in the online learning sector has been outside of distance education institutions, one prominent example being MOOC's (Quayyum & Zawacki-Richter, 2018). This thesis focuses on the intersection of online learning and distance education, what will be called *online distance learning* for the remainder. Insights and implications generated in this thesis are generally applicable to (1) distance education contexts using online learning as delivery mode, as well as (2) online learning offerings that share many of distance education's characteristics, e.g. relatively long courses (e.g. several weeks or months) with adult students using computer-mediated communication to engage with fellow students as well as instructor-facilitators.

Although we are now in an age where ubiquitous high-fidelity teleconferencing is feasible, the reality of most online distance learning experiences is different. Real-time or synchronous video conferencing is only a small part of many students' educational experience. Instead, asynchronous text-based learning activities via learning management systems (LMS) prevail (Legon & Garrett, 2018). If a student in online distance education decides to invest time into study, frequently his or her first step will be to log into the institution's LMS (e.g. Moodle or Blackboard). There, he or she can engage with the current topic of the course by viewing learning material, engaging in collaborative learning activities with peers, partaking in message board discussions with instructors and peers, joining real-time chats, as well as testing knowledge with quizzes. Thus, LMS are the de facto learning environment in which most of online distance learning takes place and through which students communicate. Being at the heart of most learning activities, it is essential for researchers to understand how these environments shape the online distance learning experience.

2.3 Online learning environments as learning spaces

The notion that learning environments should be considered in our investigation of learning processes is accompanied by the epistemological shift from a mostly cognitivist view of learning to a more constructivist one (Jonassen, 1991). It is the idea that context in which learning takes place itself affects cognition, that learning is situated (Brown et al., 1989). With earlier theorizing, this was outside of scope, processes of learning being relatively uniform and determined by cognitive architecture. And while human cognitive architecture indeed is the bottleneck of learning (Miller, 1956; Baddeley, 1994; Sweller et al., 2019), our view is now extended to consider architecture outside of the brain as well. More specifically, social constructivist accounts of learning highlight the importance of the learning environment for individual and collaborative knowledge construction (Vygotsky, 1978; Adams, 2006). Tied to this is the very similar concept of learning space, as formulated by Peters (2002):

“On the basis of general environmental assumptions [...], a learning space not only creates the preconditions for the interaction between those taking part in the learning process, it can also influence their interactions. It can even influence the contents and forms of these interactions, e.g. by inviting or challenging to a defined behavior, or deterring it, by affecting the participants aesthetically, and by enriching their experiences.” (p. 75)

Recently, there has been an increase of investigations concerned with physical learning spaces (e.g. van Merriënboer et al., 2017; Zeivots & Schuck, 2018). With scholars announcing the “spatial turn” of technology-enhanced learning (Bligh & Crook, 2017), it is important to remember that there are also learning environments that exist outside of geographical space, for example environments for online distance learning. That doesn’t mean these environments are *per se* “space-less” (Flynn et al., 2018). As in physical space, online learning environments can afford and constrain certain behaviors, and, thus, open up or limit space for learning processes. For example, an online learning environment with no communication feature constrains the learning process such that collaborative learning becomes near-impossible. On the other hand, design features can invite behavior; an easily navigable interface design, for example, affords free roaming of the environment and effortless interaction with learning materials, fellow students, and instructors. Learning space research is relatively new field of study and the literature is fragmented (Ellis & Goodyear, 2016). To guide our design of online learning environments, we can turn to ecological psychology, a long-standing research tradition concerned with designing for desired behavior.

2.4 The ecological psychology of online learning environments

Ecological psychology is a school of thought that attempts to explain how human behavior is influenced by characteristics of the external world and, consequently, how this may inform human-centered design. Pioneered by James Gibson, one of its hallmarks is the

examination of how features of the environment influence varied human behavior (Gibson, 1966; Gibson, 1986). Though occasionally misconstrued as simple stimulus-response view of behavior, ecological psychology actually aims to unravel the complex interplay of the two variables, environment and human behavior (Covarrubias et al., 2017). As an example, we can think of how a chair invites a human to sit on it, whereas a table usually does not. Now, if we imagine a change of height in the human, so that he is much taller, a table may suddenly readily invite sitting. Similarly, if we imagine a different goal, e.g. observing a presentation from the back of a room filled with tall people, a table may also lend itself more readily to sitting than a chair.

This notion of inviting behavior is formalized in the concept of affordances. Popularized in Donald Normans “The Design of Everyday Things” (1988) and extended by Chemero (2003), affordances are the relations between action-possibilities of the environment and the human actor. This relation is complex and dynamic because two human actors, depending on their abilities (e.g. tall vs. short) and goals (e.g. relaxing vs. seeing a presentation) may perceive different action-possibilities in the same object or similar action-possibilities in different objects. Alas, the so-called *invitation character* of the action-possibility can change dynamically. Designing environments for certain behaviors, thus, becomes a probabilistic rather than a deterministic endeavor, a view that fits comfortably within the social constructivist view of learning.

Extending this idea of affordances to the more specific subsets of human behavior in online learning environments, Kirschner (2002) was the first to talk about the notion of *educational* affordances. Subsequently, *social* affordances were introduced, as action-possibilities in online learning environments that allow for and invite enactment of social behavior (Kreijns et al., 2002; Kreijns, 2004; Kirschner et al., 2004). Just like a town square affords physically meeting and interacting with friends, acquaintances, or even strangers, online environments can also afford such social behavior, given they are designed accordingly. In this sense, social affordances are design features regarding interaction and communication that, together, create a set of probable uses, meanings, and practices within an environment (Postigo, 2016). Lately, the concept has become increasingly popular with relation to educational technology. Affordance-based investigations of technology for learning include for example information and communication technologies in general (Conole & Dyke, 2004), 3D virtual environments (Dalgarn & Lee, 2010), handheld devices (Song, 2011), and wearable technologies (Bower & Sturman, 2015). These investigations typically result in lists of possibilities for engaging with the technology. However, their actual *Aufforderungscharakter* (German for “invitation character”), i.e. the ability of the affordances to invite behavior is not assessed and their influence can, thus, only be hypothesized (Aagaard, 2018). Empirical evidence

for if and how affordances interact with perceptions and behavior in learning environments, and thus, may provide learning-related benefits is currently still lacking.

2.5 Psychology of Social Networking Sites

When it comes to online spaces inviting social behavior, social networking sites (SNS) are shining examples. The largest social network, Facebook, has over two billion monthly users worldwide (Statista, 2020a), 1.4 billion daily users (Statista, 2020b), and a share of 79% of American internet users. As of 2016, other social networks and social media applications, like Twitter and Instagram are smaller but still used by 24% and 32% of American internet users, respectively (Greenword et al., 2016). This makes online social networking one of the most ubiquitous human social activities (Zhu, 2019). With billions of people logging onto SNS daily and spending hours communicating and interacting with others in various forms, there must be lessons to be learned towards the design of online distance learning experiences? In order to finalize the framing of this thesis, two general characteristics of SNS' are discussed briefly: (1) social affordances and (2) social contagion.

Social affordances of SNS. With many million users interacting effortlessly and communicating seamlessly with each other, SNS' are paragons of sociability. Social affordances are the active ingredients of the lively social behavior in these environments; they are the design choices and features within an application that allow for and facilitate interaction and communication (Bucher & Helmond, 2018). To better understand social affordances of SNS, we can turn to Pereira et al.'s (2010) framework of seven elements that are characteristic of social software: identity, presence, relationships, reputation, groups, conversation, and sharing.

Depending on goals and design philosophy of the application, different elements will be emphasized and, through this, different behavior is elicited (O'Riordan, Feller, & Nagle, 2012). For example, Facebook emphasizes presence, identity, conversation, and relationships, with sharing and reputation being less important. On the other hand, Youtube primarily emphasizes sharing, with conversation and identity being secondary and presence and relationships being mostly non-existent. LMS's like Moodle with a focus on managing students, courses, and activities, again have different affordances. Thus, depending on the experience we want students to have in the learning environment, certain affordances may be used to counteract limitations of LMS's like Moodle. One straightforward way of doing this could be to appropriate design choices and functionalities from SNS, the central idea being that, the more the learning environment resembles an SNS, the more sociable it becomes.

Social contagion. Analogous to pathogen contagion, social contagion is the idea that information can spread dynamically through a network of individuals. For example, emotional states like happiness can diffuse through a social network, so that this affect is shared by people no longer connected to the cause (Fowler & Christakis, 2008). Whereas in earlier theorizing it was thought that conscious processing accounted for the phenomenon, there is research to suggest that these processes can be more automatic and unconscious than previously thought (Hatfield et al. 1993). Moreover, these effects can even be observed in the absence of direct interaction and without individuals being aware of it, as shown in a large-scale Facebook study (Kramer et al., 2014). Here, researchers slightly manipulated the prevalence of positive or negative emotions presented in the news feed of Facebook users and observed that this had an effect on the emotional valence of the users' subsequent messages. Correspondingly, Ferrara & Yang (2015) found that the emotional valence that Twitter users were exposed to influenced the content of their own post. In a study conducted on SNS' Digg and Twitter, Hodas & Lerman (2014) were able to show that dynamics of social contagion depend on exposure to information, with higher information visibility increasing "infection probability".

Although not yet researched, it is plausible that similarly to emotional valence in SNS, socio-emotional information may also be susceptible to contagion in online learning environments. In this view, more visible and connected students may engage in sociable behavior in the learning environment and thereby influence perceptions of other students, who in turn perceive the environment to be sociable and/or social interaction to be plentiful. Perceptions of sociability may, thus, be transmitted between students. In some cases they may enact similar behavior, in others they may merely indicate higher values of socio-emotional variables when prompted. Through this, socio-emotional aspects may be fostered with only small interventions because the relevant information spreads through the network of students. Of course, this may only be possible in environments which allow for sufficient visibility of behaviors and interactions, i.e. are sufficiently similar to SNS.

Concluding this framing of the research, we have seen that, along with the general societal tendency toward technological mediation, online distance learning is similarly becoming an increasingly ubiquitous mode of education. Although adopting an environment-based perspective toward understanding behavior and perceptions in learning is not necessarily new, there is now an increased interest in considering the effects of online learning spaces on learning-related behavior and processes, making this an emerging line of research. To contribute to this, we can identify largely untapped research areas like ecological psychology and concepts like social contagion from non-learning research areas in order to help us frame the more specific topics of this thesis that will be presented in the following chapters.

3 STATE OF RESEARCH

3.1 Problems of online distance learning

With online distance learning now considered mainstream and more students than ever before enrolling in distance education courses (Seaman et al., 2018), addressing trademark issues that are typical for this mode of education becomes increasingly important, the most prominent being student attrition. Attrition is when students drop out of the program before completing their degree. There are many studies identifying factors that influence dropout rates in online and distance education (Willing & Johnson, 2009; Croxton, 2014; Kauffman, 2015; Hone & El Said, 2016; Selim, 2007; Park & Choi, 2009). However, first to formalize this phenomenon in a research framework was Tinto (1975) with the Student Integration Model. Here, students individual preconditions play a role, as much as their commitment, and system factors. Although seminal for student attrition in higher education, its ability to model the phenomenon for online distance education more specifically is limited (Rovai, 2003). Thus, even though it is well documented that online distance education is in many ways demanding and requiring a large degree of persistence (Allen & Seaman, 2014), research has yielded few reasonably general recommendations to improve on this.

This may be, to some extent, due to the fact that many of the reasons for dropout do not lend themselves to easy to quantification. Scheduling conflicts, family issues, financial problems, technical issue, and academic integration may all be part of the picture (Park & Choi, 2009). Also, the degree of flexibility afforded by online distance learning may not prove beneficial or liberating for everybody (Houlden & Veletsianos, 2019). Because these attrition factors can be very individual, systematically improving them may be especially challenging, no less due to some of them being largely outside of the realm of instructional design, educational technology, or even learning research in general (e.g., family issues, financial problems). For example, so called “non-traditional” students (Schuetze & Slowey, 2002), who make up a large part of online distance education are especially at risk of dropout (Choy, 2002), most likely due to their responsibilities outside of education (e.g. jobs and family). Some groups of non-traditional students face a particularly high risk of dropping out, such as female students, students with a migrant background, and fully-employed students (Stoessel et al., 2015).

Fortunately, one straightforward variable –students’ satisfaction with the learning experience– has been consistently shown to be positively associated with persistence in online distance learning (Levy, 2007; Schreiner, 2009; Park & Choi, 2009; Joo et al., 2011; Joo et al., 2013; Reio & Crim, 2013; Lee et al., 2013). According to this large body of research, students who are more satisfied with the learning experience are also more likely to continue investing

into their learning. With regards to a psychological explanation for this phenomenon, motivation is an obvious candidate (Joo et al., 2011); students satisfied with the learning experience will likely feel more motivated to continue their efforts in the face of attrition factors and are less likely to drop out. The relationship of satisfaction with persistence is important for online distance learning research, because satisfaction is a variable easily quantified. Therefore, exploring ways to systematically improve satisfaction is a straightforward and fruitful avenue to improving persistence.

Of course, satisfaction is not synonymous with learning. In fact, we know that such affective variables may only correlate moderately with actual learning (Sitzmann et al., 2010). Still, while learning achievement should be at the heart of every learning experience, cognitive learning gains are not the only relevant variable education. As humans do, we don't always optimize for maximum utility. Instead, perceptions and emotions play a huge part in decision-making. Facing many competing duties and interests, the chances of an online distance student logging into the learning environment today is dependent on the experience he or she had yesterday. If the experience was pleasurable and satisfying, chances of this occurring are higher. So even though satisfying experience do not always lead to learning, dropping out of a program due to unsatisfying experiences surely is the antithesis of learning success. This emphasizes the importance of affective variables like satisfaction, confidence, and perceived learning in designing online distance learning experiences (Richardson et al., 2010).

So what could be a general and scalable approach to making online distance learning more pleasurable and satisfying? We know that online distance students value social connections. This is confirmed in ample research demonstrating the importance of the socio-emotional dimension for the learning experience (Yu et al., 2010; Stürmer et al., 2018; Stone & O'Shea, 2019; Kaufmann & Vallade, 2020). A recent literature review confirms this by showing that, among program-related factors, low interaction with peers and tutors, as well as feelings of isolation are the most common challenges faced by students in online distance learning (Kara et al., 2019). Indeed, due to the mediated nature of learning in online spaces, socio-emotional aspects do not always emerge by themselves but instead need to be consciously designed for (Kreijns et al., 2003).

3.2 Social Networking Sites in Education

Due to the strong portfolio of social networking sites (SNS) in the socio-emotional realm, researchers and practitioners have seen them as panaceas for online distance education. This is exemplified in the many studies in which an SNS is investigated for its capabilities as a learning environment, with intentions to improving socio-emotional aspects for students. Much of this research has been conducted with Facebook (DeSchryver et al., 2009; Deng & Tavares, 2013;

Kurtz, 2014; Jong et al., 2014; Akcaoglu & Lee, 2018; Thai et al., 2019) but also other applications like Twitter (Lowenthal & Dunlap, 2009), Ning-based SNS' (Quong et al., 2018) and Elgg (Veletsianos et al., 2013). Consistent with the strenghts of SNS, these studies generally report successes in improving students' experiences in the socio-emotional realm.

Although promising, these approaches are not without pitfalls. One such pitfall is context collapse. Context collapse occurs when people are confronted with very different social contexts in a single online space (Vitak, 2012; Davis & Jurgenson, 2014). This may occur, for example, when Facebook is used to replace a LMS. Because Facebook has a real-name, single-account policy, students will then be confronted with a blending of their personal and academic spheres, what some students may consider an encroachment (Dennen & Burner, 2017). In fact, Smith (2016) showed that students were critical of this blend (referring to it as a "double edged sword"), as the personal space distracted from academic work but also due to potential role conflicts, for example seeing faculty's private life through their Facebook profile. Another problem that may occur with approaches where SNS are used to supplement the LMS, is the problem of environment split. Students now have to switch between platforms for their study-related activities, which some may find cumbersome and wasteful (Akcaoglu & Lee, 2018). Lastly, blindly appropriating non-educational technology to educational contexts may amount to inviting a trojan horse into the educational city. Foreign (i.e. non-educational) technologies carry their developers' world views, which are "revealed through the activities supported and encouraged by the technology" (Veletsianos, 2016, S.253). These may not necessarily be the ones intended by instructors or their institution and appropriating these technologies may have undesired effects. This is why some critics suggest that educators should be wary of handing over control of assumptions and notions of good education to corporate technologies that may follow a very different agenda (Lane, 2009; Veletsianos, 2016).

It is, thus, advisable to learn from what we know of the psychology of SNS' to improve online distance learning experiences in the socio-emotional dimension without bluntly inserting them and hoping that they will work their "social magic". If we instead use the body of knowledge to better understand how social aspects emerge and can be fostered through design, we further our understanding and improve practice. But what exactly is desired, when we talk about the importance of the socio-emotional dimension of online distance education? Remembering the introduction and fate of our protagonist in the Naked Sun, we might frame this as a question of the difference between the modes of seeing and viewing. What is required is a better understanding of the psychological mechanism underlying this difference and how to manage it in online distance learning practice. As we will see, the notion of social presence is eminently suitable for this endeavor, though it is not without issues.

3.3 Social Presence

Although presciently foreshadowed by Isaac Asimov in 1957, the term social presence wasn't coined until twenty years later by social psychologists Short, William, and Christie (1976), defining it as “the degree of salience of the other person in the communication and the consequent salience of the interpersonal relationships” (p.65). An evocative concept resonating with many, social presence has since been frequently reinvented and periodically applied towards understanding emerging technologies, and, thus, undergone many changes (Lowenthal, 2010; Lowenthal & Snelson, 2017). We can distinguish three early phases of research: social presence in telecommunications, social presence in computer-mediated communication, and social presence in online learning.

Social presence in telecommunications. In the 1970's, American telephone companies were working on developing audiovisual teleconference systems. Although still costly and bulky, they were seen as profitable future-bearing technologies to be pioneered in professional business contexts and later marketed to U.S. households. With these new technologies looming on the horizon and Marshall McLuhan's (1964) influential work of the previous decade still fresh on people's minds, psychology scholars busily investigated communication media in the 70's. Speculating on McLuhanesque media effects, they tried to understand how different communication channels may influence communication; its content, duration, and success (Short, 1974; Pye & Williams, 1977; Williams, 1978). It was against this backdrop that Short et al. (1976) wrote their book *The Social Psychology of Telecommunications*, in which they compared attitudes that people held toward different means of communication (e.g. face-to-face, audio-video, video, audio). They suggested that attitudes were largely influenced by the ability of the media to convey socio-emotional cues of communicators like facial expression, gesticulation, and intonation, either limiting or enabling the relative salience of the communication partner, what they defined as social presence (p.65). Although the perceived adequacy of communication media was expected to vary depending on the context at hand, the ability of the medium to convey social presence between communicators was seen as fixed (Short et al., 1976; Williams, 1977). Thus, they posited an assumed continuum of social presence bearing media, with face-to-face being at one end and audio-only at the other. The more socio-emotional cues are available, they theorized, the more communication partners would be perceived “as real social beings, with individual personalities, wishes, feelings, and aspirations” (Williams, 1977, p. 972). Hence, the more telecommunication media approximated the “richness” of face-to-face communication, the higher the degree of social presence (Daft & Lengel, 1986).

Social presence in computer-mediated communication. In the 1980's, as text-based computers-mediated communication (CMC) like e-mail and computer conferencing

became available, scholars noticed that these supposedly lean and unsociable media did not hamper interpersonal connections to the extent that was expected from the deterministic view of the previous decade (Hiltz, 1978; Myers, 1987). Instead, it seemed that, gradually, people were able to compensate for the lack of socio-emotional cues and that communicating through CMC need not be *per se* impersonal or depersonalized. In the social information processing view of CMC (Walther, 1992), social information conveyed through the exchange of messages is processed by communicators and, over time, mentally integrated into an impression of the other person (Walther, 1993). Through this, socio-emotional aspects of communication need not be hampered and experiences of social presence are not only possible but even likely to evolve (Walther & Burgoon, 1992). This shift away from a technologically deterministic view of social presence is called the *relational view* (Kehrwald, 2008). In this view, the communication media influences but not determines how lively, personal, and unmediated social interaction can feel. Therefore, social presence is a dynamically changing and evolving sense of other salient actors, influenced by characteristics of the communication medium. This view occupies the middle ground between technological and social determinism (Kreijns et al., 2011) and dominates our current understanding of social presence (Lowenthal, 2010; Kehrwald, 2010; Kreijns et al., 2013).

Social presence in online distance learning. When in the late 1980's and early 1990's first instances of internet technology proliferated, predecessors of our modern learning management systems emerged (e.g. EKKO, Paulsen & Rekkedal, 2001). As they were largely text-based, CMC became increasingly relevant for education. Again, scholars turned to social presence to understand mediated social interaction, this time in educational contexts. As in previous decades, they estimated that mediation posed a potential psychological barrier to truly interactive learning experiences. Pioneering the notion of social presence in the context of online distance learning, Gunawardena (1995) again found, in line with the relational view of social presence, that although low on socio-contextual cues, social interaction via CMC was not perceived to be impersonal or distant. Shortly thereafter, Gunawardena & Zittle (1997) connected social presence to satisfaction, presenting empirical evidence that students perceiving more social presence were also more satisfied with the learning experience, a finding well replicated over the years (Richardson & Swan, 2003; Swan & Shih, 2005; Richardson et al., 2017). This connection was influential, as researchers and practitioners now associated tangible effects with the concept, thus, making it a design consideration for learning experiences. Then, in 1999, the now ubiquitous Community of Inquiry was introduced (Garrison et al., 1999). This framework for educational CMC famously consists of teaching presence, cognitive presence, and social presence. Through this, social presence research accelerated and became a staple of online distance learning.

Since then, it has become a truism to assume that more social presence is always better. However, when we deconstruct social presence research, we see a landscape hardly warranting certainty. Three high-level challenges can be identified in the literature: (1) identification of social presence, (2) explanation of social presence, and (3) application of social presence.

Identification. In contrast to the consensus regarding its relational nature, there is surprisingly little agreement as to what social presence *actually is*. For example, some researchers define social presence as the degree to which the communication partner is perceived to be “real” and “there” in mediated communication, whereas others focus on an affective connection or feelings of community (Lowenthal & Snelson, 2017). This is exemplified in two highly cited definitions, with Gunawardena & Zittle (1997) defining it as “the degree to which a person is perceived as a ‘real person’ in mediated communication” (p.9), while Tu & McIsaac (2002) define it as a “measure of the feeling of community that a learner experiences in an online learning environment” (p.131). These are clearly not the same. Different ideas under the same name are subject to the jingle-fallacy, where one erroneously concludes that studies examine the same construct, while actually they only carry the same name. This fallacy, first formulated by Thorndike (1904), has been found to be present in many areas of psychology (e.g. Higgs & Lichtenstein, 2010; Larsen & Bong, 2016; Weidmann et al., 2017) and does, to some extent, apply to social presence research as well.

To make matters worse, social presence definitions are not only different from each other but often convoluted within themselves. Lowenthal (2010) made this point by referring to a continuum of definitions. On one end of the continuum there are definitions limited to the “realness”-aspect. On the other end are those focusing on the affective component. However, “like most continuums the majority of researchers find themselves somewhere in the middle - placing a little bit of emphasis on an emotional connection - rather than on the ends of the continuums” (Lowenthal, 2010, p.120). Due to this, definitions of social presence tend to lack clear analytic boundaries and, thus, are hard to empirically distinguish from other related variables. One variable most often conflated with social presence, especially within the Community of Inquiry framework, is the notion of learning climate and sense of community (Kreijns et al., 2014).

Of course, differing and convoluted definitions lead to differing and convoluted, i.e. problematic measures. Indeed, social presence research abounds in ways of measuring the construct, with little consistency at its core. Kreijns et al., (2011) show that many of these measures do not exclusively measure social presence, but instead “measure varying aspects of an amorphous set of variables – including social presence, to varying degrees” (p.371). This was

also criticized by Biocca et al., (2003) when they noted the confounding of social presence with its correlates or effects. Indirectly confirming this, a recent meta-analysis showed that effects associated with social presence (i.e. satisfaction & perceived learning) strongly differed depending on the social presence measure used (Richardson et al., 2017).

Explanation. Besides identification of the central construct, social presence literature also has deficits with regards to explanation. A comprehensive explanation of social presence should proffer two strands of elaboration, effects associated with social presence as well as mechanisms, processes, and conditions governing its emergence. The former is important to estimate how beneficial and, thus, practically relevant the construct is towards known issues of online distance learning. The latter serves purpose of modeling and predicting social presence such that our understanding of the phenomenon is enhanced and design recommendations become possible.

In terms of effects associated with social presence, the literature is eclectic; variables investigated in relation to social presence are manifold. Still, there are some communalities, like social interaction (Tu & McIsaac, 2002, Hostetter & Bush, 2006; Wei et al., 2012), intimacy and immediacy (Wei et al., 2012), learning achievement (Zhan & Mei, 2013; Joksimovic et al., 2015; Maddrell et al., 2017), as well as satisfaction and perceived learning (Richardson et al., 2017). Sense of community and group cohesion are also often investigated, albeit rarely as a separate variables, but instead as implicit (affective) component of social presence (Kreijns et al., 2014).

Many of these investigations share similar problems. First, as explained above, many of these studies operate with different understandings, and, as a result, measures of social presence. Thus, despite the ostensive empirical robustness of some associations (i.e. satisfaction and perceived learning, see Richardson et al., 2017), this lack of consistency should make us cautious when appraising the literature. Second, these studies frequently fail to provide theoretical rationales for their causal assumptions. That is, there are seldom psychological mechanisms suggested to explain these relationships. An especially peculiar case is the relationship of social presence to learning achievement. Although scholars seem convinced of the efficacy of social presence with respect to learning achievement (e.g. Wei et al., 2012; Molinillo et al., 2018), there are seldom plausible theoretical accounts for this link and evidence that does not rely on self-report measures of learning is, at best, mixed (Chang, 2014; Wise et al., 2004; Zhan & Mei, 2013; Joksimovic et al., 2015; Maddrell et al., 2017, Dahlstrom-Hakki et al., 2020).

Having established effects of social presence and, thus, determining its practical relevance, it becomes increasingly important to understand the construct better in terms of the

processes governing its emergence. This means delineating factors that facilitate or hamper this particular psychological phenomenon so that we can model and predict when, how, and where students may experience social presence. This side of the equation has been mostly neglected due to the focus on effects of social presence (Cui et al., 2013). While the focus on effects has been sensible to determine the merits of social presence, as a result, knowledge of causes, conditions, and mechanisms governing social presence is still lacking (Biocca et al., 2003; Öztok & Kehrwald, 2017). This is unfortunate, as such knowledge is needed to arrive at a model of social presence that has explanatory power, one that can predict social presence in different situations. For example, what are facilitating or hampering conditions? Are there different psychological mechanisms at work? Are all students similarly prone to the phenomenon? Finally, answers to some of these questions could then be put to action by translating this knowledge into tentative and testable design recommendations.

Application. From the previous issues follows a third problem of social presence research. If scholars can't agree what they are talking about when they talk about social presence and, as a result, can neither explain how social presence comes to be nor what to expect from it, we can hardly expect empirically tested and practically relevant design recommendations. But of course, this is a necessary step to actually cash out on the decades-long research, by actually improving online distance learning in the socio-emotional realm. If we can agree what social presence is and how it is causally connected to other variables and mechanisms, then the logical next question is, how to make social presence happen.

In fact, the literature is practically silent with empirically supported design recommendations for enhancing social presence (Cui et al., 2013). Only a few papers have been concerned with strategies for establishing social presence in mediated learning environments (Aragon, 2003; Dikkers et al., 2017; Lowenthal & Dunlap, 2018). Strategies include, for example, orientations or personalized feedback (Lowenthal & Dunlap, 2018) or more general recommendations like "Cultivate connections to build community" (Whiteside et al., 2017, p.270). Although certainly a good start, these strategies are still somewhat under-researched in that their effectiveness has not been rigorously tested, for example through experimental designs. In addition, they do not flow from a theoretical model that has grappled with identification and explanation of social presence, yielding a set of testable hypothesis that together may form a theoretical account of how social presence works. As an example, there is some tentative evidence from two studies that small group size may be beneficial for enhancing social presence, but, unfortunately, Lowry et al., (2006) did not conduct their research in an educational setting and Akcaoglu & Lee (2016) measured the related but not synonymous sociability construct. In another study, the latter authors supplemented the online learning environment with Facebook groups and

found that this was beneficial for social presence (Akacoglu & Lee, 2018). Although generally interesting and practically relevant, here, too, hypotheses are not generated based on a thorough identification and explanation of social presence. Because of this, findings do not flow back to inform a model or theory, but instead remain one-off and intertwined with a specific technology, making cumulative progress difficult.

A new phase for social presence research. When introducing the concept or reviewing the state of research, scholars often talk about “social presence theory” (e.g. Gunawardena, 1995; Lowenthal, 2010; Cui et al., 2013; Dahlstram-Hakki et al., 2020). This is a euphemism at best. Regarding the relatively high bar for what can be considered a scientific theory, our understanding of social presence is a far cry from achieving this status. For example, a first step in theory development would be a reasonable degree of consensus regarding the *what* (Whetten, 1989). Researchers would have to agree on what social presence is, at least to a degree that establishes a common basis. As we have seen, this is not the case, the literature being rich with differing interpretations, convoluted definitions, and problematic measures. Next up, is the *how* and *why* (Whetten, 1989). Here, associated notions and variables should be introduced and causalities between them hypothesized, i.e. what are antecedents of social presence, what are its effects, and which mechanisms can be identified to explain these causalities? Modeling relationships and testing them empirically is a central step in theory development (Jaccard & Jacoby, 2020). Here, too, social presence research is lacking. Although effects have been investigated, we know next to nothing of the psychological processes associated with social presence. Without this, systematic testing and probing is difficult and empirical findings will be little more than one-off data points in a scattered literature, as is currently the case. From insight regarding *how* and *why*, the *who*, *where*, and *when* may slowly form. These conditions place constraints on a possible social presence theory. Social presence can’t be used to explain *everything* in mediated spaces and its emergence may depend on a set of diverse internal and external factors. Thus, sensitivity to context is important (Whetten, 1989). Acknowledging this, hypotheses flowing from a theoretical model may become more nuanced and thereby increasingly relevant for practice.

There have been numerous critiques of the eclectic state of research. Already back in 2003, Biocca et al. noted the need for a more robust theory of social presence. They, too, found problems with definitions, measurement, and explanatory power of social presence. More recently, Öztok and Kehrwald (2017) wondered if it may be time to “kill” social presence. That is, should we rid ourselves of this shapeshifting concept and find a more refined alternative to help us understand mediated learning experiences? If we are to understand socio-emotional perceptions and behavior in mediated environments, we need a theory that systematically

constrains our thinking, allows for the generation of testable hypotheses, and, ultimately, leads to a body of knowledge than can guide practice in the field. Social presence has a long history of research that is as colorful as it is replete with problems. It is at once a notion that has stood the test of time while also being so evocative that it has been repeatedly reinvented, in frequently inconsistent ways. Should we, in fact, decide to not “kill” social presence but instead move forward with it, it is time for a new phase of social presence research; one in which the concept is taken so seriously that identification, explanation, and application of social presence are tackled in a systematic and cumulative manner. First steps toward this will be taken throughout the empirical articles in this thesis. To do this, all investigations are rooted in a theoretical model that will be revised and expanded based on the empirical findings

3. 4 The SIPS Model

In this research thesis, building on the work of Kreins (2004) and Kreijns et al. (2013), social presence is understood as part of a nomological network of variables, that together explain socio-emotional aspects of online distance learning. It consists of four main variables: *Sociability*, *Social Interaction*, *Social Presence*, and *Social Space*; in short: *SIPS*. Figure 1 shows these variable, with direction of arrows indicating the hypothesized causal flow. Each of these variables will be explained in short below, while also explaining the assumed causalities and mechanisms between them.

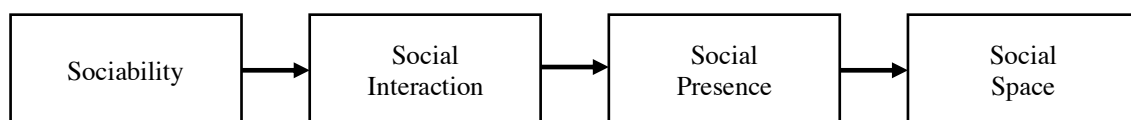


Figure 1. Central variables of the SIPS model

Sociability. Online learning environments may differ in their ability to facilitate socio-emotional aspects of the learning experience, that is, they can be more or less *sociable*. Thus, sociability is an attribute of the learning environment and sociable learning environments are designed such that they allow for and facilitate quick, easy, and informal social interaction (Kreijns et al., 2007; Abedin et al., 2012). Design elements that determine the degree of perceived sociability of an environment are called social affordances (Gao et al., 2010; Kreijns et al., 2013). For example, an interface showing which others students are currently online and willing to chat, allows for the initiation of spontaneous and informal communication. Although relational in the sense that these perceptions of sociability may differ between students or change over time, the main driver of this variable is design. That is, depending on how the environment is designed in terms of social affordances, its perceived degree of sociability can be very high or rather low. An environment lacking in this regard may be called *aloof*.

Social interaction or student-student interaction. A sociable learning environment increases the chances of social interaction occurring, because it lowers the psychological barrier for engaging with others (Kreijns, 2004). Social interaction, here constrained to student-student interaction (Moore, 1989), is the process of contacting peers and exchanging messages with them in the learning environment. Past research has often focused on cognitive or task-centered interaction between students (Abedin et al., 2012), although this restriction is considered a pitfall (Kreijns et al., 2003). An environment where social interaction feels easy and conversation flows naturally between students is desirable for both cognitive and motivational purposes (Borokhovski et al., 2012). According to the SIPS model, a third important reason is that social interaction is the main driver of social presence and a sound social space.

Social presence. In line with previous work by Kreijns et al. (2011), the SIPS model assumes a clear and narrow definition of social presence, one that is rooted in the original conception of Short et al., (1976). Accordingly, in this thesis, social presence is defined as *the psychological phenomenon that the other is perceived as ‘real’ in the communication, the subjective feeling of being with other salient social actors in a technologically mediated space*. In other words, “the sense of ‘being there, together’, when ‘being there’ does not involve a physical presence” (Öztok & Kehrwald, 2017, p.5). In this understanding, social presence is a perception of other salient actors in an environment, where actual physical presence is impossible. Thus, it is somewhat of an illusion, a semblance of non-mediation between social actors (Lombard & Ditton, 1997). As students interact socially, they will slowly gather an idea of their communication partners, how they talk and think, what they like, and what their goals are; in sum, what makes them ‘them’. This process is called impression formation and it is a natural consequence of continued social interaction (Walther, 1993). Having these impressions in the mind’s eye, a student may perceive his/her communication partner to be ‘real’ and ‘there’ even though they are not, thus constituting social presence.

Social Space. Students may perceive a network of interpersonal/social relationships between them and others in the learning environment. By slowly building connections through social interaction with other students, students may feel that they are learning in a well-connected social space (Kreijns et al., 2013). This notion relates to the second part of the original Short et al. (1976) definition and encapsulates what is often associated with concepts like sense of community, group cohesion, beneficial learning climate, or more recently, connectedness (Thai et al., 2019; Macleod et al., 2019). In contrast to many conceptualizations of social presence (e.g. Garrison, 2007 or Wei et al., 2012), the SIPS model proposes that the concepts of social presence and social space, although related, are distinct. This distinction is important, so that the purported effects of these variables, for example on desirable outcomes like satisfaction or learning

achievement, may be distinguished. Besides social interaction, social presence is also expected to be a driver of a sound social space, by evoking a more natural sense of face-to-face conversation and, thus, benefitting positive group dynamics.

These are the basic variables constituting the SIPS model and the general causalities assumed to be underlying their relationship. In later chapters, more elaborate and detailed instances of this model will be provided in order to showcase the relevant variables and relationships investigated in the research papers constituting this thesis.

SECTION B

EMPIRICAL STUDIES

4 RESEARCH QUESTIONS

As demonstrated in the literature review, social presence is a prominent and popular construct for researchers investigating socio-emotional aspects of online distance learning. But there are also many problems, one of which is proper definition and measurement of social presence. Aiming for a solid grasp of the central construct, the first empirical step of this thesis should be concerned with the establishment of a valid measure based on a precise definition. Thus, the first research question is:

Research question 1: *How valid is a measure based on a narrow and precise definition of social presence?*

This question will be addressed in chapter 1 “What social presence is, what it isn’t, and how to measure it”

Although these steps are necessary, they are not sufficient for theoretical advancement. Besides getting a grasp of the construct, a primary goal should be the establishment of what we can expect of social presence in terms of improving online distance learning. That is, we need to investigate correlates and effects of social presence. The literature is rich with such supposed associations but, critically, many of the studies have worked with convoluted definitions and measures. It is important that we distinguish effects of social presence from other related socio-emotional variables, like social interaction or social space. Then, if they are associated with meaningful effects, these variables should similarly be considered in future research alongside social presence, thus highlighting the need for a more integrative model of socio-emotional aspects. Due to this, an appraisal of purported effects under conditions of narrow definition and valid measurement is in order.

Research question 2: *What are correlates and effects of social presence and how are they different from other socio-emotional variables?*

This question will also be addressed in chapter 2 “Explaining social presence and the quality of online learning with the SIPS model” and chapter 3 “Designing sociable online learning environments: An affordance enrichment approach”

Besides establishing correlates and effects and, thus, appraising the practical relevance of social presence, an important contribution to a theory of social presence includes the investigation of determining factors. That is, social presence needs to be embedded in theoretically grounded and empirically testable nomological net of related variables. Here, relevant psychological

processes and potential limiting factors should be investigated in order to get a more complete picture of how social presence emerges and under which conditions. With empirical support for this nomological net and its causal assumptions, we can begin to speak of social presence theory.

Research question 3: *How does social presence emerge in online learning environments, i.e. what are determinants, precursors, and associated mechanisms?*

This question will be addressed in chapter 2 “Explaining social presence and the quality of online learning with the SIPS model”, chapter 4 “Pathway to Presence: An investigation of ambient awareness in online distance learning environments”, and chapter 5 “A trait-level view of social presence: Exploring personality in online distance learning environments”

Once meaningful effects are established and relevant psychological mechanisms and processes of social presence are identified, the logical next step is to find ways of fostering these variables. With regards to finding ways of improving online distance learning in the socio-emotional realm, rigorous research should eventually yield design recommendations. With empirically supported guidelines we can create learning conditions in which online distance students are more likely to have sound socio-emotional experiences. As there is a great deal of research on the psychology of social networking sites, e.g. how people interact via social networks, how their design influences behavior, and how affective aspects play a large role in this, we are able to derive testable design recommendations for improving learning environments in this regard.

Research question 4: *How can social presence and other socio-emotional variables be fostered through the design of online learning environments?*

This question will be addressed in chapter 3 “Designing sociable online learning environments: An affordance enrichment approach”

5 EMPIRICAL STUDIES

The following five subchapters will consist of the studies which, together, make up the empirical part of this thesis. Each study is preluded with a short introduction, a figure schematically representing the constructs and relationships of interest for this particular study, as well as its publication status and citation.

The following subchapters are thus (1) “What social presence is, what it isn’t and how to measure it”, (2) “Explaining social presence and the quality of online learning with the SIPS model”, (3) “Designing sociable online learning environments: An affordances enrichment approach”, (4) “Pathway to presence: An investigation of ambient awareness in online learning environments”, (5) “A trait-level view of social presence: Exploring personality in online distance learning environments”.

This order of studies was chosen according to the logic of the ideas presented in them. Due to the imponderability of manuscript preparation and publication processes, publication dates of these studies may not reflect this logic.

5.1 What social presence is, what it isn't, and how to measure it



This study is a deep dive into the social presence construct. Leaving other SIPS variables aside, the state of our understanding of social presence is discussed thoroughly. Drawing on the original definition of social presence and more recent work, sharp distinctions of what social presence is and isn't are made. As an important step for cumulative research, a measure of social presence based on a narrow definition is presented. This measure attempts capturing the breadth and depth of the social presence experience. Scale development was conducted with Item Response Theory. Although still a work in progress, this measure shows good psychometric properties and is suitable for use in social presence research.

Contents

- Introduction
- Convolved Definitions
- Correlates of social presence
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 - Achievement
- A new measure for social presence
 - Rasch Measurement Model
 - Uni-dimensional definition
- Method
 - Sample
 - Procedure
 - Rasch Analysis
- Results
- Conclusions

This study has been published as:

Weidlich, J., Kreijns, K., Rajagopal, K., & Bastiaens, T. (2018, June). What Social Presence is, what it isn't, and how to measure it: A work in progress. In *EdMedia+ Innovate Learning* (pp. 2142-2150). Association for the Advancement of Computing in Education (AACE).

Introduction

In 1976, Short, Williams, and Christie coined the term social presence, defining it as “the degree of salience of the other person in the communication and the consequent salience of the interpersonal relationships” (p.65). In educational settings that rely in large part on computer-mediated communication (CMC), like online learning and computer-supported collaborative learning, social presence is considered an important aspect of the learning experience. This is because, unlike face-to-face communication, CMC usually conveys little socio-emotional cues. This is especially true for text-based asynchronous communication (e.g. message boards), that are still a large part of the educational technology landscape, for example in learning management systems. In order to understand what makes CMC effective for students, social presence has become an important variable to consider. Researchers have linked social presence to important outcome variables of online learning, e.g. satisfaction and perceived learning (Gunawardena & Zittle, 1997; Richardson & Swan, 2003; Richardson, Maeda, Lv, & Caskurlu, 2017), online course retention (Liu, Gomez, & Yen, 2009) and online social interaction (Tu & McIsaac, 2002). In addition to this, social presence has a prominent role in the community of inquiry (COI) framework, alongside teaching presence and cognitive presence (Garrison, Anderson, & Archer, 2000; 2010). However, important issues regarding definition and measurement of social presence remain (Kreijns, Jochems, & Van Buuren, 2011; Kreijns, Van Acker, Vermeulen, Van Buuren, 2014; Weidlich & Bastiaens, 2017; Lowenthal & Snelson, 2017, Öztok & Kehrwald, 2017). These may even call into doubt some of its purported effects.

Convolved Definitions

When considering research on social presence, one is confronted with a plethora of different definitions and, consequently, operationalizations of social presence. This may be, in part, because social presence theory stems from a rather diverse line of research (Cui, Lockee, & Meng, 2013). Yet, it is a problem for cumulative science if researchers lack a clear and consistent way of talking about a phenomenon. Lowenthal’s Continuum (2010) illustrates this lack of clarity and consistency. He argues that even though most scholars will attribute their understanding of social presence to the concept proposed by Short et al. (1976) “[d]efinitions of social presence, at least for researchers of social presence and online learning, tend to fall on a continuum” (p. 120). The continuum reaches from social presence as the salience of the other person in the interaction - whether someone is perceived as being “present” and “real” - to social presence as the salience of the interpersonal relationships, including whether or not there is an interpersonal emotional connection. Because most researchers tend to fall somewhere in the middle of this continuum (Lowenthal, 2010), many definitions of social presence are not only different from each other but also convolved. That is, they lack clear analytic boundaries, that allow differentiating social presence from other related variables.

For illustration, consider operational definitions in these three classical social presence papers. For Rourke, Anderson, Garrison, and Archer (2000), social presence is “the ability of learners to project themselves socially and affectively into a community of inquiry” (p.50). For Tu & McIsaac (2002), it is a “measure of the feeling of community that a learner experiences in an online learning environment” (p.131). Gunawardena & Zittle (1997) stay closer to Short et al.’s (1976) original proposition by defining it as “the degree to which a person is perceived as a ‘real person’ in mediated communication” (p.9). one might think that these three definitions refer to different phenomena. Yet, they are all called social presence. For a more in-depth discussion on how social presence has been defined by different researchers, see Lowenthal & Snelson (2017).

Of course, these issues of definition will affect attempts at measuring social presence. There has been very little consistency, yet great quantity in the way researchers have measured social presence. For example, Rourke, Anderson, Garrison, & Archer (2001) established twelve indicators to analyze online discussions in terms of demonstrated social presence. As such, it is a behavioral measure. More recently, a COI survey instrument was developed (Arbaugh, Cleveland-Ines, & Diaz, 2008) and validated (Diaz, Swan, Ice, & Kupczynski, 2010; Carlon, Bennett-Woods, & Berg, 2012). Outside of the COI framework, other social presence surveys - old and new - are abundant. Gunawardena (1995), Gunawardena & Zittle (1997), Tu (2000), Picciano (2002), Wei, Chen, & Kinshuk (2012), Kim (2010) and Kim, Kwon, & Cho (2011) all developed different questionnaires to assess social presence. Because these measures are based on convoluted definitions of social presence, they too may be convoluted. Indeed, Kreijns, Kirschner, Jochems, & Van Buuren (2011) and Kreijns et al. (2014) show that many of these measures do not exclusively measure social presence, but instead “measure varying aspects of an amorphous set of variables – including social presence, to varying degrees” (p.371).

Correlates of Social Presence

In order to reach a clearer understanding of how social presence may emerge and in which ways it may affect learning processes in online environments, it is important to untangle social presence from its correlates. In other words, to clearly define what it is and what it isn’t. Although many scholars hold that social presence is multidimensional and multi-faceted (see chapter “unidimensional definition” below), we are skeptical of this proposition. The following sections will review concepts that are associated, and some of them conflated, with social presence. Because experimental evidence in social presence research is still rare, the causality of these correlates is not always readily established. Wherever possible, either based on theoretical grounds or preliminary data, the hypothesized causal relationship will be indicated.

Student-Student Interaction or Social Interaction. Many researchers have mentioned the close relationship between social interaction and social presence. For example, Tu & McIsaac (2002) assert that social presence will increase social interaction and Wei et al. (2012) propose a model in which social presence is a predictor variable for social interaction. In contrast, Kim et al (2011) determine social interaction (or interactivity, as it is referred to in their study) as a predictor for social presence in the regression model. Differently again, Biocca et al (2003) state, that social presence is the theory of mediated interaction and therefore includes the concept of interaction.

However, there is no support for the notion that social interaction is in any way an aspect or dimension of social presence. Because they are defined differently and have been shown to be empirically related, yet distinguishable (Kim, Song, & Luo, 2016; Weidlich & Bastiaens, 2017), there is no reason to conflate these two variables.

Regarding causality, arguments can be made for social interaction as predictor as well as a dependent variable. A predictor, because social interaction is a necessary first step in getting acquainted with and forming an impression of a communication partner. A dependent variable, because a higher degree of social presence may make mediated social interaction feel less artificial and, thus, more likely. Also, because communication in social systems is complex, a bidirectional view of social interaction and social presence influencing and reinforcing each other may be the most accurate portrayal of this relationship (Kreijns et al., 2013).

Cohesion, Climate, and Community. When talking about social presence, many researchers actually seem to talk about aspects relating to group cohesion, learning climate, and/or sense of community. This is clear when looking at the way they define and operationalize social presence, as it is common to include these aspects either implicitly or explicitly. This is also the case in the arguably most prominent appropriation of social presence, the Community of Inquiry framework. For example, Garrison (2009) defines it as “the ability of participants to identify with the community (e.g. course or study), communicate purposefully in a trusting environment, and develop interpersonal relationships by way of projecting their individual personalities” (p.352). Clearly, group cohesion, learning climate, and sense of community are dominant in this definition, with “the degree of salience of the other person in the communication” (Short et al. 1976) going largely unmentioned.

This conflating of rather different concepts and variables under the guise of social presence has been criticized extensively and ways forward have been proposed (Kreijns et al. 2014; Lowenthal & Snelson, 2017; Öztok & Kehrwald, 2017; Weidlich & Bastiaens, 2017). For

example, Kreijns et al. (2004) and Kreijns et al. (2014) introduce the concept of social space, the extent to which salient interpersonal relationships among students in the learning environment emerge. As such, it includes aspect of cohesion, climate, and community while being true to the second part of the Short et al. (1976) definition, “the consequent salience of the interpersonal relationships”.

In terms of causality, it is expected that social presence is a prerequisite for the development of a sound social space. This has also been empirically validated by Weidlich & Bastiaens (2017). In addition, it is likely that a sound social space will, in turn, reinforce social presence, which again implies a possible bidirectional relationship.

Satisfaction & Perceived Learning. A recent Meta-Analysis by Richardson, Maeda, Lv, and Caskurlu (2017) demonstrated moderately large correlations of social presence with satisfaction ($r=0.56$) and social presence with perceived learning ($r=0.51$). Yet, some of the included studies measure social presence based on a Community of Inquiry questionnaire, which is based on a convoluted definition of social presence. Other studies in this meta-analysis are based on the original Gunawardena & Zittle (1997) and adapted Richardson & Swan (2003) operationalization of social presence. A look at the items of these measures confirms that they, too, measure aspects of community to some extent, e.g. “The moderators created a sense of online community” (Gunawardena & Zittle, 1997, p. 15) and “this activity allowed me to form a sense of online community” (Richardson & Swan, 2003, p.68). Because these studies either implicitly or explicitly measure more than perceptions of social presence, the actual relationship of social presence to important outcome variables like satisfaction and perceived learning is not as clear as the meta-analysis would imply. In addition, some items of these social presence measures actually seem to tap satisfaction implicitly (“The quality of learning for this activity was excellent”, Richardson & Swan, 2003), making inference on this relationship somewhat of a tautology.

At this time, there is a scarcity of studies that assess the relationship of these subjective outcome variables with a precisely defined social presence. Although a seemingly large amount of research has been conducted on the assessment of these relationships, many more studies actually are necessary. In one instance, Weidlich & Bastiaens (2017), using the Kreijns et al. (2011) measure, demonstrated no relationship of social presence with satisfaction, yet a small but significant one with perceived learning.

Regarding causality, if there is a relationship, it is intuitive that social presence will be a predictor for these outcomes variables, but is questionable whether they will retrospectively influence the amount of social presence perceived during learning.

Achievement. Although scholars champion the importance of social presence for online learning and CSCL, there is still little evidence for an effect of social presence on actual learning achievement. The line of research concerned with the Community of Inquiry Framework proposes that the presences “can enhance or inhibit the quality of the educational experience and learning outcomes” (Garrison et al., 2000, p.92). Yet, most studies within this framework or on the subject of social presence specifically do not assess actual measures of cognitive learning or achievement. Instead they use self-report measures of satisfaction or perceived learning, which may be interesting in its own right (Richardson, Maeda, & Swan, 2010), but surely isn’t the same as an objective measure. A recent study using the Community of Inquiry survey found no evidence for a relationship of social presence (and the other presences) with any of three instructor-assessed measures of learning achievement (Maddrell, Morrison, & Watson, 2017). Another study used the behavioral indicators of Rourke, Anderson, & Archer (2001) to examine which of them had predictive capabilities for the final grade (Joksimovic, Gasevic, Kovanovic, Riecke, & Hatala, 2015). They found that some of them did, with “continuing a thread” and “complimenting, expressing appreciation” explaining 16% of the variance in the final grade. In a similar vein, a study by Hostetter (2013) found that social presence measured via a compound variable of the aforementioned indicators and the Richardson & Swan (2003) survey scored higher in an objective measure of learning.

Although Joksimovic et al. (2015) and Hostetter (2013) seem to provide preliminary evidence pointing to a relationship between social presence and achievement, it should be noted that their measures are not based on a precisely defined social presence and instead include other variables, via the Richardson & Swan (2003) survey and e.g. the indicator “complimenting, expressing appreciation”. As of now, we are not aware of any research that has successfully linked a precisely defined concept of social presence with an objective measure of learning.

In terms of causality, achievement is expected to be a dependent variable. Similar to satisfaction & perceived learning, social presence is seen as a causal factor, expected to explain differences in learning processes.

A New Measure for Social Presence

Because the literature is rife with convoluted definitions of social presence and problematic measures, we propose an alternative measure of social presence. It is based on a uni-dimensional definition of social presence that emphasizes ‘realness’ of the other in the interaction. Unlike previous social presence measures, it is based on the Rasch Measurement Model.

Rasch Measurement Model. Rasch Analysis is a psychometric approach to the development of tests and measures (Rasch, 1960). Rasch techniques allow to assess and document the measurement functioning of items within a measure (such as determining their level of difficulty; their fit with the Rasch Measurement Model) while at the same time doing the same for the respondents (such as determining their ability to endorse items; their fit with the Rasch Measurement Model). Through this, strengths and weaknesses of the test or instrument may be assessed systematically and the measure can be modified accordingly in an iterative process, leading to a measure with strong psychometric properties. It overcomes many of the weaknesses of Classical Test Theory, the still dominant paradigm of scale development (see for example Smith, Conrad, Chang, & Piazza, 2002). One important requirement of the Rasch Measurement Model is uni-dimensionality of the construct. Based on our understanding of social presence, this requirement is met and, thus, Rasch Analysis an adequate approach.

Uni-dimensional Definition. We define social presence as the psychological phenomenon that the other is perceived as ‘real’ in the communication, the subjective feeling of being with other salient social actors in a technologically mediated space. In other words, “the sense of ‘being there, together’, when ‘being there’ does not involve a physical presence” (Öztok & Kehrwald, 2017, p.5). This is based on the first part of the original definition of Short et al. (1976). Accordingly, items that assesses social presence should all tap this realness aspect and none of the items should tap correlates of social presence like social interaction, group cohesion, learning climate, sense of community, or satisfaction.

Kreijns et al. (2011) presented a social presence measure addressing the realness of the other in the communication. In this measure, a distinction was made between a synchronous and an asynchronous communication setting. We now believe this distinction should not have been made because it limited the applicability of the measure, as some learning contexts may not include synchronous communication setting or vice versa. In addition, this distinction violates the invariance assumption necessary for Rasch Analysis. Finally, the small number of items in this measure may point to a potential underrepresentation of the social presence construct (Messick, 1996). This makes necessary a new and improved social presence measure that is based on a precise definition and demonstrates sound psychometric properties.

Regarding the uni-dimensionality aspect, we deviate from many other social presence researchers who argue that social presence is a multidimensional construct. For example, Tu (2002a, 2002b) saw as dimensions of social presence 1) social context, 2) online communication, 3) interactivity, 4) system privacy, and 5) feelings of privacy. However, he actually identified these dimensions as variables affecting the degree of social presence. Wei, Chen, and Kinshuk

(2012) saw as dimensions of social presence, 1) co-presence, 2) intimacy, and 3) immediacy. These two latter dimensions were also put forward by Short, Williams, and Christie (1976) as we have seen above, but Short, Williams, and Christie declared social presence to be a factor contributing the level of intimacy and enabling immediacy.

We, thus, conclude that many social presence measures exist in the literature sometimes incorporating correlates and/or antecedents of social presence as if they are dimensions of it. Therefore, we decide to consider social presence as a unidimensional construct. As result, a 16-item raw social presence measure was constructed using ordered categorical polytomous ratings: each item has seven rating scale steps (1 = totally disagree, 2 = disagree, 3 = somewhat disagree, 4 neither disagree or agree, 5 = somewhat agree, 6 = agree, 7 = totally agree). Table 2 depicts the social presence measure. In this table, items that were found not fitting the Rasch Measurement Model are greyed, hence they are not part of our social presence measure. The mean *M* and standard deviation *SD* are given for the original 7 rating scale step measure and the collapsed 5 rating scale steps measure. Please refer to the analysis section for the modification steps of scale development.

Sample. Respondents were 324 students at FernUniversität in Hagen, Germany. This convenience sample consists of students enrolled in either B.Sc. Psychology or B.A. Educational Science. Table 1 shows the demographics for this sample. There was a total of 241 students enrolled in Educational Science, 71 in Psychology, spread over three semesters: winter semester of 2015/2016, winter semester of 2016/2017, and summer semester of 2017. Of these students, 260 were female, 55 were male. Mean age was 32.3 years. Please note that due to missing values, numbers may not add up to total N.

Table 1: Demographics of sample

	N	Educational Science, B.A.	Psychology B.Sc.	Female	Male	M _{age}
WS 15/16	134	99	34	100	26	32.2
WS 16/17	112	92	19	99	12	33
SS 17	78	50	18	61	17	31.8
Total	324	241	71	260	55	32.3

Procedure. Students were recruited for the survey through the learning management system Moodle, in which most learning activities took place. Nearing the end of each semester, they were asked to participate in the survey with no course credit or reward attached to participation. For a duration of two weeks, a link in the learning environment directed them to the survey, which was created via LimeSurvey (<http://www.limesurvey.org>). The 16 items of the raw social presence were only a small part of a larger survey concerned with student's perceptions and experiences in the learning environment. Students were informed that the questions are

concerned with their subjective experiences and preferences and were asked to answer accordingly. The survey took them a total of about 15 minutes to complete. It was administered over the course of three semesters, as mentioned earlier.

Table 2: Social Presence Measure

Nr Item	Item	7 rating scale steps			5 rating scale steps			N*
		<i>M</i>	<i>SD</i>	item measure	<i>M</i>	<i>SD</i>	item measure	
	In this learning environment ...							
SP01	... it feels as if we are a face to face group	2.93	1.47	39.30	2.22	.86	31.59	304
SP02	... it feels as if I cannot escape from the eyes of my fellow students	2.08	1.18		1.75	.70		299
SP03	... it feels as if I deal with 'real' persons and not with abstract anonymous persons	4.12	1.70	33.37	2.91	1.00	26.38	304
SP04	... I feel to be together with my fellow group students	3.22	1.51		2.38	.85		304
SP05	... I can form distinct impressions of some of my fellow students	3.66	1.51	35.90	2.59	.86	28.99	298
SP06	... it feels as if all my fellow students are 'real' physical persons	4.42	1.69	31.76	3.07	1.03	25.14	302
SP07	... I imagine that I really can 'see' my fellow students to be in front of me	2.98	1.56	39.04	2.23	.91	31.56	304
SP08	... my fellow students feel so 'real' that I almost believe that we are not virtual at all	3.01	1.62	38.91	2.24	.95	31.46	304
SP09	... I have my fellow group members in my minds' eye	3.09	1.70		2.30	.99		302
SP10	... all of my fellow students imagine that they really can 'see' me to be in front of them	2.82	1.46		2.17	.86		301
SP11	... all of my fellow students feel that I am a 'real' physical person	3.60	1.70	35.94	2.62	1.00	28.54	300
SP12	... it feels as if all my fellow students and I are in the same room	2.77	1.55	40.11	2.11	.89	32.49	304
SP13	... it feels as if I can really 'touch' my fellow students	1.86	1.12		1.61	.71		304
SP14	... it feels as if all my fellow students and I are in close proximity	2.45	1.48	41.80	1.96	.88	33.64	304
SP15	... I strongly feel the presence of my fellow students	3.09	1.68	38.54	2.29	.95	31.16	301
SP16	... all of my fellow students feel my presence	2.78	1.53		2.15	.90		302

* N is calculated by taking the total number of respondents (=324) and subtracting 1) respondents who did not answer this item, 2) respondents who had a misfitting answer on this item—the misfitting answer was marked as missing, and 3) respondents who completely misfit the Rasch Measurement Model (Nmisfit = 20).

Rasch Analysis. Rasch analyses were performed on all 16 items of the raw social presence measure (see Bond & Fox, 2015 and Boone, 2016 for the procedure). All misfitting items and respondents were removed (the 6 removed items are greyed in Table 2). Furthermore, the rating steps 3 and 4 as well as the rating steps 5 and 6 were collapsed in order to get proper probability curves required by the Rasch Measurement Model. In other words, it appeared that seven rating scale steps were causing some confusion in the respondents to select the right step suggesting the use of five rating scale steps (see Figure 2 and 3).

TABLE 21.1 All.sav
INPUT: 324 PERSON 16 ITEM REPORTED: 304 PERSON 10 ITEM 7 CATS WINSTEPS 3.90.0

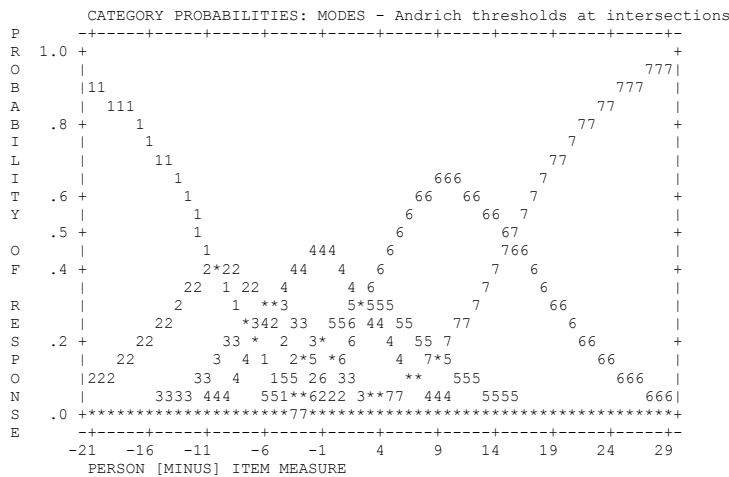


Figure 2: Probabilities of the seven rating scale steps.

TABLE 21.1 All.sav
INPUT: 324 PERSON 16 ITEM REPORTED: 304 PERSON 10 ITEM 5 CATS WINSTEPS 3.90.0

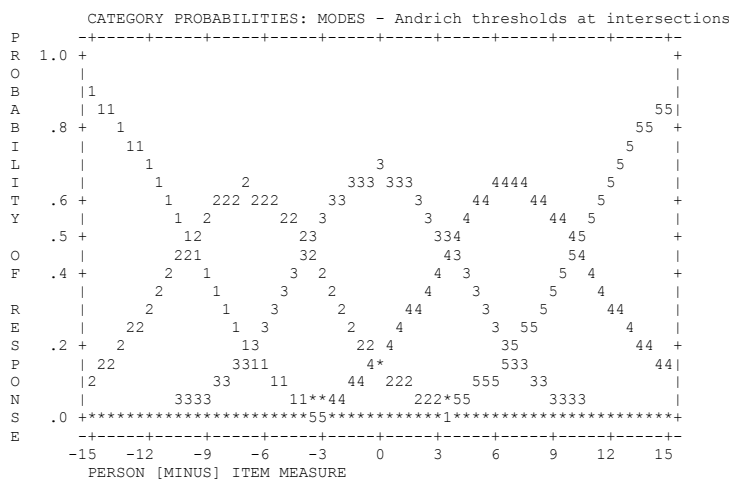


Figure 3: Probabilities of the five rating scale steps.

Results

In Figure 4, two Wright maps (Wright & Masters, 1982) are shown in which the relative difficulty levels of the items are placed next to the relative ability levels of the respondents to endorse items. That is, the left Wright map shows these difficulty levels when looking at the rating scale steps per item, whereas the right Wright map shows these levels when looking at the items themselves; thus, the Wright map at the right is a summary of the Wright map at the left. Therefore, the current social presence measure is moderate in differentiating respondents with low perceptions of the realness of the other whereas it can excellently differentiate persons with high perceptions of the realness of the other. This indicates that there is some underrepresentation

of the construct (Messick, 1996) which would undermine the statistical validity (i.e., the reliability) of the social presence measure (Baghaei, 2008). Nevertheless, item and person separation indices were very good (Boone, Staver, & Yale, 2014, p. 231): item separation index was 10.01 (should be at least 2.5 for the analysis of groups) and person separation index was 3.01 in case minimum extreme persons were included (should be at least 3.0 to represent an excellent level of separation). Classical test theory Cronbach's alpha was .93.

TABLE 12.6 All.sav										TABLE 12.2 All.sav										
INPUT: 324 PERSON 16 ITEM REPORTED: 304 PERSON 10 ITEM 5 CATS WINSTEPS 3.90.0																				

MEASURE PERSON - MAP - ITEM - 50% Cumulative probabilities (Rasch-Thurstone thresholds)										PERSON - MAP - ITEM										
<more> <rare>										<more> <rare>										
44		+						SP14	.5			+								
43		+						SP12	.5			+								
42		+						SP07	.5			+								
								SP15	.5											
41	.	+										+								
40		+										+								
								SP05	.5											
39		+						SP11	.5			+								
38		+										+								
37	.#	+						SP14	.4			+								
	.##								SP03 .5											
36	T+							SP12	.4			T+								
	## T								SP06 .5			T								
35	.#	+						SP08	.4			+								
								SP15	.4											
34	##	+										+								
	.##																	SP14		
33	##	+										+								
	S							SP05	.4			S						SP12		
32	.###	+						SP11	.4			+								
	.###	S										S						SP01	SP07 SP08	
31		+										+						SP15		
	.###																			
30	.#####	+M							SP03 .4			+M								
								SP14	.3											
	#####																			
29	.	+						SP12	.3			+						SP05		
	.#####								SP06 .4									SP11		
28	.#####	+						SP08	.3			+								
		S						SP15	.3			S								
27	#####	+										+								
	.#####	M										M						SP03		
26	.#####	+										+								
	#							SP05	.3											
25	#####	+T						SP11	.3			+T						SP06		
	#####																			
24	#####	+										+								
	.							SP14	.2											
23	.#####	+										+								
	####								SP03 .3											
								SP12	.2											
22		S+										S+								
	#####							SP01	.2											
								SP08	.2											
21	.#	+						SP15	.2			+								
20	.##	+										+								
	##																			
19		+						SP05	.2			+								
	###							SP11	.2											
18		+										+								
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17	.#	+										+								
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16		+										+								
	.#																			
15		+						SP06	.2			+								
14		+										+								
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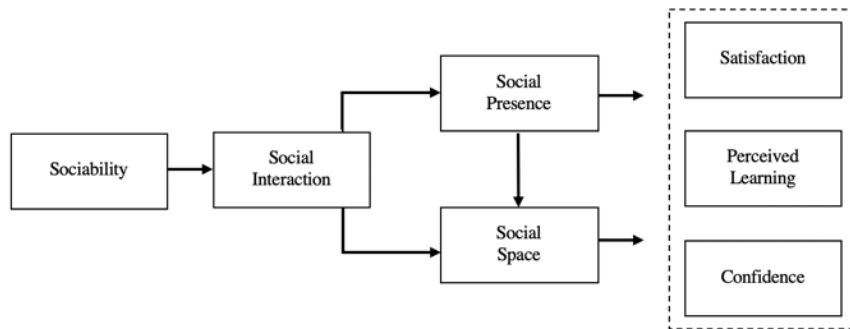
Figure 4: Wright-maps.

Conclusions

In this paper we've outlined problems surrounding research on the concept of social presence. We conclude that there are many different conceptions of social presence in the literature and most of them are convoluted to some extent. That is, they include other related variables as if they were a part of, or dimension of social presence. Because every conception is different, the effect of social presence on outcome variables is not as unambiguous as the literature might suggest.

We believe that a precisely defined construct that is measured in a psychometrically sound way is the necessary groundwork for cumulative research on social presence. Therefore, we presented a work in progress on an alternative social presence measure. It is based on a precise and unidimensional definition of social presence and developed through Rasch Measurement Model. Our analysis lead to a preliminary 10-item measure showing solid psychometric properties that may be used to assess social presence. However, because our analysis suggests that the construct of social presence isn't yet fully represented in our measure, it is still a work in progress.

5.2 Explaining social presence and the quality of online learning with the SIPS model



This study is an introduction to the SIPS model, which attempts to address drawbacks of earlier social presence research. Social presence is defined narrowly and distinguished from other related variables, measured based on this definition, and embedded in a model explaining surrounding variables. In addition, desirable outcome variables are investigated in relation to the model. Using Partial-Least-Squares Structural Equation Modeling, central assumptions of the SIPS model are supported by the data. Associations with satisfaction, perceived learning, and confidence are more nuanced than previously expected from the literature.

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Introduction

Through the development of increasingly sophisticated technology and the simultaneous increasing prevalence of the Internet, the use of online technology to deliver instruction has continually increased over the past two decades (Allen & Seamann, 2013; Snyder, Dillow & Hoffmann, 2009, Forum Distance Learning, 2013). However, there are issues concerning the nature of mediated communication in learning. A common concern is, for example, that the mediated nature of online learning may hamper a sense of interpersonal connection and relatedness between students. Where face-to-face-learning offers a rich social context and opportunities to interact in a non-mediated fashion, online learning makes use of technology to mediate social interaction.

Researchers have turned to social presence theory in order to understand the mechanisms governing beneficial learning climates and interpersonal connections among online learners. As a consequence, social presence is now considered an important factor of successful online and distance learning. However, many unresolved issues remain.

This study attempts to advance the SIPS (*Sociability, Social Interaction, Social Presence, Social Space*) model that deals with some of these issues by distinguishing social presence from other relevant social aspects of online learning. The model is analyzed with Partial Least Squares Structural Equation Modeling (PLS-SEM) and its predictive capabilities regarding the outcome variables satisfaction, perceived learning and confidence are assessed. The results suggest that creating a sociable learning environment is a viable approach to fostering socioemotional aspects that ultimately benefit the quality of the learning experience.

Social Presence Research

In 1976, Short, Williams, and Christie coined the term social presence, defining it as “the degree of salience of the other person in the communication and the consequent salience of the interpersonal relationships” (p.65). Two concepts associated with social presence are Wiener & Mehribian’s (1968) immediacy and Argyle & Dean’s (1965) intimacy. Immediacy refers to the psychological closeness between communicators, whereas intimacy refers to the degree of affiliation between communicators.

As online learning has become more popular and technically sophisticated, research began considering socio-emotional aspects of online learning. Seeking to understand the relevance of social interaction and interpersonal relationships between online learners, scholar turned to social presence theory. Social presence has since been successfully linked to important outcome variables of online learning, e.g. satisfaction (Gunawardena & Zittle, 1997), perceived

learning (Richardson & Swan, 2003), online course retention (Liu, Gomez, & Yen, 2009) and online social interaction (Tu & McIsaac, 2002). In addition to this, social presence has a prominent role in the community of inquiry (COI) framework, alongside teaching presence and cognitive presence (Garrison, Anderson, & Archer, 2000; 2010). The COI framework posits that meaningful learning takes place when an online learning experience has sufficient amounts of all three presences.

As a consequence, social presence is now considered an important factor of successful online and distance learning (Richardson, Swan, Lowenthal, & Ice, 2016; Kim, 2010; Garrison & Arbaugh, 2007). However, several issues regarding the conception and measurement of social presence remain (Kreijns, Jochems, & Van Buuren, 2011; Lowenthal, 2010; Lowenthal & Dunlap, 2014; Kreijns, Van Acker, Vermeulen, Van Buuren, 2014).

Convoluting Definitions and Problematic Measurements. As social presence theory stems from rather diverse lines of research (for reviews, see Lowenthal, 2010; Cui, Lockee, & Meng, 2013), it is not surprising that the research literature has yielded a similarly diverse assortment of conceptions regarding social presence. Even though most scholars will attribute their understanding of social presence to the concept proposed by Short et al. (1965), there is no agreed upon and universally used definition. As Lowenthal (2010) points out: “Definitions of social presence, at least for researchers of social presence and online learning, tend to fall on a continuum” (p. 120). The continuum reaches from social presence as the salience of the other person in the interaction, whether someone is perceived as being “present” and “real”, to social presence as the salience of the interpersonal relationships, including whether or not there is an interpersonal emotional connection. Note that these opposite ends of the continuum both represent an extension of the original Short et al. (1965) definition. Because most researchers tend to fall somewhere in the middle of this continuum (Lowenthal, 2010), many definitions of social presence are not only different from each other but also convoluted. Thus, Biocca, Harms, & Burgoon (2003) summarize that definitions of social presence are (a) vague, overly broad, or circular, and (b) blur the distinction between the psychological state of social presence and the behavioral effects thereof. This is a problem, because different and convoluted definitions of a construct will yield different and convoluted ways of measuring it, respectively.

Accordingly, there has been very little consistency in the way researchers have measured social presence. Seeking behavioral evidence, Rourke, Anderson, Garrison, & Archer (2001), analyzed text-based online discussion in search of indicators demonstrating social presence. They established twelve indicators, grouped into three categories (affective response, interactive response, and cohesive response) as evidence of social presence in online discussions. More

recently, a COI survey instrument was developed (Arbaugh, Cleveland-Ines, & Diaz, 2008) and validated (Diaz, Swan, Ice, & Kupczynski, 2010; Carlon, Bennett-Woods, & Berg, 2012). However, this instrument is reported to still have issues regarding factor structure (Carlon et al., 2012) and its alignment with the social presence indicators (Lowenthal & Dunlap, 2014).

Outside of the COI framework, other social presence surveys - old and new - are abundant. Gunawardena (1995), Gunawardena & Zittle (1997), Tu (2000), Picciano (2002), Wei, Chen, & Kinshuk (2012) and Kim, Kwon, & Cho (2011) all developed different questionnaires to assess social presence. Because these measures are based on convoluted definitions of social presence, they too may be convoluted. Indeed, Kreijns, Kirschner, Jochems, & Van Buuren (2011) and Kreijns et al. (2014) show that many of these measures do not exclusively measure social presence, but instead “measure varying aspects of an amorphous set of variables – including social presence, to varying degrees (p.371). This a problem when trying to empirically link desirable outcome variables (e.g. satisfaction) to social presence. How can researchers reliably rule out that these effect aren’t in fact a result of the hidden variables in social presence scales?

Untangling Social Presence and Social Space. Alternatively, there are strong arguments for distinguishing two important variables in online learning. These two variables can be directly traced back to the two-step definition of Short et al. (1965), namely (1) the salience of the other in the communication and (2) the consequence of this, the salience of the interpersonal relationship. This means restricting social presence to the first part of this definition, the “presentness” and “realness” of the other, as proposed by Kreijns, Kirschner, Jochems, & Van Buuren (2011). In this understanding, social presence is the illusion of non-mediation; although participants are aware of using mediated communication, they may perceive the communication partner as actually real and present, not unlike face-to-face communication. This sensation may come about by a process called impression formation (Walther, 1993). Connected but not identical to social presence is the second part of the definition, in which a learner may feel that there is an interpersonal relationship with a communication partner. As a perceived relationship creates a tie between communication partners, a network of interpersonal relationships may develop between learners in online learning. This network is called a social space (Kreijns, Kirschner, Jochems, & Van Buuren, 2004).

Besides the basic argument, that researchers, in order to establish meaningful empirical and theoretical connections, should strive for validity in their measurements, there are three additional arguments for the distinction of social presence and social space.

(1) Research should build on the existing body of literature concerning social aspects like group cohesion, sense of community and learning climate in online learning that has established relationships to satisfaction (Wu, Tennyson, & Hsia, 2010; Dewiyanti, Brand-Gruwel, Jochems, & Broers, 2007) and perceived learning (Rovai, 2002). These constructs are similar or possibly even identical to social space. By confounding these variables with social presence, researchers have done little to achieve clarity.

(2) A recent meta-analysis by Richardson, Maeda, Lv, and Caskurlu (2017) has shown that the relationship between social presence and satisfaction is highly variable, with the scale type being a significant moderator of this relationship. This is unsurprising, when considering that these measures are mostly based on convoluted definitions of social presence. If future research does not differentiate these social constructs in a clear and transparent fashion, research results will continue to be highly variable, thereby hindering a solid foundation of research.

(3) There is an obvious pragmatic benefit. It is much easier to explain and measure two distinguishable concepts, rather than a single concept with multiple, still widely contested dimensions. Although multidimensional constructs are not uncommon in psychological research, it seems that in social presence research, the explanatory potential of this complexity falls short of being an improvement. Instead, it seems to have caused confusion, as illustrated in the differences of operationalizations between researchers.

Social Presence and Social Interaction. Many researchers have mentioned the close relationship between social presence and social interaction. However, similar to the epistemic status of social presence, the relationship between these two constructs remains contested. Tu (2002) holds that interaction contributes to social presence. In another study, Tu & McIsaac (2002) state that “an increase in the level of online interaction occurs with an improved level of social presence” (p.131), thereby steering clear of causal assumptions. Wei et al. (2012) however, propose a causal model in which social presence is a predictor variable for learning interaction. Similarly, Hostetter & Busch (2006) also argue that social presence directly leads to social interaction. In contrast, Biocca et al. (2003) state, that social presence theory is the theory of mediated social interaction and therefore includes the concept of interaction. Gunawardena (1995) is less concrete in detailing the role of interaction by stating that “social presence is a factor of the medium, as well as that of the communicators and their presence in a sequence of interaction” (p.151).

Different wordings and causal assumptions aside, it should be noted that social presence is primarily concerned with student-student interaction, according to Moore’s (1989) classic

distinction of types of interaction in online and distance learning. This type of interaction should be called social interaction to encapsulate all possible types of interaction between students, not only interactions regarding the content or learning task, but also informal, personable types of social interaction. Kreijns, Kirschner, & Jochems (2002) propose a classification of social interaction, in which types of interaction may be distinguished by context (task vs non-task) and dimension (cognitive vs socio-emotional). Depending on which type of social interaction is dominant, the learning experience and learning environment will be different.

This research is not concerned with a specific kind of social interaction, but instead with all types of interactions that may occur between students, for example asynchronous communication via message board or synchronous communication in a chat session. Because social interaction is always the first step in forming an impression of another person (i.e. there is simply no contact without social interaction), it is also an important precursor for the emergence of social presence. Hence, in this research social interaction is understood to lead to social presence, instead of being a result of or being interleaved with social presence.

The SIPS Model for Social Aspects of Online Learning. Because there is little agreement on many basic aspects surrounding social presence research, it is difficult to investigate reliable design recommendations to facilitate social aspects in online learning (Cui et al., 2013). Hence, a coherent model or framework encompassing the relevant socio-emotional variables and their relationships is in order.

Kreijns et al. (2004; 2013) propose one such framework for computer-supported collaborative learning (CSCL) environments. It contains the relevant social variables and their relationships to each other. The framework has advantages over other frameworks by actually suggesting an explanation of how these aspects come to evolve. Some but not all of these relationships have been empirically supported through previous research (for a detailed discussion, see Kreijns et al., 2013). Since much of social presence research has been conducted without specifically having a CSCL context in mind, an adaptation to explain and predict socio-emotional aspects for online and distance learning is still needed. This adaption, called the SIPS model (for *Sociability*, *Social Interaction*, *Social Presence*, and *Social Space*), is shown in Figure 5. The relevant variables and their definitions are listed in Table 4.

The model is based on an ecological approach to fostering social aspects. Here, the sociability of the learning environment is expected to be a predictor for how much social interaction will take place because learning in sociable environment will facilitate and promote social interaction (H1). This will in turn foster the emergence of social presence (H2-H3) and the

development of a sound social space (H4-H5). The emergence of social presence is determined by a process called impression formation. When social interaction takes place, communicators will form an increasingly distinct impression of one another (H2). As these impressions become increasingly salient through ongoing social interaction, a perception of non-mediation -social presence- will emerge (H3). When meaningful social interaction takes place, learners will come to demonstrate positive communicative behavior that is indicative of a positive climate and strong group cohesion (H4). Perceiving these behaviors, learners will attest to the development of a sound social space (H5). Additionally, feelings of social presence will facilitate the development of a sound social space by evoking a sense of face-to-face conversation (H6).

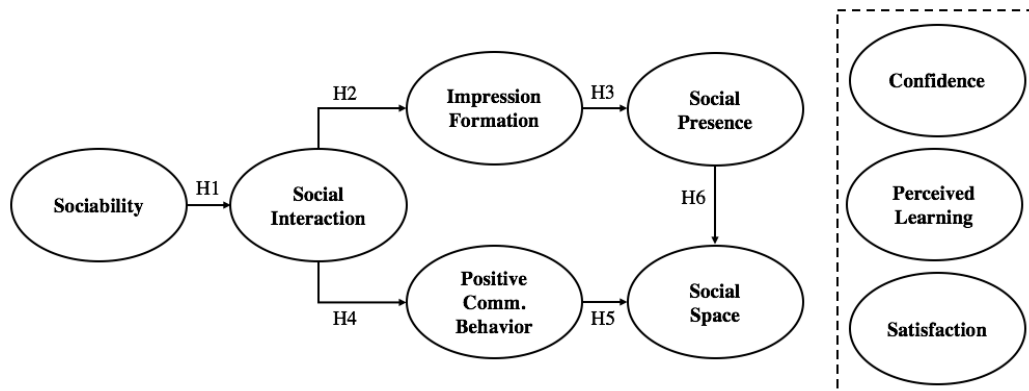


Figure 5. The SIPS model for Online Learning, adapted from Kreijns et al. (2013)

This model is streamlined in three ways. (1) It excludes the variables social affordances and pedagogical techniques, as these factors are beyond the research interest of this study. (2) Also, the model doesn't account for bidirectional relationships, but instead focuses on the expected primary causal flow of constructs. (3) In addition, it departs from Kreijns et al. (2004; 2013) by not expecting sociability to be a direct precursor to social presence. The original model holds that sociability will influence social presence directly, as well as through social interaction. However, sociability as the perceived quality of the learning environment to support social interaction (Kreijns et al. 2004), by definition is not expected to influence social presence directly, but only through social interaction (see chapter 2.3). In this conception, sociability will only be perceived by the learner to the extent that he may have a higher incentive to interact socially with his peers.

Method

This method sections describes the research questions and hypotheses, the data collection procedure, as well as the instruments used.

Research Questions and Hypotheses. The SIPS model has not been subject to validation attempts. For this reason and because it has been adapted in the previously mentioned ways, the first research question concerns the validity of the model. The model is valid if the hypothesized relationships are consistent with the data (H1, H2, H3, H4, H5, H6). This is true if the path coefficients are significant. In addition to this, the predictive capabilities of the model will be examined. This is done via effect sizes (f^2) and explained variance (R^2).

Because theoretical considerations make the often established effect of social presence on satisfaction questionable, this connection will be tested (H7). In addition, two other outcome variables will be investigated regarding their relationship to social aspects of online learning. As there is no established theory of how exactly the outcome variables are related to each other and to the model, there is only one other hypothesis. Common sense suggests that perceived learning is a predictor for satisfaction and confidence (H8).

Research question 1: *Is the SIPS-model of online learning valid and what are its predictive capabilities?*

H1: Sociability is a predictor for social interaction in the learning environment

H2: Social Interaction predicts impression formation

H3: Impression formation predicts social presence

H4: Social interaction predicts positive communicative behavior

H5: Positive communicative behavior predicts a sound social space

H6: Social Presence predicts a sound social space

Research question 2: *How do the outcome variables relate to social aspects of online learning?*

H7: Satisfaction is predicted by a sound social space instead of social presence

H8: Perceived learning is a predictor for confidence and satisfaction

Data collection. The data was collected in an Educational Science Bachelor's class at the largest German distance education university, FernUniversität Hagen. The class's primary learning activities take place in Moodle. Typical methods of communication are threaded message boards and bi-weekly non-mandatory chats. Data was collected over the course of one semester resulting in 162 responses. Online data collection via Limesurvey accounted for 42% of the data. Data collection via print survey accounted for the 58%. 26% of participants were enrolled in a

psychology bachelors program, 74% in an educational science bachelors program. Also, 75% of respondents were female, 25% male.

Results for Little's test for MCAR (missing completely at random) are not significant, Chi-Square = 3322,805, DF = 3401, $p = ,828$. Therefore, deletion of cases with too many missing variables is not expected to introduce bias (Allison, 2001; Graham, 2009). A threshold of 10% missing values per case was selected, resulting in the exclusion of 27 cases. The final amount of participants for this study is $n=135$.

Sensitivity power analysis with G*Power 3.1 (Faul, Erdfelder, Lang, & Buchner, 2007) for linear multiple regression, fixed model R^2 deviation from zero, alpha level 0.05, power 0.8, $n=135$, maximum number predictors 2, calculates a sensitivity for effect sizes as low as $f^2= 0.07$.

Instruments. Scales that have been previously validated were entered directly into the structural model. Therefore, social presence, social space, sociability and social interaction were assessed in the context of measurement model evaluation. However, some small modifications were necessary to account for the context of this sample. The social space scale was originally developed in a CSCL context. The items relating specifically to this context were excluded from data collection. Also, the Social Presence scale consists in part of items that directly relate to synchronous chat activities. As only 33% of participated in these non-mandatory chat activities during the semester, these items were also excluded from further analysis.

New items that have not yet been validated, were analyzed regarding their factor structure, using principal axis factoring, direct oblimin rotation with an eigenvalue of 1. Impression formation was excluded from this, because it is a formative measure (see chapter 4.1). Table 3 shows the results of the analysis. In all cases, the Kaiser Mayer-Olkin Measure of Sampling Adequacy exceed 0.5, which is considered the threshold. Bartlett's Test of Sphericity is also significant at the 0,01 level in every case. Items correlating lower than 0.3 and factor loadings of less than 0.7 resulted in exclusion of the item.

Table 3. Factor Analysis of new items

	KMO-Sampling Adequacy	Variance accounted for	Items deleted
Satisfaction	.824	59,2%	3, 6
Perceived Learning	.756	66,8%	-
Pos. Comm. Behav.	.786	62,2%	4
Confidence	.666	72,5%	2

To assess the normality of distribution in the data, Kolmogorov-Smirnov and Shapiro-Wilk test were applied. Significant values for the following scales suggest a possible non-normality: social presence, social interaction, perceived learning, positive communicative behavior, confidence. Therefore, additional graphical inspection of normality is warranted. Upon inspection of histograms and Q-Q plots of the variables, non-normality was confirmed for social presence (positively skewed), social interaction (Kurtosis) and perceived learning (negatively skewed). Positive communicative behavior and confidence show no substantial deviation from normality.

Table 4. List of variables for this research

Variable	Description	Measure	Example Item
Sociability	The perceived quality of the learning environment to facilitate social interaction.	Kreijns et al. (2007), adapted for non-CSCL context;	"This learning environment allows for spontaneous informal conversation"
Social Interaction	The amount and frequency of communication between a given student and his peers.	Wei et al. (2012)	"I often discussed learning issues with others"
Impression Formation	The degree to which students have gathered a distinct impression of their peers.	New measure, similar to Walther (1993)	"I have a distinct impression of the appearance of the other student"
Social Presence	The psychological sensation of the other being "there" and "present".	Kreijns et al. (2011), adapted for non-CSCL context	"When I have asynchronous conversations, I feel that I deal with very real persons and not with abstract anonymous persons"
Pos. Comm. Behavior	The degree to which positive and constructive communications are perceived.	New measure	"The discussions and conversations that I was aware of were always friendly and constructive"
Social Space	The network of interpersonal relationships between students.	Kreijns et al. (2004), adapted for non-CSCL context	"Students took the initiative to get in touch with others"
Perceived Learning	The extent to which the student feels he has gained knowledge about the class subject.	New measure	"When thinking about my learning in this online class, I would consider it a success"
Satisfaction	The extent to which the student is content regarding all aspects of the learning experience.	New measure	"This online class met my expectations"
Confidence	The extent to which the student has an optimistic view regarding his/her mastery of the subject.	New measure	"I feel that I am able to competently answer questions about the content of this online class"

Analysis

The analysis was conducted via Partial Least Squares Structural Equation Modeling (PLS-SEM). PLS-SEM is a flexible and powerful approach to multivariate analysis and is widely used in many social science disciplines. Lately, the method has also become increasingly popular in educational research (e.g. Wu, Tennyson, & Hsia, 2010; Fillio, Limayem, Laferriere, & Martha, 2009, Caio, Aijna, & Hong, 2013; Hamari, Shernoff, Rowe, Collier, Asbell-Clarke, & Edwards,

2016). It differs from covariance-based approaches (CB-SEM) by maximizing the explained variance of the dependent latent constructs. In this, it is most suitable to assess predictive capabilities of a hypothesized causal model. Hence, researchers should choose PLS-SEM over CB-SEM in situations in which the substantive theory is not yet fully developed and the primary goal is prediction of target constructs (Chin, 1998; Hair, Ringle, & Sarstedt, 2014). As detailed in Chapter 2, social presence theory is still eclectic and the relationship of its key constructs to dependent variables is not yet fully understood. In addition to this, three major advantages of PLS-SEM are applicable in this research. (1) PLS-SEM works well with smaller sample sizes. Note, however, that the present sample size is well above any rule of thumb (Hair et al., 2014) (2) PLS-SEM makes no distributional assumptions, allowing for parameter estimation with non-normal data. As not all variables are normal in this research (see Chapter 3.3), a non-parametric approach is appropriate (Chin, 1998). (3) PLS-SEM is able to include formative measurement models into the structural model. As shown in Chapter 4.1, impression formation is measured formatively.

The assessment of a PLS model is a two-step process. In a first step, the measurement model is assessed in regard to reliability and validity. Once the measurement models adequacy is established, the second step is the assessment of the structural model for its capacity to predict outcome variables. Data was analyzed with SmartPLS 3.0 (Ringle, Wende, & Becker, 2015)

Measurement Model Evaluation. Reliability of scales is assessed via Composite Reliability (Table 5). All scales are well above the threshold of 0.7. The validity of the measurements is assessed through convergent and discriminant validity. The average variance extracted (AVE) satisfies the requirement for convergent validity of 0.5 for all constructs, after weaker loading indicators (<0.6) of the sociability and social space scale have been removed. The Fornell & Larcker (1981) criterion for discriminant validity requires that the values in the matrix diagonals are all greater than the off-diagonal elements (Table 5).

Table 5. Discriminant Validity, Average Variance Extracted, and Composite Reliability

	1	2	3	4	5	6	7	8	AVE	CR
Sociability (1)	0.72								0.52	0.90
Social Interaction (2)	0.54	0.93							0.86	0.97
Social Presence (3)	0.45	0.34	0.91						0.82	0.90
Pos. Comm. Behav. (4)	0.49	0.46	0.43	0.79					0.62	0.87
Social Space (5)	0.49	0.42	0.44	0.70	0.76				0.57	0.82
Perceived Learning (6)	0.32	0.25	0.31	0.30	0.37	0.81			0.66	0.91
Satisfaction (7)	0.42	0.18	0.32	0.52	0.48	0.64	0.77		0.59	0.90
Confidence (8)	0.29	0.32	0.32	0.46	0.37	0.49	0.33	0.86	0.73	0.90

Whereas reflective measures are said to be caused by the underlying construct, formative measures represent instances in which the indicators themselves form the underlying construct (Hair, Ringle, & Sarstedt, 2011). In contrast to reflective measures, the items in a formative measure are not interchangeable, might relate to very distinct aspects of perception, and therefore do not necessarily intercorrelate. As a result, the assessment of a formative measure differs from assessment of a reflective measure.

In the present model impression formation is measured formatively. We suggest that impression formation consists of gathering a rich set of impressions other learners. Taken together, these impressions constitute a mental model that one learner can have of the other. This approach is based on Walther's (1993) approach to measuring impression formation. The items were derived from characteristics that a student could gather of his peers through interacting in the learning environment, for example personality, competence, and age.

Convergent validity is demonstrated through significant correlations between the formative indicators and a global item ("I was able to gather rich impressions of my peers"). Table 6 shows that all formative indicators correlate with the global item for impression formation at the .001 level. Problems with collinearity can be assessed through Variance Inflation Factor (VIF). A VIF value lower than 0.2 or higher than 5 is said to be indicative of potential issues in this respect (Hair et al., 2014). Table 6 shows that no indicator approaches these values. Lastly, the absolute contribution of an indicator to its construct is given by outer loadings. As can be seen in table 6, every outer loading exceeds the threshold of .50 with $p < 0.001$, thereby showing a large contribution in absolute terms.

Table 6. Correlations with criterion item, outer loadings of indicators, and variance inflation factor

	Correlation with criterion	Outer Loadings	VIF
Name	0.37***	0.66***	1.3
Appearance	0.41***	0.67***	1.5
Age	0.43***	0.73***	2.3
Personality	0.49***	0.75***	1.8
Profession	0.47***	0.78***	2.5
Learning Style	0.45***	0.75***	1.6
Competence	0.44***	0.76***	2.3
Interest	0.45***	0.75***	2.0

*** = $p < 0.001$

Structural Model Evaluation. After the appropriateness of the measurement model has been established, it is now possible to assess the structural model in order to provide evidence for the proposed theoretical relationships. This is done by examining the relationships between constructs, using the path coefficients β , effect size f^2 and the predictive capabilities of the model, using R^2 (Chin, 2010).

As there is no strong substantive theory of how the outcome variables are related, three separate models regarding the prediction of the outcome variables perceived learning, satisfaction, and confidence are tested. Table 7 shows the results of these model estimations. The significant paths of these models will be used to extend the basic model to include the outcome variables. Figure 6 shows the extended SIPS model.

Table 7. Path coefficients of all possible predictor variables on outcome variables

Predictor Variables	Dependent Variables		
	Perceived Learning	Satisfaction	Confidence
Perceived Learning	1	0.552***	0.407**
Satisfaction	-	1	-0.042
Confidence	-	-0.028	1
Social Interaction	0.071	-0.094	0.132
Social Presence	0.189*	0.056	0.102
Social Space	0.258*	0.300***	0.135

*** = $p < 0.001$; ** = $p < 0.01$; * = $p < 0.05$

An important step in evaluating the model is the assessment of the coefficient of determination (R^2) of the endogenous (dependent) constructs. R^2 is the amount of variance explained in a given dependent construct by its predictor. Because R^2 increases even if a non-significant but correlated construct is introduced to the model, R^2 adjusted should also be considered, as this penalizes increasing model complexity. Table 8 shows that these two measures converge on approximately the same values. All variables show a higher R^2 than the recommended minimum of 0.10 (Falk & Miller, 1992) and are therefore adequately explained by the model.

Table 8. R^2 and R^2 adjusted

	R^2	R^2 adj.
Social Interaction	0.29	0.28
Impression Formation	0.32	0.31
Social Presence	0.22	0.21
Pos. Comm. Behav.	0.21	0.20
Social Space	0.52	0.51
Perceived Learning	0.17	0.16
Satisfaction	0.48	0.47
Confidence	0.24	0.23

The path coefficients represent the strength of the hypothesized relationships between constructs. Both path coefficients β and R^2 are shown in Figure 6, along the arrows and in the circles, respectively. To assess the significance of the path coefficients, a bootstrapping procedure was run with the recommended subsample value of 5000 (Hair et al., 2014). The results of this are shown in Table 9.

To determine the effect sizes, Cohen's f^2 is assessed. This value shows the magnitude of the relative effect a predictor construct has on its dependent construct. Table 9 shows these effect sizes in parentheses. According to Cohen's (1988) guidelines for effect sizes, values of 0.02, 0.15 and 0.35 indicate a small, medium and large effect, respectively. Applying this, there are four large effects on dependent constructs, three medium effects and four small effects.

Table 9. Path coefficients (and effect size in f^2)

Predictor Variables	Dependent Variables							
	1	2	3	4	5	6	7	8
Sociability	0.54*** (0.41)							
Social Interaction (1)		0.56*** (0.46)	0.46*** (0.27)		Indirect effects: 0.27*** 0.33***			
Impression Formation (2)				0.46*** (0.27)				
Pos. Comm. Behavior (3)					0.63*** (0.68)			
Social Presence (4)					0.16* (0.05)	0.19* (0.04)		
Social Space (5)						0.29** (0.08)	0.27** (0.12)	
Perceived Learning (6)							0.54*** (0.49)	0.49*** (0.31)
Satisfaction (7)								
Confidence (8)								

*** = $p < 0.001$; ** = $p < 0.01$; * = $p < 0.05$

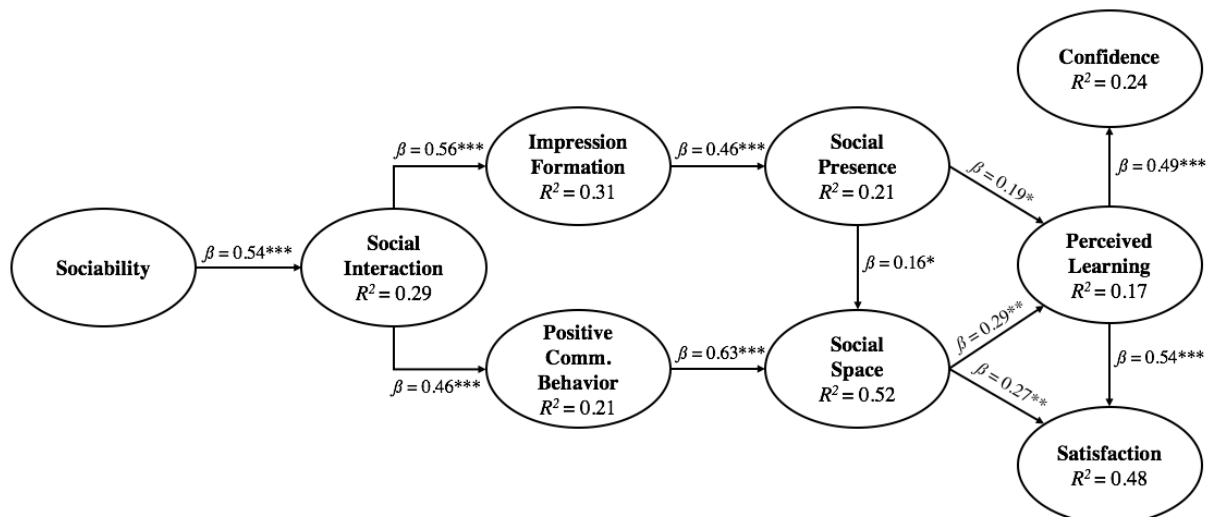


Figure 6. The SIPS model with path coefficients in β , variance explained in R^2 , and outcome variables

Interpretation of Results

Results indicate that all hypotheses are supported. The results of data analysis, its interpretation, as well as further results will be discussed in detail below. Refer to Table 10 for a summary.

Research Question 1. *Is the SIPS-model of online learning valid and what are its predictive capabilities?*

H1: The sociability of the online learning environment has a large effect (.41) on the degree to which social interaction is perceived. Because there are conceivably many other more immediate factors influencing social interaction, sociability explains only about a third (29%) of its variance. This does however lend strong support to the notion that an indirect, ecological approach via the design of the learning environment is an effective means to foster social interaction in online learning. Therefore, creating sociable learning environments, for example by implementing tools and functions that facilitate easy, seamless and quick communication might be worthwhile consideration when designing online learning experiences.

H2+H3: Social interaction has a large effect (.46) on impression formation, explaining about a third of its variance (.31). This supports the hypothesis that interacting with peers allows a learner to form rich and individuating impressions based on certain characteristics like appearance, age, and personality. Impression formation, in turn has a medium to large effect (.27) on social presence, accounting for about a fifth of its variance (21%). This indicates that distinct impressions of other learners are indeed an important precursor to social presence, but that there may be additional factors governing the emergence of social presence outside of this model. A significant indirect effect of social interaction on social presence (.27) further supports the relationship of these three variables. This means that, if teachers and instructional designers want to foster the experience of social presence, they have to take measures to increase social interaction, as these interactions yield distinct impressions between communicators.

H4+H5: Social Interaction has a medium to large effect (.27) on positive communicative behavior, accounting for about one fifth (21%) of its variance. This supports the idea that social interaction is necessary to demonstrate and perceive positive communicative behavior between students. This, in turn has a very large effect (.68) on the development of a sound social space. Together with social presence, it accounts for about half (52%) of social space's variance. In order to perceive a sound social space, positive communicative behavior needs to be demonstrated by students. Only then will this positive social climate lead to the development of salient

interpersonal connections that form a sound social space. A significant indirect effect of social interaction on social presence (.33) supports this relationship. Here too, teachers and instructional designers should consider social interaction as a main driver in the emergence of a sound social space.

H6: Social Presence has a small (.05) effect on social space. Because of its low path coefficients, social presence is expected to contribute only a fraction to the explanation of social space's variance. Although the hypothesis is supported, this relationship is much weaker than expected, implying that the psychological sensation of the other being "there" and "present" contributes marginally to the development of salient interpersonal relationships. This means, that fostering social presence is not enough to achieve a beneficial learning climate and interpersonal connections that form a sound social space. Therefore, separate measures to achieve this should be taken.

Research Question 2. *How do the outcome variables relate to social aspects of online learning?*

H7: The data shows that satisfaction is predicted by a sound social space with a small to medium (.12) effect, accounting for about half of its variance (48%) together with perceived learning. Social Presence, however shows no significant relationship to satisfaction. This implies that, although social presence may be desirable for other reasons, its direct connection to satisfaction is at least questionable. The prominent link between social presence and satisfaction seems to disappear after untangling social presence from social space. Instead, a sound social space is predictive of satisfaction with the learning experience. In order to increase the perceived quality of the learning experience, teachers and instructional designers should therefore focus on facilitating the emergence of a sound social space.

H8: Common sense suggests that perceived learning will predict confidence and satisfaction. This is supported by a medium to large effect (.31) and a large effect (.49), respectively. Perceived learning accounts for about one fourth (24%) of the variance of confidence. This means that, although it is a strong predictor, there are other factors influencing a student's confidence in his learning gains. Because of its strong path coefficient to satisfaction, perceived learning is expected to contribute to most of the variance in satisfaction. This means that, even though socio-emotional aspects are important for a satisfying online learning experience, perceived cognitive learning gains remain most essential. Social Presence and Social Space, being rather subjective socioemotional perceptions, have little impact on perceived learning (.04 and .08, respectively).

This supports the notion that although socioemotional aspects are important in online learning, they can hardly influence actual cognitive learning gains (or even a perception thereof). Similarly, confidence is not predicted by any other variable than perceived learning, demonstrating that social aspects are far outweighed by cognitive aspects when considering the subjective quality of the learning experience. This can be taken to mean, that social aspects, especially social space, may be helpful auxiliary variables in improving the perceived quality of the online learning experience, but that (the perception of) cognitive learning gains still are the main driver of satisfaction and confidence. Interestingly, social interaction too, does not predict perceived learning. Although social interaction was operationalized to encompass all kinds of student-student interaction, task or non-task, cognitive or socioemotional, there was no connection to perceived learning.

Table 10. Summary of hypotheses tests

Hypotheses	This study
H1 Sociability is a predictor for social interaction in the learning environment	supported
H2 Social interaction predicts impression formation	supported
H3 Impression formation predicts social presence	supported
H4 Social interaction predicts positive communicative behavior.	supported
H5 Positive communicative behavior predicts a sound social space	supported
H6 Social Presence predicts a sound Social Space	supported
H7 Satisfaction is predicted by a sound social space instead of social presence	supported
H8 Perceived learning predicts Confidence and Satisfaction	supported

Discussion

This study was a first step in empirically validating the distinction between social presence and a sound social space. As the data shows, the classic effect of social presence on satisfaction was not found in this study. Instead, a sound social space seems to be the dominant socioemotional factor driving satisfaction. This result is not in line with previous social presence research, in which the relationship to satisfaction is rather prominent (Richardson et al., 2017). However, as suggested in chapter 2.1 und 2.2, this relationship may be the result of convoluted definitions and measures of social presence. If most social presence scales do indeed measure the extent of a sound social space (Kreijns et al., 2014), the frequency of studies supporting the relationship to satisfaction is not surprising. The finding that a sound social space, not social presence, may be the main driver of satisfaction needs to be replicated in different contexts and with larger samples. Only then can the confusion surrounding social presence and its effects give way to a clearer understanding. We suggest using the Kreijns et al. (2011) scale for this purpose,

as it seems to be one of the few validated scales that aim to strictly distinguish social presence from other social variables. As shown in this research, the theoretical and empirical distinction of social presence and social space has explanatory power and should consequently be considered in further research on social aspects of online learning.

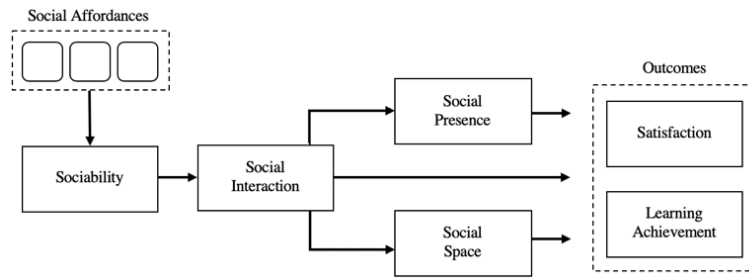
In addition, this research is a first attempt in validating the SIPS model, adapted from Kreijns et al. (2004; 2005; 2013) for a non-CSCL context. The predictive capabilities of the model demonstrate that it is a viable framework for understanding and predicting social aspects of online learning. However, this study also shows that the socioemotional variables, albeit important for the online learning experience, are outweighed by the importance of actually perceiving individual learning progress.

These findings emphasize the importance of designing and implementing sociable learning environments, supporting the ecological approach to fostering social interaction. Because social interaction, although considered essential, does not always develop automatically (Kreijns, Kirschner, & Jochems, 2003), the sociability of a learning environment should be a major consideration. Although there has been research to this end (Gao, Dai, Fam, & Kang, 2010; Abedin, Daneshgar, & D'Ambra, 2012), further research determining factors of sociability and finding ways to create sociable learning environments is needed. Deliberately designing a sociable learning environment and comparing its effects with a non-sociable learning environment on the emergence of social presence, the development of a sound social space as well as outcome variables is an important next step in validating the model and advancing the ecological approach. So long as there are no established design recommendations to improve the sociability of the learning environment, practitioners may rely on fostering social interaction directly in order to facilitate social presence and the emergence of a sound social space.

This study has several limitations. The analyzed model was streamlined in ways to depict the hypothesized primary causal flow of the relevant variables. This was done in accord with operational definitions of the variables. However, it is possible that bidirectional relationships do indeed exist and their investigation may be an important step in getting the full picture. As of now, the PLS-SEM algorithm does not allow for specification of causal loops and therefore may not be suitable for a more comprehensive analysis of these social variables. For this research, a rather typical distance education class was chosen. Students communicated predominantly via asynchronous threaded message boards and there was little synchronous activity. In addition, the class size is large (>500), suggesting relative anonymity among students. Although this context was explicitly chosen to validate the model and test the hypothesis in a rather exemplary distance education online class, a more ambitious instructional design may lead to stronger effects and

higher amounts of explained variance. Only 32% of participants took part in the synchronous chat activities. Because of this, the items of the social presence scale that directly relate to chat activities were excluded from analysis. This reduced the scale considerably, possibly limiting its validity. Also, this study is of correlational nature. The causal claims of the model are rooted in theory, but not strictly falsifiable via this research design. Future studies should try to experimentally manipulate sociability and other antecedent variables in order to reliably identify causal mechanisms that actually lead to the emergence of social presence and a sound social space.

5.3 Designing sociable online learning environments: An affordance enrichment approach



This study focuses more specifically on the sociability construct of the SIPS model and investigates an approach to enhancing the online learning environment in this respect. An experiment was conducted to test the effectiveness of social affordances toward this goal. Furthermore, the remaining socio-emotional aspects were expected to be positively influenced as a result of this intervention, so that students would report more social interaction, perceive their peers to be socially present and find the environment to be a sound social space. The data was consistent with these expectations. Lastly, outcome variables were assessed. Satisfaction was found to be increased through our intervention, whereas learning achievement was not.

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This study has been published as:

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Introduction

Today, distance education programs rely on online technology to deliver instruction and mediate learning activities. Where face-to-face learning offers a rich social context and opportunities to interact in a non-mediated fashion, online and distance learning typically doesn't provide these hallmarks. Here, social interaction with peers and instructors always relies on technology for mediation (Hillman, Willis, & Gunawardena, 1995; Weidlich & Bastiaens, 2018) and under these conditions, a sound social experience is often more difficult to realize (Kreijns, Kirschner, & Jochems, 2003; Boling, Hough, Krinsky, Saleem, & Stevens, 2012; Symeonides & Childs, 2015). As a result, concepts and theories relating to the social realm are found in different lines of research in online distance education.

Though not the only relevant socio-emotional variable, social presence is by far the most researched concept on how students interact through and relate to others in computer-mediated communication (CMC) and is considered to be an essential factor in online learning (Garrison & Arbaugh, 2007; Richardson, Maeda, Lv, & Caskurlu, 2017). The SIPS model, consisting of sociability, social interaction, social presence, and social space (Weidlich & Bastiaens, 2017) attempts to explain under which conditions social presence is facilitated. It posits social interaction as the main driver, which in turn is fostered through a sociable learning environment. Kreijns, Kirschner, and Vermeulen (2013) suggest that certain environmental characteristics are essential to making a learning environment sociable instead of aloof. These characteristics are called social affordances (Kreijns, 2004). Experimental evidence for the effectiveness of social affordances for enhancing the learning experience in the social realm is still lacking and, thus, constitutes a gap in research.

To this end, an *in vivo* pre-/post-test quasi-experiment ($n = 143$) was conducted, in which the online learning environment was enriched with three social functions, hypothesized to serve as social affordances. The results are important in lending empirical support to the ecological approach to designing sociable learning environments, as well as showing ways to easily foster desirable socio-emotional variables, as per the SIPS-model (Weidlich & Bastiaens, 2017), explained below.

Literature review

The SIPS model. Kreijns et al. (2004) and Kreijns et al. (2013) propose a framework for social presence in computer-supported collaborative learning (CSCL), detailing the relevant socio-emotional variables. These variables are called socio-emotional as they are (1) concerned with the social realm, more specifically the sociality between fellow students and (2) affective in

that they generally relate to non-cognitive aspects of the learning experience. Some but not all of the relationships have been empirically supported through previous research (for a detailed discussion, see Kreijns et al., 2013). Since most social presence research has been conducted without specifically having CSCL in mind, an adaptation of this framework for general online and distance learning contexts was still needed. This adaptation, called the SIPS model (for its central variables, Sociability, Social Interaction, Social Presence, Social Space, see Fig. 1 the most basic version of the model) was introduced in detail in Weidlich & Bastiaens, (2017).

The SIPS model differs from the original framework in three ways. (1) It excludes the variable pedagogical techniques, because this is not a socio-emotional variable. (2) The model assumes a primary causal flow from left to right, whereas the original framework depicts many bidirectional relationships. This step was necessary to formulate testable causal hypotheses from the model. (3) Sociability is not a direct precursor to social presence, but instead exerts effects only through social interaction, because impressions are chiefly formed through direct interaction (Weidlich & Bastiaens, 2017). A first empirical examination through Partial Least Squares Structural Equation Modeling (PLS-SEM) has shown evidence consistent with this model (Weidlich & Bastiaens, 2017). The following paragraphs will give a brief overview of the central socio-emotional variables in the SIPS model in this order: Social presence, social space, social interaction, and sociability.

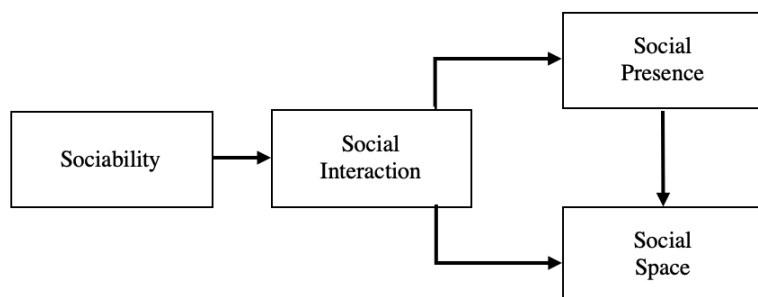


Figure 7. SIPS model as proposed by Weidlich & Bastiaens (2017), adapted from Kreijns et al. (2013).

Social Presence. In 1976, Short, Williams, and Christie coined the term social presence, defining it as “the degree of salience of the other person in the communication and the consequent salience of the interpersonal relationships” (p.65). It is considered an important aspect of the learning experience, because unlike face-to-face communication, CMC usually conveys little socio-emotional cues. This is especially true for text-based asynchronous communication (e.g. message boards) in learning management systems (LMS), that are still a large part of the educational technology landscape (Legon & Garrett, 2018). Social presence has been linked to important concepts of online and distance education, e.g. satisfaction and perceived learning (Gunawardena & Zittle, 1997; Richardson & Swan, 2003; Richardson, Maeda, Lv, & Caskurlu, 2017), online course retention (Liu, Gomez, & Yen, 2009) and online social interaction (Tu & McIsaac, 2002). When appraising research surrounding social presence, one is confronted with a

plethora of different definitions (Lowenthal & Snelson, 2017) and, consequently, operationalizations of social presence (Caspi & Blau, 2008; Kreijns, Kirschner, Jochems, & Van Buuren, 2011). This may be, in part, because social presence theory stems from a rather diverse line of research (Cui et al., 2013). For this study we adopt a clear and narrow definition of social presence, that is rooted in its original conception by Short, Williams, & Christie (1976). Here, social presence is defined as *the psychological phenomenon that the other is perceived as ‘real’ in the communication, the subjective feeling of being with other salient social actors in a technologically mediated space*. In other words, “the sense of ‘being there, together’, when ‘being there’ does not involve a physical presence” (Öztok & Kehrwald, 2017, p.5). Despite much research about how social presence relates to desirable outcome variables like satisfaction and perceived learning (Richardson et al., 2017), there is still a lack of empirically tested design recommendations for fostering a sense of social presence among students in online learning environments (Cui, Lockee, & Meng, 2013). Recently, Lowenthal & Dunlap (2018) provided a set of instructional strategies to establish and maintain social presence. Yet, besides these and other direct approaches (e.g. mandatory message board postings or group forming sessions), there is still a gap in the literature concerning the question of which properties and qualities of the learning environments are conducive to social presence and other socio-emotional variables.

Social space. Students may perceive a network of interpersonal/social relationships between them and others in the learning environment, constituting a sound social space (Kreijns et al., 2013). This notion relates to the second part of the original Short et al. (1976) definition and encapsulates what is often associated with concepts like sense of community, group cohesion, and beneficial learning climate. Regrettably, it is also often conflated with social presence (Kreijns, Van Acker, Vermeulen, & Van Buuren, 2014; Lowenthal & Snelson, 2017). For example, in the following definition of Garrison (2009): “the ability of participants to identify with the community (e.g. course or study), communicate purposefully in a trusting environment, and develop interpersonal relationships by way of projecting their individual personalities” (p.352) or in the model of social presence in Wei, Chen, and Kinshuk (2012). The SIPS model proposes that the concepts of social presence and social space, although related, are rather distinct. When social interaction takes place, students demonstrate communicative behavior that is indicative of a sound social space (Weidlich & Bastiaens, 2017; Molinillo, Aguilar-Illescas, Anaya-Sanchez, Vallespin-Aran, 2018). Additionally, feelings of social presence will facilitate the development of a sound social space by evoking a sense of face-to-face conversation.

Social interaction or student-student interaction. This is the process of communicating and exchanging messages with peers in the learning environment. Past research has often focused on cognitive or task-centered interaction between students (Abedin et al., 2012), although this

restriction is considered a pitfall (Kreijns et al., 2003). As students interact with each other they will (1) perceive social presence of their peers (Oyarzun, Stefaniak, Bol, Morrison, 2018), via a process called impression formation (Walther, 1993; Weidlich & Bastiaens, 2017) and (2) perceive the learning environment to be a sound social space. The SIPS model suggests that sociable learning environments facilitate an increase in spontaneous yet sustained social interaction. This has also been called *contextual* (vs. *designed*) social interaction (Borokhovski, Tamim, Bernard, Abrami, & Sokolovskaya, 2012; Lu & Churchill, 2012).

Sociability. Online learning environments may differ in their ability to facilitate social interaction, so that they can be more or less sociable. Sociability is, thus, an attribute of the learning environment, as perceived by the students (Kreijns, Kirschner, Jochems, & Van Buuren, 2007; Abedin, Daneshgar, & D'ambra, 2012). Sociable learning environments have tangible elements that allow for and facilitate quick, easy, and informal social interaction, especially in the socio-emotional dimension (Gai, Dai, Fan & Kang, 2010; Kreijns et al., 2013).

Environments lacking these characteristics may be called *aloof*. For example, in a qualitative study, pre-service teachers found their Moodle learning environment not to be an interactive space to connect and communicate (Deng & Tavares, 2013). They found this to be in stark contrast to Facebook, which the sampled students used readily for informal and free-flowing social interaction. Similar sentiments are echoed in Kurtz (2014), where students were asked to compare Facebook with the course website and reported the former to be substantially more conducive to social interaction and collaboration. In Sadowski, Pediatitis, and Townsend (2017), Australian students saw much potential in SNS's to overcome the social isolation of studying online, but also had strong reservations towards blending their personal sphere with the more formal educational sphere. In a survey of Taiwanese students, Facebook was judged to be overwhelmingly more suitable for sharing material and resources, for receiving updates, as well as overall course interaction than more traditional LMS solutions (Jong, Lai, Hsia, Lin, & Liao, 2014). These studies seem to uniformly point to the sociability of SNS', a quality that although important, is not necessarily designed into traditional LMS like Moodle or Blackboard. So, what is it that makes SNS so sociable and how can this be used to enhance online distance learning in the social realm? We suggest that social affordances are key to understanding sociability.

Social Affordances and Social Networking Sites. James Gibson (1977) proposed the term affordance to refer to action possibilities in the physical environment that are objectively measurable but only become manifest in relation to an agent. This was later extended by adding the prefix *perceived* affordance, distinguishing between objective action possibilities from those perceivable by an agent (Norman, 1999). This is known as perception-action-coupling (Withagen,

de Poel, Araujo, & Pepping, 2012). Kirschner (2002) defined *educational* affordances as those characteristics of an artifact that determine if and how a particular learning behavior could possibly be enacted within a given context. Similarly, *social* affordances are characteristics of an environment that allow for and invite enactment of social behavior (Kirschner, Strijbos, Kreijns, Beers, 2004). Just like a town square affords meeting and interacting with people in the physical space, online environments can also afford social behavior. This indirect or ‘ecological’ (Kreijns, 2004) approach to designing sociable environments via affordances may be preferable to more direct approaches like mandatory message board postings because it emphasizes the notion of user agency, as he/she will use the environment in a way that is afforded both by the capabilities of the environment and his/her individual needs (Withagen et al., 2012).

Social Network Sites (SNS) are paragons of social affordances and sociability. The largest social network, Facebook, has over two billion monthly users worldwide (Statista, 2018), 1.4 billion daily users (Statista, 2018b), and a share of 79% of American internet users. As of 2016, other social networks and social media applications, like Twitter and Instagram are smaller but still used by 24% and 32% of American internet users, respectively (Greenword, Perrin & Dugga, 2016). With billions of people logging onto SNS daily and spending hours communicating and interacting with others in various forms, there must be ways to learn from and make use of their mechanisms to improve online distance learning experiences in the social realm?

To better understand characteristics of SNS, Pereira, Baranauskas, and da Silva (2010) present a framework of seven elements, or affordances, that are characteristic of social software: identity, presence, relationships, reputation, groups, conversation, and sharing. Depending on the application, its values and goals, different elements will be emphasized. For example, Youtube primarily emphasizes sharing, with conversation and identity being secondary and presence and relationships being mostly non-existent. On the other hand, Facebook emphasizes presence, identity, conversation, and relationships, with sharing and reputation being less important. LMS like Moodle with a focus on managing courses and activities may not have these same affordances (Deng & Tavares; Kurtz, 2014; Jong, Lai, Hsia, Lin, & Liao, 2014). We suggest that, depending on the experience we want students to have in the learning environment, certain affordances may be emphasized to counteract these limitations of an LMS like Moodle.

Social Networking Sites in Online Distance Education. Since the advent of SNS and social media, scholars and practitioners of online distance education have investigated their social potential by integrating them into the educational realm (Tess, 2013). However, there are different ways of approaching this. Our review of the literature suggests that there have been two

main approaches to salvaging SNS in educational contexts, the *substitute approach* and the *supplement approach*.

Substitute approach. This approach consists of using a SNS as the primary learning environment, usually replacing a more traditional LMS like Moodle or Blackboard. In a quasi-experimental study, Thoms and Eryilmaz (2014) found an online social networking environment to be superior to a Blackboard environment in terms of social interaction and course community. They identify typical SNS affordances like online/offline status and detailed profiles as essential for this advantage. Recognizing risks and large costs of institution-wide switches to new systems, they instead suggest adapting and enriching existing LMS's to make them more sociable. Similarly, a closed SNS ("Ning") was reported to effectively reduce transactional distance and foster social interactions when used as the central platform for learning activities, instead of Blackboard (Quong, Snider, & Early, 2018). Qualitative inquiry further suggested that although students liked the platform and found it conducive to social interaction, they sometimes found it distracting and had troubles managing their time. Veletsianos, Kimmons, and French (2013) found that instructors faced numerous frustrations with open source SNS Elgg when using this as central learning environment instead of the previous LMS, one being the incongruency between the technology's implicit values and the pedagogical beliefs of the instructors.

Supplement approach. This approach is characterized by not replacing the traditional LMS, but instead supplementing it with a SNS. Therefore, investigations using this approach essentially make use of two platforms. In an experimental study, DeSchryver, Mishra, Koehler, and Francis (2009) assigned their students to either learning in Moodle or Moodle plus Facebook. They found no effect overall social presence. In contrast to this, Akcaoglu & Lee (2018) found that students provided with a supplementary Facebook group did score higher in social variables like co-presence and sociability than their peers lacking this provision. Additional qualitative data, however, indicates that some students found it strenuous and wasteful of their time to visit another platform for study related content. In a similar vein, Thai, Sheeran, and Cummings (2019) also found that supplemental Facebook groups enhanced social connectedness, course engagement and satisfaction.

Possible Pitfalls. Both of these approaches may bring difficulties. By introducing non-educational technology in educational contexts, they also introduce the developer's world views, which are "revealed through the activities supported and encouraged by the technology" (Veletsianos, 2016). These may not necessarily be the ones intended by instructors or their institution and appropriating these technologies may have undesired effects. Additionally, students may find it difficult to navigate the blending of their academic sphere and their social

space, a phenomenon called *context collapse* (Dennen & Burner, 2017; Sugimoto, Hank, Bowman, & Pomerantz, 2015). Smith (2016) investigation shows that students were critical of this blend, referring to it as a “double edged sword”, for one because the social space distracted from academic work but also due to potential role conflicts, for example seeing faculty’s private life through their Facebook profile. Lastly, we find fault in the general approach of bluntly inserting a comprehensive SNS like Facebook into an existing educational program, as this amounts to using a sledgehammer to crack a nut. Instead of carefully adapting existing learning environments according to their strengths and weaknesses, past research has simply appropriated them in hopes of working their ‘social magic’. This technological determinism (Oliver, 2011) is a pitfall in designing learning experience that are effective in the socio-emotional realm.

The present research

To combat the aforementioned problems, we suggest a third approach: Enrichment of existing learning environments. This approach differs from the previous two approaches in that no proprietary software is integrated but instead existing learning environments are enhanced with those features of SNS that are expected to be conducive to sociability, i.e. social affordances. This approach has hardly been investigated in the context of SNS in online distance learning, although it has been proposed as an effective approach by different scholars (Du, Fu, Zhao, Liu, & Liu, 2012; Garmendia & Cobos, 2013; Anderson & Dron, 2018).

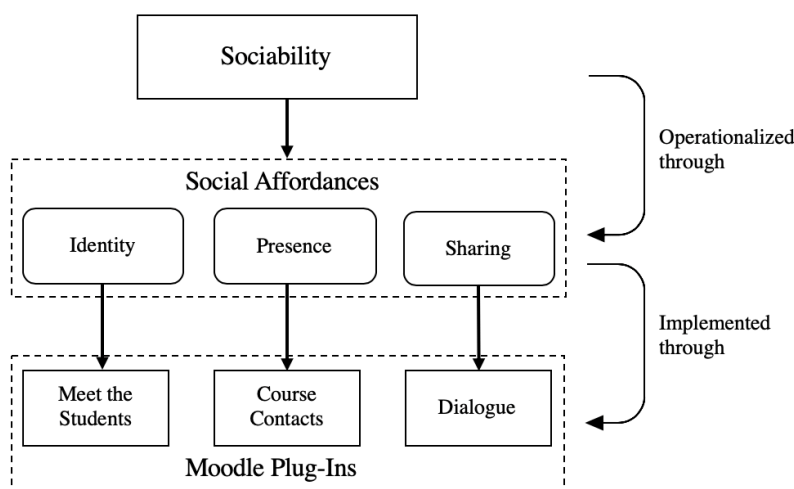


Figure 8. Sociability is operationalized through three social affordances, which in turn are implemented through three Moodle plug-ins.

In this study, we investigate the effectiveness of this enrichment approach. Using the SIPS model as a guide, we assess if introducing social affordances inspired by SNS features are effective. On the implementation level, we chose three plugins, relating to three affordances: identity, presence, and sharing (see chapter 6.2). These three affordances are expected to be effective operationalizations of the sociability variable (see Figure 8). We consider them effective if they make for a more sociable online learning environment. This is our first research question:

Research question 1: *Do social affordances impact students' perceptions of the sociability of the learning environment?*

H1: Social affordances will lead to students perceiving the environment as more sociable.

The SIPS model proposes that a more sociable learning environment will be conducive to social interaction and this, in turn, will lead to more social presence among peers and the development of a sound social space (Weidlich & Bastiaens, 2017). Our second research question is, thus:

Research question 2: *Is the sociable learning environment effective in fostering social interaction, social presence, and social space?*

H2: The sociable learning environment will result in more social interaction, higher social presence, and social space.

Additionally, these changes should also be reflected in outcomes of the learning experience. To assess this, we look at one affective and one cognitive outcome variable respectively, satisfaction and learning achievement. Past research has mostly shown a strong relationship of social presence and satisfaction, as demonstrated in a recent meta-analysis (Richardson et al., 2017). However, because many studies may have operated with convoluted definitions of social presence, these results are not entirely undisputed (Weidlich & Bastiaens, 2017). One much less investigated but arguably more important relationship is the relationship to learning achievement. Although Joksimovic, Gasevic, Kovanovic, Riecke, & Hatala (2015) and Hostetter (2013) seem to provide preliminary data pointing to a relationship between social presence and achievement, it should be noted that their measures are not based on a precisely defined social presence and instead include other variables (see Weidlich, Rajagopal, & Kreijns, 2018 for a more in-depth discussion). It will be instructive to add to this evidence base by also looking at the effects of the intervention on learning achievement. Thus, our third research question is as follows:

Research question 3: *Do these socio-emotional variables impact affective and cognitive outcomes, i.e. satisfaction with the learning experience and learning achievement?*

H3.1: Students in the more sociable learning environment will report more satisfying learning experiences.

H3.2: Students in the more sociable learning environments will not perform better in the learning achievement test.

In order to better understand how students perceived these features and engaged with them, we also ask students to report on visibility and engagement with these features. This is our fourth research questions.

Research question 4: *How did students perceive the social features in terms of visibility and how frequently did they engage?*

Exploratory: No hypotheses

Lastly, we wanted to know if differing degrees of visibility and engagement with these features had differing effects on socio-emotional variables of the SIPS model. This is our final research question:

Research question 5: *How does visibility of and engagement with a specific social feature influence focal socio-emotional variables?*

Exploratory: No hypotheses

Method

To investigate our research questions, we conducted an *in vivo* pre-/post-test quasi-experiment with two conditions, one being the online course with an SNS-enhanced Moodle, the control condition being the same online course with a non-enhanced Moodle. In the winter semester of 2016/2017, students that took part in the facultative course were assigned to the control condition. Students that applied for the course one year later, winter semester of 2017/2018, were assigned to the experimental condition. To account for potential baseline differences due to non-randomization, a pre-test with the focal variables was administered before students entered the course. Through this, both populations can be compared prior to analysis in terms of equivalence and their pre-test scores can be used as a covariate for a more robust comparison. See Table 12 for an overview of the instruments in this study, along with internal consistency for this sample, as calculated via Cronbach's Alpha.

Sample. This sample consists of a total of 143 students, 101 Educational Science, B.A. and 42 Psychology, B.A. students at Germany's largest distance education provider, FernUniversität in Hagen. Mean age is higher than usual in higher education with 38.55, but not atypical for Distance Higher Education. With 21 men and 120 women, there is asymmetry in terms of gender distribution (Table 11).

Table 11. Demographics of Sample

	Age	Gender		Course of study	
		Male	Female	Edu Science	Psych.
Control	38.4	12	72	55	31
Experimental	38.7	9	48	46	12
Mean/Total	38.55	21	120	101	42

Course Content & Design. The experiment was conducted in conjunction with an Educational Science, B.A. class called „Allgemeine Didaktik und Mediendidaktik“ (Instructional Design and Educational Technology) at FernUniversität in Hagen, Germany. Obligatory for Educational Science students and facultative for Psychology students, this class is concerned with introducing different concepts of pedagogy and instruction design models. At its heart is the 4C/ID model by Van Merriënboer & Kirschner (2018), on which students are to write an end-of-term paper. In this paper, students conceptualize a 4C/ID blueprint in a given content domain and outline an implementation for this instructional blueprint.

Table 12. List of variables and their measures

Variable	Description	Items	Measure	Cronbach's Alpha
Sociability	The perceived quality of the learning environment to facilitate social interaction.	10	Kreijns et al. (2007), adapted for non-CSCL context;	0.85
Social Interaction	The perceived amount and frequency of communication between a given student and his peers.	4	Wei et al. (2012)	0.93
Social Presence	The psychological sensation of the other being “there” and “present”.	10	Weidlich, Rajagopal, & Kreijns (2018)	0.90
Social Space	The perceived network of interpersonal relationships between students.	13	Kreijns et al. (2004), adapted for non-CSCL context	0.78
Satisfaction	The extent to which the student is content regarding all aspects of the learning experience.	6	Weidlich & Bastiens (2017)	0.90
Learning Achievement	Multiple Choice items assessing content knowledge	10	New (scores ranging from 0 to 10)	0.69

The actual intervention took place in an additional elective course, “Focus course 4C/ID”, that aimed to provide a more in-depth engagement with the 4C/ID model, placing an emphasis on preparing students for the end-of-term paper. Students were recruited via e-mail and postings in Moodle. Students choosing to enroll in this four-week Moodle course gave consent to academic use of their data and were informed of the workload as well as the requirements of this course. Once enrolled, they completed a questionnaire prior to entering the online learning environment, which served as a pre-test. The course was unsupervised, in that no teacher led the activities. Although based on a weekly structure, the course was self-paced, in that students could choose to engage with all materials and activities from the very beginning. It consisted of three content

domains relevant for their paper, with an additional week at the very end to wrap up discussions and finish learning activities. In order to successfully complete the course, students were asked to also take part in the post-test questionnaire at the end, as well as a ten-item multiple choice test.

The course included several facultative assignments relating to learning activities, e.g. collaboratively answering questions about provided literature or giving peer feedback to provisional blueprints of fellow students. These activities were designed to (1) accomplish the learning goals associated with this course and, more importantly, (2) to provide incentives to interacting with peers, so that social capabilities of the learning environment will be recognized as useful and engaged with to reach learning goals or satisfy social need.

Both versions of the course, control and experimental, were nearly identical in content and design, with the only difference being three social affordances that were introduced in the experimental condition as well as a document introducing these three plugins to participants with a brief description of their functionalities. All three plugins were screened for review ratings and quality to ensure a smooth user experience without technical issues. The selected plugins were:

Identity. Building and displaying an online identity is a common feature in SNS design (Bouman et al. 2008, Pereira et al., 2010). This is usually fostered through a powerful profile feature with profile pictures, biographies, and other identifying information. In contrast, although technically present, profiles in LMS are not always emphasized sufficiently as to represent this affordance. In Moodle, for example, profile pictures are very small and mainly displayed in message board threads. Upon logging into the environment, a student will at first glance not be presented with any indication of other humans in the environment. In order to strengthen this social affordance, we chose the plugin “Meet the students” (moodle.org/plugins/block_meet_the_students). This plugin prominently displays enlarged profile pictures of students on the side bar of the Moodle environment (Figure 9). Clicking on the picture takes the student to the profile page.

Presence. This affordance is usually represented through features of the environment that allow a user to know if he is sharing the virtual space with somebody else, i.e. whether someone is online (Pereira et al. 2010). Through this, users can spontaneously strike up a conversation. Strictly speaking, some LMS already have a feature resembling this affordance. For example, Moodle features a function to display currently online students. However, it is notably lacking compared to what many SNS offer, where you have a constantly updating list of potential communication partners, according to their online status, as well as means to connect with them. In light of this, we aimed to strengthen the presence affordance with the plugin “Course contacts”

(moodle.org/plugins/block_course_contacts). This plugin displays a list of users on the side bar, indicating their online/offline status and indicating quick ways for communicating with them (Figure 10).

Sharing. In learning environments that want to allow for collaborative learning activities, the ability to share documents and material with fellow students is essential. If students are to decide whether to approach a learning activity solo or with fellow students, they might find coordination of activities and sharing of documents too cumbersome if the environment isn't conducive. Moodle, for example does not allow for multi-user message exchange with document sharing. In order to emphasize the affordance of sharing we implemented the “Dialogue” plugin (moodle.org/plugins/mod_dialogue) in the experimental condition. This plugin adds a new course activity that allows students to start a private conversation, with a powerful attachment function allowing for easy sharing of documents and materials (Figure 11). Although, also including conversation functionality, this plugin was integrated to emphasize sharing, for example of materials, but also personal things like photos, to make for an environment with more potential for interactive and collaborative learning experience.

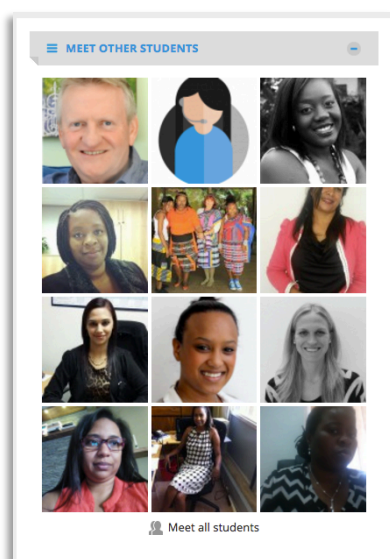


Figure 9. “Meet the Students”

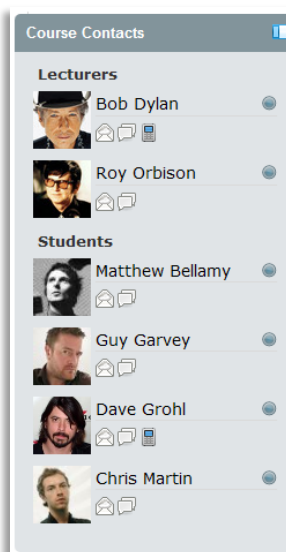


Figure 10. “Course Contacts”

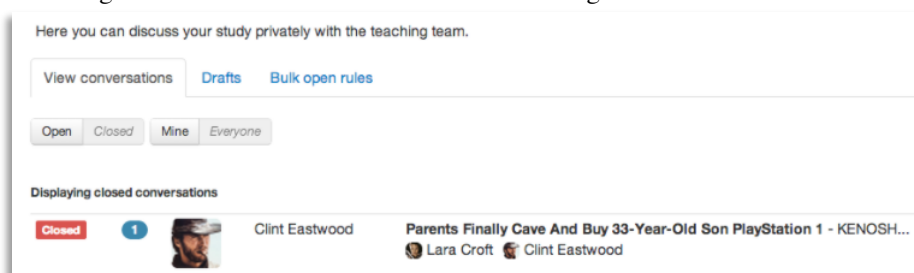


Figure 11. “Dialogue”

Sensitivity Analysis and Assumption Checks. For main effects of condition on socio-emotional variables, sensitivity analysis using G*Power (Faul, Erdfelder, Lang, & Buchner, 2007) for ANCOVA with alpha-level of .05 and power of .8 with total sample size of 143, two groups, and one covariate results in a smallest detectable effect size $f = .236$. This equals $\eta^2_p = .0137$. For main effects of condition on learning, sensitivity analysis for t-test with alpha-level of .05, two-tailed, and power of .8 with a sample size of 79 and 54, respectively, results in a smallest detectable effect size $d = .498$. For analysis of visibility of social affordances on dependent variables, sensitivity analysis for t-test with alpha-level of .05, two-tailed, and power of .8 with a sample size of 16 and 41, respectively, results in a smallest detectable effect size $d = .84$. For bivariate correlations of engagement with social affordances and focal variables, an alpha-level of .05, two-tailed, power of .8 with a sample size of 58 results in a smallest detectable effects size of $r = .35$.

Comparing pre-test results along condition yields no significant differences. Both groups were approximately equal in terms of focal dependent variables upon entering the experiment. This suggests that the assumption of independence of covariate and treatment effect has been met for all variables. Levene's test was also carried out to check for linearity and homoscedasticity. There were no significant group differences for any of the variables. Furthermore, the interaction terms of condition and pretest scores were all non-significant, indicating homogeneity of regressions slopes. Therefore, major assumptions of ANCOVA are met (Table 13). All statistical analyses were conducted with IBM SPSS version 25 and JASP 0.8.4.

Table 13. Descriptive data and p-values for (1) t-test for pretest scores, (2) Levene's test, and (3) interaction terms condition*pretest.

	Descriptives				Assumption Checks		
	Cond.	N	Mean	SD	p (1)	p (2)	p (3)
Sociability	C	86	2.68	0.63	.78	.25	.06
	E	58	2.71	0.60			
Interaction	C	86	2.19	1.00	.64	.97	.51
	E	58	2.28	1.11			
Social Presence	C	86	2.51	0.78	.26	.96	.14
	E	58	2.66	0.74			
Social Space	C	86	2.51	0.54	.10	.29	.90
	E	58	2.36	0.49			
Satisfaction	C	86	3.51	0.83	.20	.84	.37
	E	58	3.32	0.84			

Note. C = control condition, E = experimental condition

Results

Effects of Condition.

RQ1: *Do social affordances impact students' perceptions of the sociability of the learning environment?* This research question is addressed with Analysis of Covariance (ANCOVA), condition as fixed factor with two levels (control and experimental), pretest scores of sociability as covariate and posttest scores of sociability as dependent variable. Results of condition on sociability is considered the manipulation check in this study, as this shows if the implemented social affordances actually result in a sociable learning environment, as perceived by students.

Results of ANCOVA yield a statistically significant effect of the treatment on sociability $F(1, 140) = 9.90, p = .002, \eta^2_p = .066$. This medium size effect (Cohen, 1988) indicates that between six and seven percent mean difference can be attributed to condition. This supports the hypothesis (H1) that merely emphasizing social affordances leads to the learning environment being perceived significantly more sociable.

RQ2: *Is the sociable learning environment effective in fostering social interaction, social presence, and social space?* This research question is also addressed with ANCOVA, condition as fixed factor with two levels (control and experimental), pretest scores of focal variables (social interaction, social presence, social space, and satisfaction) as covariate, respectively, and posttest scores of the same focal variables as dependent variables. The manipulation being effective (RQ1), this analysis answers the question if a sociable learning environment will increase (1) social interaction, as well as (2) perceptions of social presence and facilitate the (3) emergence of a social space (H2).

Regarding (1) social interaction, results of ANCOVA yield a statistically significant effect of the treatment on social interaction $F(1, 140) = 7.175, p = .008, \eta^2_p = .049$. This small-to-medium size effect (Cohen, 1988) indicates that five percent mean difference can be attributed to condition. Regarding (2) social presence, results of ANCOVA yield a statistically significant effect of the treatment on social presence $F(1, 140) = 8.49, p = .004, \eta^2_p = .057$. This small-to-medium size effect (Cohen, 1988) indicates that more than five percent mean difference can be attributed to condition. Regarding (3) social space, results of ANCOVA yield a statistically significant effect of the treatment on social space $F(1, 140) = 9.11, p = .003, \eta^2_p = .061$. This small-to-medium size effect (Cohen, 1988) indicates that six percent mean difference can be attributed to condition. Regarding (4) satisfaction, results of ANCOVA yield a statistically significant effect of the treatment on satisfaction $F(1, 140) = 6.16, p = .014, \eta^2_p = .042$. This small-to-medium size effect (Cohen, 1988) indicates that four percent mean difference can be

attributed to condition. This pattern of results lends support to the hypothesis (H2), that a sociable learning environment will indeed result in perceiving more social interaction, higher social presence, a sound social space, as well as a higher degree of satisfaction (Table 14).

RQ3: *Do these socio-emotional variables impact affective and cognitive outcomes, i.e. satisfaction with the learning experience and learning achievement?* The part of the question concerning satisfaction is also addressed with ANCOVA, condition as fixed factor with two levels (control and experimental), pretest scores of satisfaction as covariate and posttest scores of satisfaction as dependent variables.

Regarding satisfaction, results of ANCOVA yield a statistically significant effect of the treatment on satisfaction $F(1, 140) = 6.16, p = .014, \eta^2_p = .042$. This small-to-medium size effect (Cohen, 1988) indicates that four percent mean difference can be attributed to condition. This pattern of results lends support to the hypothesis (H2), that a sociable learning environment will indeed increase satisfaction with the learning experience (Table 14).

The part of the question concerning learning achievement is addressed with Welch's t-Test, condition as fixed factor with two levels (control and experimental) and the results of the multiple-choice achievement test as dependent variable. Welch's t-Test is preferred over the more common Students t-Test is chosen here, because it provides better Type 1 error control and is more robust when assumptions are violated (Delacre, Lakens, & Ley, 2017).

Results of Welch's t-Test yields no significant difference, $t(127.7) = 1.013, p = .31, d = .16$. Descriptive data shows a higher achievement score in the control condition, with a mean difference of .23.

Table 14. Results of ANCOVA with pretest scores as covariate and posttest scores as dependent variable, Welch's t-Test with learning achievement as dependent variable.

	Cond.	Descriptives			ANCOVA			Welch's t-Test		
		N	Mean	SD	F	p	η^2_p	t	p	d
Sociability	C	86	2.90	0.67	9.90	.002	0.066			
	E	57	3.24	0.64						
Interaction	C	86	2.45	1.11	7.18	.008	0.049			
	E	57	2.96	1.13						
Social Presence	C	86	2.47	0.75	8.49	.004	0.057			
	E	57	2.91	0.92						
Social Space	C	86	2.60	0.50	9.11	.003	0.061			
	E	57	2.83	0.63						
Satisfaction	C	86	3.81	0.79	6.16	.014	0.042			
	E	57	4.07	0.72						
Learning	C	79	5.29	1.49				1.01	.313	0.18
	E	54	5.06	1.19						

Note. C = control condition, E = experimental condition

Visibility of and Engagement with Social Functions.

RQ4: *How did students perceive the social features in terms of visibility and how frequently did they engage?* This research question is addressed by inspecting descriptive data of items relating to the visibility of and engagement with social affordances, as implemented via plugins. Results show that although the majority of students were aware of the features in the online learning environment, about one third of students were not (Table 15). In the case of “Dialogue” almost 40% of students were not aware of the tool or did not notice it. “Course contacts” was the most prominent plugin in terms of visibility, with more than 72% of students being aware of the function in the learning environment.

In terms of frequency of engagement, descriptive results show that in many cases, social features were not engaged with at all. This is especially salient in the case of “Dialogue”, where half of students never used it and nobody used it very frequently. These values are somewhat lower for “Meet the Students” and “Course Contacts” but still above 40%. Between 18.2% and 22.4% indicated occasional engagement and very few students indicated frequent or very frequent engagement.

Table 15. Descriptive data of visibility of and frequency of engagement with social features.

	Visibility: “I was aware of Moodle tool XY”		Engagement: “I engaged with Moodle tool XY”				
	Yes	No	very frequently	frequently	occasionally	rarely	never
„Meet the Students“	39 (67.2%)	19 (32.8%)	2 (3.6%)	5 (9.1%)	10 (18.2%)	14 (25.5%)	24 (43.6%)
„Course Contacts“	42 (72.4%)	16 (27.6%)	1 (1.8%)	5 (8.8%)	11 (19.3%)	15 (26.3%)	26 (45.6%)
„Dialogue“	35 (60.3%)	23 (39.7%)	-	4 (6.9%)	13 (22.4%)	12 (20.7%)	29 (50%)

RQ5: *How does visibility of and engagement with a social affordance influence focal socio-emotional variables?* This research question is addressed by assessing the relationship of “visibility” and “engagement” with the dependent variables. As “visibility” is a nominal variable with two levels (yes/no), Welch’s t-Test is used. In order to control Type 1 error rates, a Holm-Bonferroni correction of p-values is applied. “Engagement” is a continuous 5-likert scale, so bivariate correlations are computed to see if there is a relationship between engaging with social affordances and the focal variables.

Results of Welch’s t-Tests with visibility as group designator (no = 0 / yes = 1) for “Meet the Students” yields a significant result on social presence, $t(31.40) = 4.158$, $p = .006$ (after Holm-Bonferroni Correction), $d = 1.20$. This large effect (Cohen, 1988) suggests that students aware of “Meet the Students” had much higher perceptions of social presence than those that were not. There is no statistically significant difference for any other dependent variables.

Results of Welch’s t-Tests with visibility as group designator (no = 0 / yes = 1) for “Course Contacts” yields a significant result on sociability, $t(26.48) = 3.582$, $p = .006$ (after Holm-Bonferroni Correction), $d = 1.06$. This large effect (Cohen, 1988) suggests that students aware of “Course Contacts” perceived the learning environment to be substantially more sociable than students that were not. There is no statistically significant difference for any of other dependent variables.

Results of Welch’s t-Test with visibility as group designator (no = 0 / yes = 1) for “Dialogue” yields no statistically significant result on dependent variables. This can be taken to mean that the visibility of this social feature is not associated with any effects on dependent variables.

Results of bivariate correlations between frequency of self-reported engagement with plugins and focal variables show significant relationships for “Meet the Students”

with Interaction ($\rho = .27, p = .045$), “Course Contacts” with Interaction ($\rho = .26, p = .039$), as well as “Dialogue” with satisfaction ($\rho = .28, p = .033$) (Table 16). However, because all p-values are just below the cut-off criterion of 5% and not adjusted for multiple comparisons, these correlations are largely uninterpretable.

Table 16. Bivariate Correlations in Spearman’s rho

	Engagement with...		
	“Meet the Students”	“Course Contacts”	“Dialogue”
Sociability	.17	.21	.05
Interaction	.27*	.26*	.11
Social Presence	.24	.16	.17
Social Space	-.02	.11	.21
Satisfaction	-.16	-.13	-.28*
Learning	.09	-.13	.02

Note. * $p < .05$;

Discussion

Designing and providing learning experiences that take socio-emotional aspects into account is a worthwhile goal for online and distance education providers, as this is expected to improve the overall learning experience, resulting in student satisfaction and course retention. This study reports on a unique and innovative approach to creating sociable learning environments and fostering socio-emotional variables like social presence.

The results of this study suggest that social affordances do indeed have an effect on perceptions of sociability. This means that enhancing the environment with certain social affordances, inspired by SNS, makes for a sociable learning environment, in that it is then perceived to allow for and facilitate spontaneous and informal social interaction. These results are in line with research showing that SNS are perceived to be especially capable of sparking and maintaining social interaction among students (Sutcliffe, Gonzalez, Binder, & Nevarez, 2011; Deng & Tavares, 2013; Kurtz, 2014; Thomas & Eryilmaz, 2014; Quong et al., 2018). More than that, this study is a first step in empirically validating the ecological approach to fostering sociable learning environments. Researchers and practitioners alike may benefit from these findings in that there is now empirical evidence suggesting that social affordances inspired by SNS’ social features can make for a more sociable learning environment.

But a sociable learning environment is not enough. In order to actually reap the benefits of this approach, downstream effects of other variables need also be observed. According to the SIPS model, sociability facilitates student-student social interaction, which in turn enhances social presence and allows for the emergence of a sound social space. Importantly, the data in this study is consistent with this model. We found higher perceptions of social interaction in the more

sociable learning environment, as well as a higher degree of social presence and indication of a sound social space. Although group differences were rather small, this pattern of results is extremely unlikely under the null hypothesis and, thus, suggestive of a reliable mechanism underlying these effects. Because we accounted for potential pre-treatment differences, we can conclude with some certainty that there is a causal mechanism.

Our findings empirically validate the approach of ecologically manipulating socio-emotional aspects of the online learning experience. Because these socio-emotional variables are related to satisfaction, which in turn is associated with student retention (Levy, 2007; Schreiner, 2009; Park & Choi, 2009; Joo, Lim, & Kim, 2011; Joo, Lim, Kim, 2013, Lee & Choi, 2013), these findings have important implications for designing satisfying and ‘sticky’ learning experiences. Researchers and practitioners concerned with the relative social isolation of online distance education and its possible detrimental effects may consult the SIPS model for guidance on how to overcome these drawbacks. Although there are other, more direct strategies to fostering socio-emotional variables (Stürmer et al., 2018; Lowenthal & Dunlap, 2018), we argue that the ecological approach may be an exceedingly low-cost and relatively easy way to do this. Thus, we suggest that practitioners may consider sociability as an important socio-emotional design consideration. Besides Moodle, other popular learning management systems like Canvas and Blackboard, too, provide a large array of plug-ins to extend the features of a given installation. Guided by the notion of affordances, carefully choosing plugins to strengthen the desired affordances may enrich the environment in ways that enhance the learning experience in the socio-emotional dimensions. Researchers may extend this line of research by validating the model in different contexts and with different social affordances to see if our findings prove robust. Outside of the socio-emotional realm, there may also be educational affordances and hedonic affordances (Kreijns & Kirschner, 2018) to consider when looking to improve the quality of learning experience in online distance learning.

We found no statistically significant difference in learning achievement, as measured by the end-of-course multiple-choice test. Although intuition might suggest that social aspects impact the effectiveness of online learning, this relationship has not been consistently empirically supported, the evidence here is still mixed (Joksimovic et al. 2015; Hostetter, 2013; Maddrell et al., 2017). We suggest that this may be the case because these studies work with different and possibly convoluted definitions of social presence (Weidlich, Rajagopal, & Kreijns, 2018). Results of this present study suggests that socio-emotional perceptions regarding the learning environment and of peers do not actually impact learning. Future research should investigate this question more thoroughly, for example by using precise definitions of their socio-emotional independent variables and employing multiple ways of assessing learning achievement.

With respect to visibility of the implemented social affordances we found that, although there was information material provided at the beginning of the course, not all students were aware of them during learning activities in the environment. Accordingly, a substantial share of students did not engage with these functions; up to 50% of students indicated that they never engaged with a given function. These results are surprising because they indicate that, even though there was a robust pattern of group differences in terms of socio-emotional aspects, for some students, these affordances did not seem to have any perceived benefits.

Effects of visibility of the implemented plugins on socio-emotional variables were also assessed. We found a large effect for visibility of “Meet the Students” on social presence. It seems that students aware of this feature also perceived a much higher degree of social presence. This can be explained by considering the definition of social presence and the function of the affordance associated with this plugin. “Meet the students” prominently shows enlarged profile pictures of students in the learning environment, thereby allowing for more distinct impressions of peers. Through the process of impression formation, by which student form an individuating picture of the other, social presence, the perception of interacting and communicating with other ‘real persons’ in the environment, emerges. Hence, being aware of this functionality in the learning environment seemed to greatly enhance perceptions of social presence. There was also a large effect for visibility of “Course Contacts” on sociability. This suggests that students aware of functionality perceived the environment to be much more sociable. This may also be explained by considering how this plugin works and how it relates to sociability. “Course Contacts” displays a list of students, sorted by last online activity, signals their current online-offline status and indicates quick ways for communicating. This aligns nicely with our definition of sociability, as the capacity of the learning environment to facilitate quick and informal social interaction. Hence, this affordance greatly influenced perceptions of sociability. The third plugin, “Dialogue”, chosen to represent the affordance “sharing” seems not to have exerted any significant influence on dependent variables.

Although these individual effects are large, they are hard to reconcile with the general treatment effect that was shown to apply to all dependent variables. It is somewhat puzzling that, although these affordances seems to have had an effect across the board, the visibility of these features did not influence all dependent variables analogously. This may be explained by low statistical power of these sub-group analyses (see limitations section).

In addition to visibility, actual engagement with these plugins may be expected to be related with socio-emotional variables. However, as zero-order correlations show, there were only

few weak relationships. Engagement with “Meet the Students” and “Course Contacts” was positively associated with interaction, engagement with “Dialogue” was negatively associated with satisfaction. These relationships are not statistically significant after Holm-Bonferroni Correction. It can be concluded, that there was no statistically robust connection between actually engaging with these features and socio-emotional variables.

These findings are surprising and run somewhat against intuitions of a deep social interactional fabric being a necessary to fostering socio-emotional variables. Instead, it seems that a *veneer of sociability*, as realized by introducing social affordances, may be enough to benefit social interaction, feelings of social presence, as well as the emergence of a sound social space, and ultimately lead to more satisfying learning experiences. Although surprising and in need of replication, this notion may be tentatively interpreted as good news, especially for practitioners looking for easy ways to improve their online distance offerings.

Limitations

With respect to methodology, some limitations apply. Although we aimed to counteract non-randomization through a pre-/posttest design, it is not impossible that there were differences in the groups that we did not account for. Unfortunately, due to practical constraints, there was no pretest for learning achievement. Also, reliability for this measure was slightly below the common threshold of .7. For these reasons, the results for learning achievement need to be interpreted with caution. If feasible, a randomized experiment is the next logical step to replicate and extend these findings.

Though statistical power was appropriate for our main research questions, it was rather low for our exploratory subgroup analysis, as these tests were only powered to detect large ($d > .8$) mean differences and medium sized ($r > .3$) correlations. This may have resulted in Type II errors as well as inflated effect sizes for those effect sizes that were significant (Gelman & Carlin, 2014). Therefore, these analyses, too, should be interpreted with caution. Future research along these lines may make sure that the experimental condition itself is large enough to conduct well-powered analysis within.

Another shortcoming is that we relied on self-report data for our conclusions. Though adequate for highly subjective perceptions like social presence, the validity of self-report for observable behavior like social interaction or engagement with plugins may be considered questionable.

In terms of measuring our key constructs, we relied on previously validated instruments. Our measure of social presence has also been shown to be psychometrically sound, but still is a

work in progress. Scale development via Rasch analysis showed that it still has shortcomings in representing respondents with low perceptions of social presence (Kreijns, Weidlich, & Rajagopal, 2018). We can not rule out that this may have introduced a bias in responses, although effects of this are expected to be small.

The pedagogical specifics of this course are also important in that they may limit a broad applicability of our findings. It should be noted that this was a relatively short, self-paced course without instructors; as such, findings generated from this study may not be readily applied to educational interventions that differ substantially from the context at hand. This may be especially true for courses with an active instructor, as past research has shown this to be a major affordance in leveraging social presence and/or instructor social presence (Swan & Shih, 2005; Richardson & Lowenthal, 2017; Oyarzun, Barreto, & Conklin, 2018).

Lastly, students self-selected into this study. We have no way of ruling out that this may have introduced a bias that limits the validity of these findings.

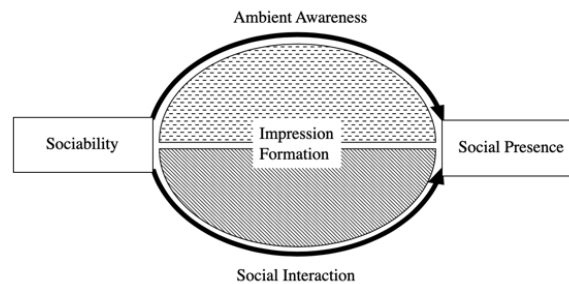
Conclusion

This study lends empirical support to the ecological approach to designing sociable online learning environments. Our intervention, although subtle and indirect, yielded small but consistent effects on our dependent variables. Students not only perceived the environment to be more sociable, but downstream effects were also observed, thus supporting the SIPS model.

The results of this study further suggest that social affordances do not necessarily need to be engaged with in a deep way in order for them to have an effect on perceptions of students. Instead, the mere availability and/or occasional engagement may be enough to turn an aloof environment into a relatively sociable one.

Because this was pre-/posttest quasi experiment in a field setting with high ecological validity, this general finding is not only academically relevant, but may also be readily translated into design recommendations for learning environments in the field. Practitioners and researchers may want to emphasize different social affordances in the learning environment by introducing plugins and observe how they are used by students and how this affects the social experience of online learning.

5.4 Pathway to presence: An investigation of ambient awareness in online learning environments



Ambient awareness is a recently suggested construct pertaining to interpersonal perceptions in Social Networking Sites (SNS). Although possibly relevant for online learning environments and theoretically related to social presence, there has been no research of this concept in learning contexts. In a first sample, results suggest that ambient awareness may indeed develop in online learning environments and that impressions formed through ambient awareness differ in some ways from impressions formed through social interaction. Moreover, ambient awareness may serve as a pathway to social presence, as suggested by results of mediation analysis in a second sample.

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Introduction

Online technology, especially social media and social networking sites (SNS) like Twitter and Facebook, provide us with a continual stream of information about other individuals. Through this, we have an unprecedented amount of insight into behaviors and perceptions of other social actors in our network. Even by simply browsing social media and SNS, a non-directed behavior, we are confronted with this plethora of information.

Even though distance and online learning has become mainstream (Seaman, Allen, & Seaman, 2018), there are still concerns leveled at many of these educational offerings, one of them being the relative social isolation of online and distance students (Rovai & Wighting, 2005; Boling, Hough, Krinsky, Saleem, & Stevens, 2012; Symeonides & Childs, 2015; Stürmer, Ihme, Fisseler, Sonnenberg, & Barbarino, 2018). How to help students feel psychologically close, whilst being physically distant, is a long-standing research agenda in distance and online learning, with seasoned and prominent concepts capturing the notion of psychological distance, like social presence (Gunawardena & Zittle, 1997). Despite advances in our understanding of the social realm in mediated learning contexts, many questions still remain. For example, what are the mechanisms of how important socio-emotional variables emerge and how can they be fostered to improve educational offerings in the socio-emotional realm? One fruitful but not yet well-trodden path may be to leverage our understanding of the psychology of social technology like social media and SNS and to use this knowledge base to better understand and ultimately improve socio-emotional aspects of online and distance learning.

This study investigates if and how ambient awareness, a form of peripheral awareness of others in online environments like SNS, can play an important role in how impressions of other social actors emerge and how this leads to perceptions of social presence among students. Through this, we close an important gap in our understanding about the relevant mechanisms that govern social presence in online distance learning.

Theoretical background

Social presence and the SIPS model. The term social presence goes back to the seminal work of Short, Williams, & Christie (1976) who defined it as “the degree of salience of the other person in the communication and the consequent salience of the interpersonal relationships” (p.65). Though not the only relevant socio-emotional variable, social presence is by far the most researched concept on how students interact with and relate to others in computer-mediated communication (CMC) and is considered to be an essential factor in online learning (Garrison & Arbaugh, 2007). It is also associated with beneficial variables like satisfaction and perceived learning (Richardson, Maeda, Lv, & Caskurlu, 2017). Its importance stems from the

fact that, unlike face-to-face communication, CMC usually conveys little socio-emotional cues. This is especially true for text-based asynchronous communication (e.g. message boards) in learning management systems (LMS), that are still a large part of the online learning landscape (Legon & Garrett, 2018). For this study, we adopt a clear and narrow definition of social presence rooted in the original work of Short et al. (1976), as elaborated in Weidlich, Rajagopal, and Kreijns (2018): *the psychological phenomenon that the other is perceived as 'real' in the communication, the subjective feeling of being with other salient social actors in a technologically mediated space*. In other words, “the sense of ‘being there, together’, when ‘being there’ does not involve a physical presence” (Öztok & Kehrwald, 2017, p.5).

Building on the work of Kreijns, Kirschner, and Vermeulen (2013), social presence is now embedded in the SIPS model which consists of four central variables: (S) the *sociability* of the learning environment, (I) social *interaction* among students, (P) feeling that other students are socially *present*, (S) leading to a sound social *space*. The SIPS model has recently been investigated on the structural level (Weidlich & Bastiaens, 2017) and in terms of its central propositions (Weidlich & Bastiaens, forthcoming). It has also recently been extended by Kreijns & Kirschner (2018).

Perceptions of social presence emerge through a process called impression formation (Walther, 1993). When students form salient impressions of other social actors (i.e. fellow students), for example in terms of personality, appearance, or competence, they may perceive them to be ‘real’ and ‘there’ when communicating in a computer-mediated fashion, thus constituting social presence (Weidlich & Bastiaens, 2017). In its current form, the SIPS model holds that these impressions are formed chiefly through direct social interaction, as this is expected to be the most straightforward way of forming these impressions. However, in learning environments with sufficient identifying information (e.g. profiles) and behavioral traces (e.g. posts, threads, and status updates), it is imaginable that salient impressions of other social actors may form without any kind of direct interaction or even an intention of gathering impressions, merely by browsing the environment and picking up on fragments of information ‘on the fly’. This mechanism is captured by a recently defined and investigated concept called ambient awareness (Leonardi & Mayer, 2015; Levordashka & Utz; Krämer et al., 2017). The following two sections will briefly review research on online impressions formation as well as what is currently known about ambient awareness.

(Online) Impression Formation. Solomon Asch (1946) is credited with being the first to experimentally study how people form impressions of others. His pioneering work was highly influential in that it not only identified some traits as central in forming impression (vs.

peripheral), but also in that his experimental paradigm was used and adopted widely in the following decades. Building on this work, Rosenberg, Nelson, and Vivekananthan (1968) found regularities in people's judgement of others in that they followed a seemingly underlying structure, organized around the dimensions of "warmth" and "competence". This was later termed the "Big Two" by Judd, James-Hawkins, Yzerbyt, and Kashima (2005), hinting at the Big Five personality dimensions.

Applying this research into the realm of technological mediation, there is now a large literature investigating how people form impressions through CMC. For example, Hancock & Dunham (2001) found that individuals rated other peoples' personality profiles differently, depending on if the previously interacted face-to-face or via CMC. More specifically, impressions formed in the CMC condition were less detailed but more intense, suggesting that with less social cues to go on, individuals exaggerated certain attributions. Governing these processes are inferential heuristics (e.g. stereotyping) to generalize from social cues and markers (Fiske & Taylor, 2013). Similarly, in a widely cited series of studies, Walther (1992; 1993; 1996), found that despite qualitative differences, CMC was not per se less personal or intimate than face-to-face, but instead that communicators adapt to and compensated for the lack of social cues over time. The mechanisms governing these processes were formulated in Social Information Processing Theory (Walther, 1992) and extended with the hyperpersonal model of CMC (Walther, 1996). These and other theories and frameworks are extensively reviewed in Walther (2011).

Moving away from dyadic exchange via CMC, this research tradition has now been applied to the domain of social media and SNS. For example, Van der Heide, D'Angelo & Schumaker (2012) showed that textual self-disclosure had a strong impact on impressions of others regarding their social orientation (i.e. introverted vs extraverted). However, when introducing profile pictures valenced in social orientation, these showed a visual primacy effect in that these had a much stronger impression effect, effectively undermining textual self-disclosures. Consistent with the hyperpersonal model, social media users spontaneously generated rich first impressions of others via social media profiles, in a study by Bacev-Giles and Haji (2017). Although profiles were sparse in detail, impressions were favorable, the target being perceived similar and likeable. Consistent with the notion of inferential stereotyping (Fiske & Taylor, 2013), unsubstantiated gender role stereotypes (e.g. male = athletic, female = friendly) were also found.

Taken together, the literature suggest that interpersonal impressions of others in the technologically-mediated realm are governed by attribution processes that are complex and

nuanced. Crucially, raters will use all social information available to form impressions of others, when prompted. There seems to be a plethora of interactions and dependencies in how these impressions are formed specifically. Although our understanding has grown substantially, much work needs to be done in order to reach a comprehensive understanding of these processes. Although social presence is a popular idea in research on online learning, impression formation, as an essential aspect of social presence, is hardly ever mentioned. Clearly identifying these relationships empirically constitutes, thus, a gap in research.

As in the investigations reviewed above, it is common in impression formation research to ask participants to judge others (Uleman & Kressel, 2013). Thus, there is an impression formation goal underlying these processes. But what about non-deliberate evaluations that are spontaneous and generated ‘on the fly’, as would be the case when merely browsing social media? A recently proposed and still sparsely researched concept, we suggest ambient awareness as a powerful driver of interpersonal attributions in online environments.

Ambient Awareness. Ambient awareness is a peripheral social awareness of others in an online environment, resulting from exposure without direct contact or communication (Levordashka & Utz, 2016). Governed by a process called spontaneous trait inference, there is ample evidence that people spontaneously attribute traits to others based on behavioral traces and person information (see Uleman, Saribay, & Gonzales, 2008 for an extensive review). Recently, Levordashka & Utz (2017) have shown that spontaneous trait inferences also take place in social media, where conditions are different than in many experimental paradigms. For example, classic experiments in spontaneous trait inference present one pair of actor and behavior at a time. In contrast, social media displays an array of different social actors and their behaviors in parallel and users browse through these information undirected and often inattentive. It is, thus, not straightforward to assume that distinct impressions form under these conditions. Nevertheless, similar inferential processes were observed in this first investigation by Levordashka & Utz (2017).

Ambient awareness is a special type of spontaneous trait inference in that impressions are formed spontaneously and goal-free, as well as without direct interaction. It is the sense of knowing the other through posts, statements, activities, which together form salient impressions. The concept suggests that traits like personality or competence of others can be inferred without interaction and although a single bit of information may not be telling, taken together they can paint a picture of the other person. Through this, individuals may infer traits without (1) having met the other person and (2) without even having communicated with the other person. In this

sense, ambient awareness would be present if a person can indicate an impression of others without having actually interacted with them.

As a rather novel concept, research is still limited. So far, ambient awareness has only been investigated in SNS. Pioneering the notion of ambient awareness, Leonardi (2015) first found that simply through being exposed to communication of others in an enterprise social network, employees became aware of crucial metaknowledge (e.g. “who knows what”) and accurately so. Applying this concept to everyday social media use, Levordashka & Utz (2016) found evidence for ambient awareness in Twitter users. Not only was there a general feeling of knowing their Twitter network as a whole but also did they report awareness of specific information in target individuals, for example their personality, career, and humor. Crucially, this knowledge emerged without direct interaction or communication. It was not predicted by Twitter use, relationship duration, and network size. Subsequently, Krämer et al. (2017) showed that the concept also resonated with Facebook users. They, however, did find a relationship between intensity of Facebook use and self-reported ambient awareness. Based on their data, they go on to suggest that especially for people that are less socially connected, ambient awareness may be a powerful mechanism and contribute to well-being. It is important to note that there are other awareness variables in the literature. For example, social network awareness (Cadima, Ferreira, Monquet, Ojeda, & Fernandet, 2010) and group awareness (Kirschner, Kreijns, Phielix, & Fransen, 2015). These are of a related nature but differ crucially from ambient awareness in that they (1) are not peripheral and indirect (i.e. ambient) and (2) they do not relate to interpersonal impressions of other students.

As of now, ambient awareness has not been investigated in an technology-enhanced learning context. Considering that the online and distance education landscape is still dominated by asynchronous activities in learning management systems (Legon & Garrett, 2018), we suggest that ambient awareness may be key to a better understanding of interpersonal processes and social aspects in these contexts. To the extent that learning management systems and SNS share properties and social mechanisms like message boards, profiles, and updates, we propose that ambient awareness will also emerge in online learning environments implemented through LMS. Moreover, we suggest that this process may be key to understanding how impressions of other salient social actors are formed that then lead to feelings of social presence. As this connection has not been investigated, we attempt to fill this gap in research to better understand how social presence emerges in online learning environments. Understanding these mechanisms has been shown to be non-trivial in the past (Kreijns, Kirschner, & Vermeulen, 2013; Weidlich & Bastiaens, 2017). Consequently, design recommendations for socio-emotional aspects of online

learning are still lacking (Cui, Lockee, & Meng, 2013). Ambient awareness may be a key concept to better understand these complex and nuanced processes.

Research questions and general method

Research question 1: *Do students perceive ambient awareness of others in an online learning environment?*

Research question 2: *Is there a relationship between ambient awareness and activity in the learning environment?*

Research question 3: *How many impressions and which types of impressions emerge through ambient awareness (versus social interaction)?*

Research question 4: *Does ambient awareness explain how social presence emerges from a sociable learning environment, i.e. is ambient awareness a mediator controlling for social interaction?*

To investigate ambient awareness in online learning environments, in a first sample (N = 51), we adopt the paradigm of Levordashka & Utz (2016) to assess if there is a general perception of ambient awareness, as well as specific awareness of certain impressions about target individuals. We go beyond this by comparing impressions formed through ambient awareness with impressions formed via direct social interaction like communication. Research questions one through three will be answered through this paradigm (see section “Impressions formed through ambient awareness”). In order to embed the concept of ambient awareness in a more general framework of social aspects in online learning (the SIPS model), in a second sample (N = 169), we test ambient awareness as mediator between the sociability of the learning environment and social presence (Figure 12). This would then be a second pathway to social presence, one that hasn’t been explicitly considered in social presence research thus far. This will answer our fourth research question (see section “Ambient awareness as a pathway to social presence”).

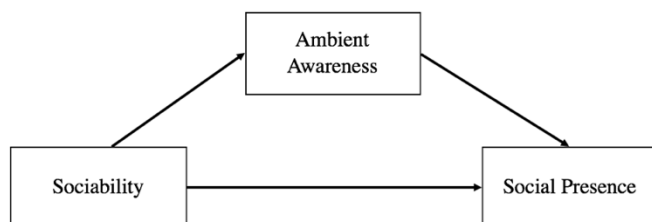


Figure 12. Conceptual mediation model (RQ4)

Impressions formed through ambient awareness

Participants and procedure. Students took part in an optional four-week course on instructional design and the 4C/ID model by Van Merriënboer & Kirschner (2018) in the winter semester of 2017/2018. This course was implemented via the LMS Moodle and featured typical activities like continuous asynchronous message board discussions, regular synchronous chats, individual and collaborative learning tasks. The environment also featured typical properties that are similar to SNS, like profiles with profile pictures, online status, different ways to connect with others and the possibility to browse through the history of message board postings. After concluding the course in week four, students were lead to a survey via a URL in the learning environment. The survey took approximately 15 minutes to complete and students received no compensation for their participation. The survey was open for a duration of one week. Participants were B.A. Educational Science students at the largest distance university in Germany, FernUniversität in Hagen. Sample size is $n = 51$ (41 women) with a mean age of 36.3 years ($SD = 11.9$).

Measures. Our methods of measuring ambient awareness were inspired by the paradigm of Levordashka & Utz (2016) but adapted in content for the online distance learning context at hand. We first asked participant to indicate their general experience of ambient awareness in the learning environment with a single item, taken from Levordashka & Utz (2016), but adapted for the present context and translated into German: *“It is possible that in some online learning environments, you can form distinct impressions of other students, even without communicating with them. Even with only a few message board postings and some profile information, together they might give you an idea of the person who posts them - what they are like, what they do, etc. Do you experience such general awareness of the people in the learning environment? And to what extent?”* Students rated this item on a continuous scale from 1 (not at all) to 5 (to a great extent).

Our second measure sought to identify awareness of individual targets. Participants were presented with a set of five randomly chosen fellow students, one at a time. They were shown name and profile picture of that fellow student and were then asked to indicate if they had engaged in any kind of direct social interaction in the learning environment. After this, they were asked to indicate the impressions they had of the fellow student. This was prompted with the following instruction: *“Even without direct contact, it is possible that you’ve indirectly learned something about that person. Please indicate the category/categories of which you were able to gather an impression. Please listen to your gut feeling and try to answer this instinctively. It’s not important*

that you've gathered deep and accurate impression. Instead it is enough if you feel that you have an inkling, a general idea of that impression category." If students indicated no social interaction, but still indicated that they had formed impressions of the target, this was counted as an instance of ambient awareness. If students indicated social interaction, the reported impressions were counted as a result of their interaction. Our suggested impressions are such, that they are not directly observable but need to be either inferred through social interaction or attributed from identifying information or behavioral traces (in contrast to Weidlich & Bastiaens, 2017, where impression formation included to an extent directly observable impressions like name, appearance, and age). Through this modification we make sure that these are indeed 'processed' and not merely observed characteristics of other individuals. These suggested impressions were: Personality, warmth, competence, learner type, humor, interest in learning subject, motivation for learning, and 'other'. Personality and humor was chosen because these are well documented attribution traits under conditions of spontaneous trait inference (Uleman et al. 2008). Warmth and competence were chose as they are considered the "Big Two" of impression formation (Judd et al. 2005). Additionally, learner type, interest in learning subject, and motivation for learning were chosen as learning-related impressions that were expected to be inferable from behavioral traces in the learning environment. Ambient awareness score was calculated as the sum of reported impressions of an individual toward a given target. As participants could indicate up to eight impressions per person, this resulted in a maximum of forty possible impressions.

We also asked participants to indicate their general degree of activity in the learning environment via the item: *"I regularly took part in activities in the learning environment"*. Students rated this item on a Likert scale from 1 (strongly disagree) to 5 (strongly agree).

In addition, we calculated an activity score to account for engagement with the activities in the learning environment. This composite score consisted of self-reported (1) participation in chats, (2) activity in chats, (3) activity in message boards, and (4) percentage of postings read in message boards. Sub-scores were added, resulting in maximum of 16 points in activity score.

Analyses. Our first research question will be answered through a descriptive look at the data. We suggest that evidence for ambient awareness comes from self-reported agreement to the general ambient awareness items and target ambient awareness, the ability to report impression of a given student in the learning environment without having engaged in social interaction with this student. Thus, participants indicating that they have formed an impression without interaction were counted as an instance of impression formation. Our second research question will be answered through bivariate correlation between measures of ambient awareness and of activity. In addition, we will conduct linear multiple regression to further investigate the relative predictor

influence on ambient awareness. Research question 3 will be accounted for by adding the total amount of impressions formed through ambient awareness and social interaction and by looking at the prevalence of different impression types generated through ambient awareness and social interaction. Research question four will be answered in section “Ambient awareness as a pathway to social presence”.

Results

Regarding the first research question (“Do students perceive ambient awareness of others in an online learning environment?”), we found a mean agreement with the ambient awareness single-item measure of 2.75 (of five, SD = 1.13), indicating that the concept does to a certain degree resonate with students in this sample (Table 17). In addition, students reported a mean of 1.78 target impressions gathered through ambient awareness, with a minimum of zero and a maximum of five. Taken together, this descriptive data provides evidence for the existence of ambient awareness in this online learning environment.

Table 17. Descriptive Data

	N	Min.	Max.	Mean	Median	SD
Ambient awareness general	49	1	5	2.75	3	1.13
Ambient awareness target	51	0	5	1.78	2	1.29
Activity general	48	1	5	2.56	2	.99
Activity score	48	1	16	7.93	7.5	3.53

To answer the second research question (“Is there a relationship between ambient awareness and activity in the learning environment?”), we calculated zero-order correlations through Pearson’s r (Table 18). Results show that, indeed, both general ambient awareness and target ambient awareness correlate highly with general activity and activity score, with overall large effect sizes of $r \sim 0.5$ (Cohen, 1988), $p < .001$. To further investigate these relationships, we conducted two linear multiple regressions, with both general activity and activity score as predictor and general ambient awareness as dependent variable in the first and target ambient awareness as dependent variable in the second. Results indicate that both predictors explained 36.7% of variance of general ambient awareness ($R^2 = .367$, $F(2,45)=13.02$, $p < .001$). We found that activity score significantly predicted general ambient awareness ($\beta = .50$, $p < .01$), but general activity did not ($\beta = .14$, $p > .05$). Results of our second regression indicate that both predictors explained 31.4% of variance of target ambient awareness ($R^2 = .314$, $F(2,45)=10.31$, $p < .001$). We found that, again, activity score significantly predicted target ambient awareness ($\beta = .35$, $p < .05$), but self-reported general activity did not ($\beta = .25$, $p > .05$). Thus, our data suggests converging evidence of a relationship between activity in the online learning environment and the emergence of ambient awareness. However, we found that the calculated activity score, albeit also self-reported, performed better as predictor for ambient awareness than the self-reported general activity.

Table 18. Bivariate Correlations

	1	2	3	4
Ambient awareness item (1)	-			
Ambient awareness score (2)	.723***	-		
Activity item (3)	.487***	.500***	-	
Activity score (4)	.598***	.530***	.698***	-

Note. *** < .001

With regards to the validity of the single-item general ambient awareness measure, we found evidence for construct validity. As shown in zero-order correlations, there is a larger correlation ($r = .723$) between general ambient awareness and target awareness than between general ambient awareness and general activity as well as activity score ($r = .487-.598$). Moreover, the large correlation between general ambient awareness item and the more specific behavioral ambient awareness score is evidence for convergent validity.

Regarding research question 3.1 (“How many impressions emerge through ambient awareness (versus social interaction)?”), we found that thirty-five students had experienced target ambient awareness and they formed an average of 1.93 impressions of others in the learning environment, with a maximum of five impressions (Table 19). On the other hand, twenty-seven students reported having formed impressions through social interaction with an average of 2.62 impressions, with a maximum of six impressions. This suggests that social interaction is a more effective way to form impression of others, thus a more powerful pathway to knowing fellow students. However, it was also less common as a means to form impressions. In learning environments with low degrees of social interaction, ambient awareness might play an increasingly large role in terms of interpersonal perceptions.

Table 19. Amount of impressions gathered per type of impression formation

	N	Min	Max	Mean	Median	SD
Ambient Awareness	35	1	5	1.93	2	.90
Social Interaction	27	1	6	2.62	2.5	1.18

Note. Does not add up to $N = 51$, because a given participants could form impressions through both ambient awareness and social interaction.

Regarding research question 3.2 (“Which types of impressions emerge through ambient awareness (versus social interaction)?”), we found that certain types of impressions were more likely to be formed through ambient awareness than social interaction and vice versa. For example, 83.3% of students that reported having gathered impressions through social interaction, indicated that they had some impression of the targets personality. This was much less prevalent in ambient awareness impression formation, with only 39.5%. On the other hand, about half of students (50.9%) that had formed impressions through ambient awareness reported having an indication of the targets competence. This was somewhat lower under conditions of social interaction, with 39.8%. Figure 13 shows the prevalence of impression types for both ambient awareness and social interaction (does not add up to 100% because students could gather more

than 1 impression per peer). The small percentage of ‘other’ impressions suggests, that our suggested impression categories did well in capturing the breadth of impressions.

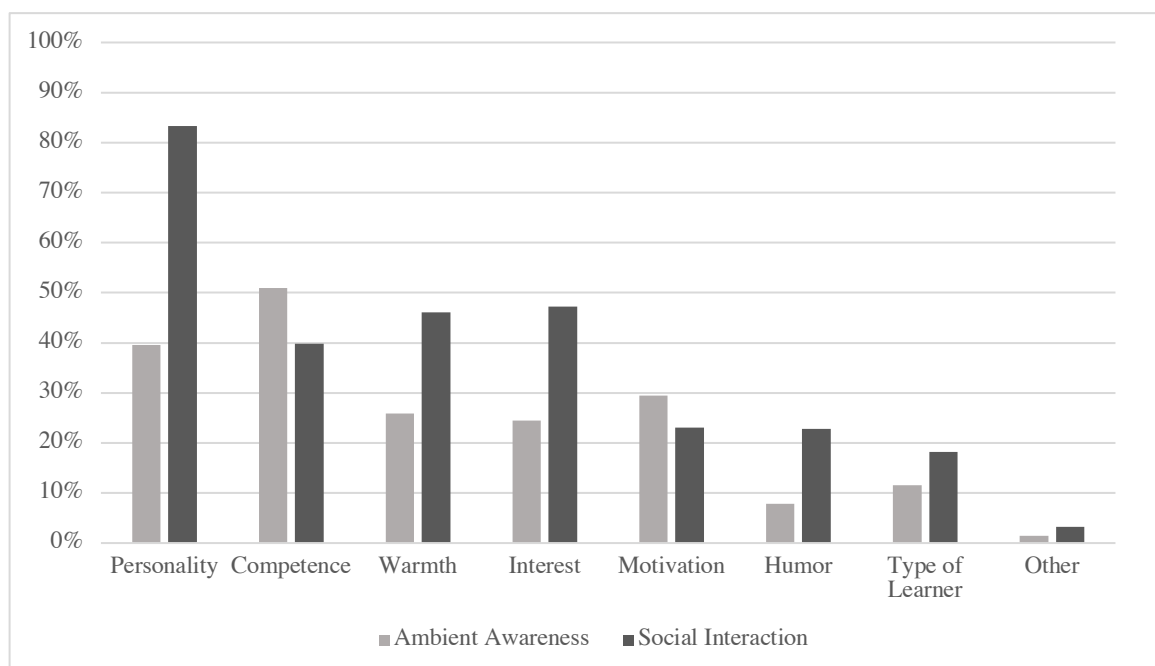


Figure 13. Prevalence of impression types through ambient awareness and social interaction, sorted from most prevalent (left) to least prevalent (right).

Ambient awareness as a pathway to social presence

Participants and procedure. Participants in this second sample were again B.A. Educational Science students at FernUniversität in Hagen, Germany. Final sample size is $n = 169$ (147 women) with a mean age of 35.5 years ($SD = 8.47$). Students were recruited during summer semester of 2018 in the regular bachelor’s class instructional design and educational technology as well as the optional four-week course on instructional design, focusing on the 4C/ID model. Toward the end of both classes, students were lead to the survey via a URL in the learning environment. The survey was open for a period of ten days. To account for potential double participation, students generated an individual code for the survey based on their birthday and mother’s maiden name. Through this, five duplicate entries were deleted from the final dataset.

Measures. Sociability of the online learning environment was measured with a ten-item measure, presented in Kreijns, Kirschner, Jochems, Van Buuren (2007), and slightly adapted in wording to be more generally applicable to online learning environments that are not necessarily collaborative. Cronbach’s Alpha for our sample was .85. Recently developed and improved upon previous measures, we used the ten-item social presence measure (Weidlich, Rajagopal, & Kreijns, 2018; Kreijns, Weidlich, & Rajagopal, 2018). Cronbach’s Alpha in the present sample was .91. Social Interaction was measured with a four-item scale that was taken from Wei, Chen,

& Kinshuk (2012). In our sample, we found a Cronbach's Alpha of .92. Lastly, we used the general ambient awareness item from Levordashka & Utz (2016), assessed in our first sample.

Table 20. Zero-order correlations between SIPS variables in this study and ambient awareness

	1	2	3	4	5
Sociability (1)	-				
Social Interaction (2)	.343***	-			
Social Presence (3)	.380***	.367***	-		
Social Space (4)	.510***	.402***	.451***	-	
Ambient Awareness (5)	.357***	.236***	.456***	.192*	-

Note. * < .05 *** < .001

Analysis. Classical tests of mediation are the Baron and Kenny (1986) procedure and Sobel test (Sobel, 1986). Because these methods have been associated with shortcomings (Hayes & Scharkow 2013; Shrout & Bolger, 2002), a bias corrected bootstrapping procedure was used instead. The PROCESS macro for IBM SPSS is a tool for path analysis-based moderation and mediation, as well as conditional process models (e.g., moderated mediation and mediated moderation) (Hayes, 2013). For our mediator analysis, the sociability of the learning environment was used as X variable in the mediator model, social presence the Y variable, with ambient awareness being the mediator. We ran the analysis twice, once (1) not controlling for social interaction and again (2) with controlling for social interaction.

Results. Our first model (not controlling for social interaction) explains 26% of variance in social presence, with $R^2 = .261$, $F = 29.34$, $p < .001$, suggesting partial but substantial mediation of ambient awareness with an indirect effect of .151 [.061; .255]. This suggests that even though ambient awareness explained some of the relationship between the sociability of the learning environment and social presence, this mechanism does not account for all of the relationship. According to the SIPS model, some of this relationship should also be explained by social interaction, as a sociable learning environment is conducive to social interaction, which in turn is a driver of impression formation and, thus, social presence (Weidlich & Bastiaens, 2017). Therefore, we tested a second model with the same conceptual relations, but added social interaction as a control variable. Our second model (controlling for social interaction) is slightly better at explaining variance of social presence with 30%, $R^2 = .306$, $F = 24.19$, $p < .001$. Here, too, we found a partial mediation of ambient awareness with an indirect effect of .122 [.042; .220].

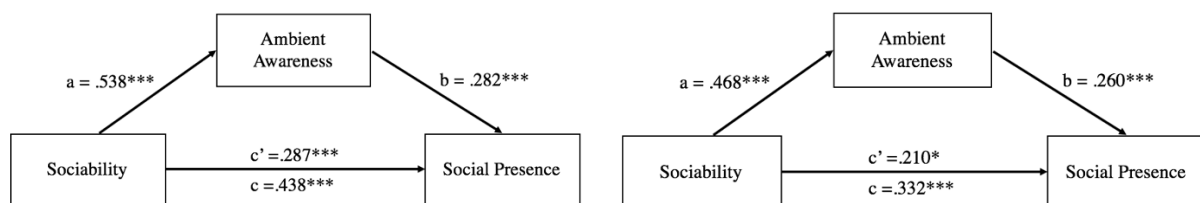


Figure 14. Mediation model results, not controlling for social interaction (left) and with controlling for social interaction (right).

Limitations

Our findings are likely not applicable to all online learning settings. It may be important that there are mechanisms in the learning environment that allow for impression formation without social interaction. These mechanisms (or social affordances, see Weidlich & Bastiaens, 2019) can be found in social media and SNS. With regards to the current evidence base, we hypothesize that ambient awareness only arises in environments that are sufficiently similar to these rather sociable environments. Thus, generalizability of our results may be limited to these kinds of learning environments.

Ambient awareness was measured with a single-item for answering our fourth research question. Of course, there are constraints in validity associated with single-item measures and unfortunately reliability could not be estimated in this study. However, there are single-item measures that have been proven to be valid and reliable, job-satisfaction being a prominent case (Wanous, Reichers, & Hudy, 1997). In our case, we present some compelling evidence of convergent validity (i.e. correlation between general ambient awareness and awareness score) as well as construct validity (i.e. ambient awareness mediates between sociability and social presence, as predicted by the SIPS model). This suggests adequate reliability (Drost, 2011).

The design of this study did not allow for assessing impression formation effects of social interaction whilst explicitly excluding ambient awareness effects. Therefore, our comparison of these two impression formation modes may be asymmetrical in that we compare awareness vs. social interaction + ambient awareness. Although a limitation in terms of clean differentiation, this would actually strengthen the evidence for ambient awareness effects in that the share of ambient awareness impressions would be even higher.

Another limitation is that we had no experimental manipulation in our second sample and therefore our causal assumptions in the mediator model are merely theory-driven but not empirically derived. Therefore, results should be interpreted with caution until our causal assumptions are proven substantiated.

Conclusion and Discussion

In this study, we found evidence for ambient awareness in online learning environments. Both our general ambient awareness item as well as target ambient awareness suggest that under the conditions of our study, by virtue of being exposed to behaviors of their peers, students formed impressions ‘on the fly’, possibly without impression formation goals and definitely without direct social interaction. In our first sample of 51 students, 35 participants formed at least one

impression through ambient awareness, with most forming two impressions. Our evidence for ambient awareness is in line with work by Levordashka & Utz (2016) and Krämer et al. (2017).

We also found a relationship between measures of activity and ambient awareness. However, there seem to be nuances to this relationship. We found that participant's general estimation of regularity of activity did not predict ambient awareness. In contrast to this, the activity score, calculated via self-reported engagement with learning activities (e.g. discussions in message boards, synchronous chats), did indeed predict ambient awareness. This aligns with the results of Krämer et al. (2017), who found that intensity of Facebook use (e.g. time spent on Facebook per day, number of Facebook friends, etc.) was associated with ambient awareness. However, our results diverge from Levordashka & Utz's (2016) findings in that general activity in the learning environment was not a predictor for ambient awareness, but their Twitter use variable was. Here, differences between SNS and online learning environments like LMS may come into play. Where SNS use is mostly synonymous with browsing of feeds and, thus, provides the user with a constant stream of information about others, online learning environments are often different in design. Similarly, social affordances will differ, too (Weidlich & Bastiaens, 2019). Here, students don't have feeds but instead engage with the environment in a more goal-oriented way, for example by looking for relevant information or interacting with peers in learning activities. Therefore, logging in regularly may be a necessary but not sufficient condition for ambient awareness to emerge. Instead, a certain degree of intensity of use and investment into the learning experience may be necessary. Besides activity, a worthy direction of investigation may be how group size (e.g. small classes versus MOOCs) may further moderate the relationships investigated in this study. A first step in understanding social presence with differing groups sizes has been made by Akcaoglu & Lee (2016).

Our results lend support to the notion of ambient awareness in online learning environments, as students were indeed able to form impressions of their fellow students in the learning environment peripherally. This mechanism was comparable to social interaction. Although mean amount of formed impressions was lower than through social interaction, therefore a less powerful pathway to rich interpersonal impression, the higher prevalence of instances of ambient awareness may counteract this to an extent (Metzger, Wilson, & Zhao, 2018).

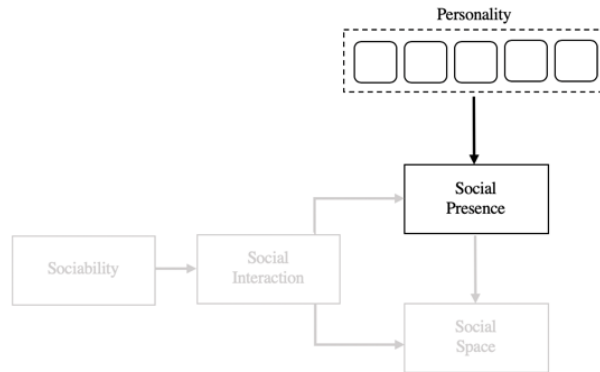
Interestingly, we found differences in types of impressions that were formed through ambient awareness versus social interaction. Only two types of impressions were more prevalent in ambient awareness, competence and motivation. One interpretation of this may be, that impression formation through ambient awareness may transport different kinds of information

about others. Here, it seems that the academically related impressions of competence and motivation are more readily transported than more personal impressions like personality and warmth. However, this needs to be investigated in more detail in future studies. Should differential impression formation effects prove to be a reliable phenomenon in online learning environments, design principles specifically accounting for this may be warranted.

Lastly, we were able to show that ambient awareness has explanatory potential in shedding light on the processes that govern the emergence of social presence. More than that, our analysis shows that ambient awareness explains, to an extent, how sociable learning environments actually lead to a higher degree of social presence. We, thus, propose ambient awareness as a pathway to social presence and worthy of further exploration in the context of socio-emotional aspects of online and distance learning.

Designers of online learning environments that take socio-emotional aspects of online learning into account may want to design for ambient awareness. This may be especially useful in learning contexts, that don't necessitate a high degree of social interaction between peers. In sufficiently sociable learning environments, students may be able to form impressions of others and perceive them to be socially present through ambient awareness alone.

5.5 A trait-level view of social presence: Exploring personality in online distance learning



Little is known about how individual differences may influence social presence experiences. Past research on personality factors' relationships with perceptions and behavior in online environments like Social Networking Sites though suggests that there may be similar relationships with perceptions in online learning environments. In this study, associations between Big Five personality dimensions and social presence are examined. Results suggest that certain personality configurations are indeed associated with social presence, whereas single personality dimensions are not. These findings highlight the importance of considering individual differences in our theorizing on social presence.

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Social presence and the SIPS model
Personality
Literature review
Personality in online distance learning
Personality in social networking sites
Research questions
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Sample
Procedure
Measures
Results
Discussion
Limitations
Conclusion

This study is currently in preparation as:

Weidlich, J., & Bastiaens, T. J. *A trait-level view of social presence:
Exploring personality in online distance learning*
[Unpublished: Manuscript in preparation]

Introduction

Distance Education enrollments are steadily increasing. Presently, students taking at least one distance education course make up 31.6% percent of total enrollments in the US, continuing the yearly trend of an approximately 1% increase since 2012 (Seaman, Allen, & Seaman, 2018). An instantiation of distance education, massive open online courses (MOOC) have also been growing, with more courses every year and about 20 million new users in 2018 (Shah, 2018). As these educational offerings make up an increasingly large part of the market share, their unique characteristics need to be properly understood in order to ensure high-quality online learning experiences.

Today, distance education programs rely in large part on online-technology to deliver instruction and mediate learning activities; hence, the term online distance learning. But there are concerns leveled at this mediated nature of learning. One such concern is that online and distance students may feel socially isolated or disconnected from fellow students (Rovai & Wighting, 2005; Boling, Hough, Krinsky, Saleem, & Stevens, 2012; Symeonides & Childs, 2015; Stürmer, Ihme, Fisseler, Sonnenberg, & Barbarino, 2018). Where face-to-face situations offer a rich social context and manifold avenues for non-mediated communication, online and distance learning often doesn't provide these perks. In online distance learning, social interaction with peers and instructors completely relies on technology for mediation (Hillman, Willis, & Gunawardena, 1995; Weidlich & Bastiaens, 2018). Communication technologies in these learning contexts are often text-based and, thus, constrained in the amount of social cues they can convey, compared to face-to-face settings. Due to this, rich socio-emotional experiences are more difficult to develop and sustain in online distance learning. As a result, concepts and models relating to interpersonal aspects are found in different lines of research in online distance education, for example Transactional Distance (Moore, 1993) or the Community of Inquiry Framework (CoI) (Garrison, Anderson, & Archer, 1999).

Social presence is a prominent concept to explain the challenges associated with the socio-emotional realm of online distance learning. It is by far the most researched concept on how students interact with and relate to others in computer-mediated communication (CMC) and is considered to be an essential factor in online learning (Garrison & Arbaugh, 2007; Richardson, Maeda, Lv, & Caskurlu, 2017). Social presence is originally defined as “the degree of salience of the other person in the communication and the consequent salience of the interpersonal relationships” (Short, Williams, & Christie, 1976), although today many different definitions are used (Lowenthal & Snelson, 2017). In line with many calls for a more precise definition and usage of social presence (Biocca, Harms, & Burgoon, 2003; Öztok & Kehrwald, 2017; Weidlich, Kreijns, Rajagopal, & Bastiaens, 2018), in this study, social presence is understood as the salience

of the other person in mediated communication, thus leaving out more expansive conceptions that have been associated with drawbacks (Öztok & Kehrwald, 2017).

Due to the perceived importance of the construct, social presence is frequently considered an imperative for good learning experience design. However, there are two arguments against this imperative. Firstly, it is not self-evident that all online/distance programs benefit from (very) high perceptions of social presence. For example, short courses and/or courses that do not rely on social interaction or collaboration for learning activities may be just fine without. There are also cost-benefit tradeoffs to be considered, as direct strategies for fostering and maintaining social presence (for an overview, see Lowenthal & Dunlap, 2018) can be time intensive and more indirect strategies are still emerging and may have smaller effects (Weidlich & Bastiaens, 2019). Secondly, not all students may be equally prone to engrossing themselves socio-emotionally in the online learning experience. The possibility of consequential individual differences for social presence, for example, certain personality characteristics that account for how students interact with and perceive others in the learning environment, may play an important role. If so, the imperative of social presence may penalize or benefit students differentially, for things they have no control over.

This second proposition, the possibility of individual personality differences that may be consequential for social presence experiences constitutes a gap in research, as there are hardly any studies examining personality (or other individual difference constructs, for that matter) and its relation to social presence. This exploratory study aims to address this gap in research by investigating how personality, as measured by the most prominent model, the Big Five (John & Srivastava, 1999), interacts with social presence. Results of this research are relevant for researchers and practitioners by examining whether individual differences in personality need to be considered when designing and researching socio-emotional aspects like social presence in online distance learning experiences.

Theoretical Foundations

Social Presence and the SIPS model. In 1976, Short, Williams, and Christie coined the term social presence in their book “The Social Psychology of Telecommunications” in order to explain how different telecommunication media affect communication. They defined it as “the degree of salience of the other person in the communication and the consequent salience of the interpersonal relationships” (p.65), positing that social presence is a quality of the communication medium itself. Accordingly, perceptions of social presence would be low in computer-mediated

communication (CMC) because unlike face-to-face communication, CMC usually conveys little socio-emotional cues.

Today, social presence is understood as less technologically determined and more as a function of social context as it is mediated by technology (Gunawardena, 1995). In time, the concept has been used to better understand all kinds of CMC, especially in the realm of online and distance education (for reviews, see Lowenthal, 2010; Cui, Lockee, & Meng, 2013). It is now considered an important aspect of the learning experience, especially for educational context that rely on text-based asynchronous communication (e.g. message boards) in learning management systems (LMS), that are still a large part of the educational technology landscape (Legon & Garrett, 2018). Social presence has since been linked to important other variables of online distance education, e.g. satisfaction and perceived learning (Gunawardena & Zittle, 1997; Richardson & Swan, 2003; Richardson, Maeda, Lv, & Caskurlu, 2017), online course retention (Liu, Gomez, & Yen, 2009) and online social interaction (Tu & McIsaac, 2002). In addition to this, social presence has a prominent role in the community of inquiry (COI) framework, alongside teaching presence and cognitive presence (Garrison, Anderson, & Archer, 1999; 2010).

As a more recent development, social presence plays an important role in the SIPS model, which consists of four central variables: the *sociability* of the learning environment (S), social *interaction* among students (I), perceptions of other students being socially *present* (P), leading to a sound social *space* (S) (Weidlich & Bastiaens, 2017). Together, these variables explain interpersonal socio-emotional aspects of the online learning experience. It is a generalized and modified adaptation of the Kreijns, Kirschner, & Vermeulen (2013) framework for social aspects of CSCL. The SIPS model has recently been investigated on the structural level (Weidlich & Bastiaens, 2017), in terms of its central propositions (Weidlich & Bastiaens, 2019) and has been further elaborated in Kreijns and Kirschner (2018).

As a construct rooted in social psychology, it is implicitly assumed that social presence is highly context-dependent and situational and, thus, a variable to be fostered and maintained. However, this may only be one side of the coin. As our literature review will show, there is ample research showing how trait-level personality differences interact with perceptions and behaviors in social networking use as well as online and distance learning. We contend that in order to get the full picture of how social presence and related socio-emotional variables emerge in online learning, the trait-level view may need to be considered.

Personality. The personality of an individual reflects a relatively enduring and automatic patterns of thoughts, feelings and behaviors (Roberts, Lejuez, Krueger, Richards, & Hill, 2014). Consequently, personality differences are between-subject discrepancies of these

patterns in trait-evoking situations. As a starting point for investigating these differences, the lexical hypothesis suggests that the most salient and relevant of ways in which people may differ from each other are encoded in natural language (Boudreaux & Ozer, 2015). The systematic extraction of all personality-relevant terms from the dictionary is thus called the lexical approach (Goldberg, 1993). Pioneering work was done in German by Baumgarten (1933) and in English by Allport and Odbert (1936). Over the years, scholars have gradually condensed these descriptors into taxonomies with different overarching dimensions. Although this approach is not without criticism (lexical fallacy, see Fiske, 2018), it has been extraordinarily successful in producing empirically supported psychological entities like emotions and personality dimensions.

After decades of research, the field of personality psychology has converged on a general and overarching taxonomy of personality traits (John & Srivastava, 1999). In their seminal work of the 1980's and 90's, Costa & McCrae developed the NEO Personality Inventory, which first consisted of three dimensions (1985), later adding two dimensions (1992), yielding the now widely known Five Factor model of personality (FFM) or Big Five. Moreover, they found six facets representing each personality dimension, making up the NEO PI-R, as it is known today (Costa & McCrae, 1995). The five personality dimensions are:

Openness to experience is the individuals interest in novel ideas and experiences. Individuals with high scores in openness are often creative, curious, flexible, and imaginative. Individuals with low scores tend to prefer familiar and more conventional experiences (McCrae, 1996).

Conscientiousness is the individuals tendency to engage in self-regulation. Individuals with high scores in conscientiousness are often orderly, dutiful, thorough, deliberate. Individuals with low scores tend to demonstrate a lack of direction and self-control (Roberts, Lejuez, Krueger, Richards, & Hill, 2014).

Extraversion is the individuals inclination towards the social realm. Individuals with high scores in extraversion are often sociable, assertive, outgoing, and energetic. Individuals with low scores tend to direct their interests inwards (Watson & Clark, 1997).

Agreeableness is the individuals disposition to smooth interpersonal relationships. Individual with high scores in agreeableness are often trusting, modest, compliant, and compassionate. Individuals with low scores tend to be more antagonistic (Graziano & Tobin, 2019).

Neuroticism is the individuals tendency to think and react in maladaptive ways. Individuals with high scores in neuroticism are often anxious, moody, self-conscious and vulnerable. Individuals with low scores tend to be more emotionally balanced (Ng, 2015).

Research has produced ample evidence of relationships between behaviors and perceptions with personality. Meta-analytically supported links with FFM traits include job performance (Barrick & Mount, 1991), resilience (Saeed, Oshio, Taku, & Hirano, 2018), academic performance (Vedel, 2014; Schneider & Preckel, 2017), motivation (Judge & Ilies, 2002), creative self-beliefs (Karwowski & Lebuda, 2016), self-efficacy (Stajkovic, Bandura, Locke, Lee & Sergeant, 2018), academic dishonesty (Giluk & Postlethwaite, 2015), physical activity (Wilson & Dishman, 2015), internet addiction (Kayis et al., 2016), and even humor style (Mendiburo-Seguel, Paez, & Martinez-Sanchez, 2015).

This history of well-supported relationships to perception and behavior suggest that personality may also have explanatory power in domains that are yet understudied in regards to individual differences, like online distance learning. We suggest taking FFM as a starting point to investigate these relationship as this the most widely used and best supported model of personality. For these reasons, the following literature review only includes research that has used the FFM. Other frameworks or models pertaining to individual differences like the Myers-Briggs Type Indicator or Learning styles will not be considered due to their well-documented problematic epistemic status (McCrae & Costa, 1989; Boyle, 1995; Kirschner, 2017).

Literature Review

Because there is ample evidence for relationships between personality traits and academically relevant variables (e.g. academic performance, motivation, self-efficacy), one might be tempted to generalize these results to online distance learning contexts. A comparative study by Varela, Cater, & Michel (2012), however, demonstrates that this may be a pitfall. The authors were able to show, that, depending on delivery mode of instruction (e.g. face-to-face versus online delivery), personality traits were differentially related to learning outcomes. For example, conscientiousness was a significant predictor for learning in online settings, but not in face-to-face settings. Similarly, gregariousness, a facet of extraversion, was negatively related to learning in online setting, but not face-to-face. Similarly, one sample may display divergent personality traits, depending on if the items are worded as to relate to face-to-face or online communication (Blumer & Döring, 2012). This can be explained through consideration of the differential properties of these trait-evoking contexts. For example, the circumstance of permanent technological mediation that is characteristic of online distance learning makes connecting with other people more challenging (Weidlich & Bastiaens, 2018), thereby possibly benefitting extraverted students, as they might go the extra mile to interact with others. Conversely, one could

also argue that behavioral patterns of extraverted students have less leverage in online settings and, thus, negate their usual advantage.

Therefore, the existing evidence of personality differences in experiences with online distance learning will be reviewed in section 3.1. Additionally, relationships between personality and social presence in these settings need to be reviewed. Because this latter literature is especially sparse, with only very few scattered examples, we additionally reviewed more broadly perceptions and behaviors on Social Networking Sites (SNS's) in relation to personality. As SNS' are exemplary in terms of social interaction, sociability, and interpersonal relationships, we then synthesize findings in terms of their relevance for online distance learning in section 3.2.

Personality in Online Distance Learning. Chen & Caropreso (2004) investigated how personality affected discussions in an online collaborative learning setting by grouping students high, neutral, low, and high+low groups regarding Extraversion, Agreeableness, and Openness. Interestingly, they found more task-related communication in the high groups than in neutral and low groups, but most in high+low groups, indicating beneficial group dynamics emerging from these personality differences. Looking at online course impressions of students, Keller & Karau (2013) found conscientiousness to be the best predictor of self-report engagement, value to career, overall evaluation, anxiety, and preference for online over face-to-face instruction. They concluded that conscientiousness may be an especially important trait for the getting the most out of online courses. Shih, Chen, Chen, Chen, & Wey (2013) investigated motivation, satisfaction, and personality in online learning and found extraversion and conscientiousness to be predictive of motivation as well as satisfaction. Finally, looking at students of a distance vocational school, Randler, Horzum, and Vollmer (2014) found that personality predicted opinions towards distance learning. More specifically, openness to experience predicted distance learning willingness and extraverted students reported less distance learning anxiety. In our literature research we found only one study specifically looking at relations between personality and social presence (Hingorani, 2008). It concluded extraversion to be positively related to perceptions of social presence. The same was true for the dimension thinking/feeling. However, as this was investigated with the Myers-Briggs Type Indicator, results need to be interpreted with caution.

Although the literature is still sparse, evidence of the importance of personality for online distance learning experiences is accumulating. Preliminarily, it appears that conscientiousness emerged as the most consistent important personality dimension, indicating that conscientiousness students seem to benefit the most from learning in these settings. Extraversion and openness to experience seem to also have some predictive value, but the evidence is still

somewhat inconsistent. Neuroticism and Agreeableness showed no pattern of relationship with students' experiences.

Personality in Social Networking Sites. As many online offerings for communication and social interaction are still relatively new, behaviors and perceptions in these novel social spheres are still being negotiated. Social Networking Sites (SNS) as highly social online platforms are, thus, perfect settings for expressing individual differences in personality; that is, they are expected to be highly trait-evoking (Stopfer et al., 2013). To the extent that online learning environments share some of these trait-evoking properties, results of personality influencing behavior may be transferable to these contexts, too.

In an early study on personality and Facebook behavior, Ross et al. (2009) found that openness to experience predicted prosocial behavior like posting on other's wall, commenting, and sending private messages. Similarly, Amichai-Hamburger & Vinitzky (2010) reported that openness to experience positively predicted sharing of personal information, whereas extraversion negatively predicted sharing of personal information but positively predicted number of friends. Gosling, Augustine, Vazire, Holtzmann, and Gaddis (2011), too, found extraversion to be related to different kinds of prosocial behavior on Facebook, like viewing pages of others, commenting, adding photos of oneself, number of friends, and hours spent on Facebook. Openness to new experience also correlated with some of these behaviors, but was a less salient predictor.

Comparing Facebook users to non-users, Ryan & Xenos (2011) found that extroverted individuals were generally more likely to be Facebook users and that they displayed more active social contribution, more real-time social interaction, and less passive engagement when using Facebook. Hughes, Rowe, Batey, and Lee (2012) were able to show that individual who preferred Facebook over Twitter were also more likely to be high on extraversion and neuroticism. They go on to suggest this difference may be due to different characteristics of the SNS', with Facebook being more personality- and identity-based, whereas Twitter is more anonymous and content-based. Correspondingly, Seidmann (2013) found that extraversion was associated with self-presentation and neuroticism with self-disclosure and emotional disclosure on Facebook.

Focusing on self-disclosure, Hollenbaugh and Ferris (2014) reported that extroverted Facebook users displayed more depth in self-disclosure whereas users with high scores on openness to experience display more breadth in self-disclosure. In a recent study comparing investigating Facebook and Pinterest, Lin, Lee, Yin, and Gilbreath (2017) found that here, too, extraversion and openness to experience were related to motivations for using these platforms.

Recently, scholars have turned to predicting personality traits from user-behavior data on SNS's. A meta-analysis by Azucar, Marengo, and Settanni (2018) showed that prediction coefficients obtained from data sources like textual posts and images largely corresponded to coefficients from offline behavior and that, in line with previous research, extraversion and openness to experience were predicted best by these digital footprints.

In conclusion, extraversion consistently emerges as the best predictor for pro-social behavior on SNS's like Facebook. As scholars have suggested, this may be because extroverts mirror their offline behavior in online setting and use SNS's as another way of satisfying social needs (Gosling et al, 2011). Openness to experience also emerged as a personality trait relevant for SNS's behavior, but less consistently so.

Depending on how online distance learning environments are designed and implemented, they may mirror properties and mechanisms that can be found in SNS' (Du, Fu, Zhao, Liu, & Liu, 2013; Garmendia & Cobos, 2013; Anderson & Dron, 2017, Weidlich & Bastiaens, 2019). Granting this, we hypothesize that when looking at social presence and other socio-emotional variables in online learning, there may be also be differences in perceptions due to personality differences. More specifically, we chiefly expect extraversion and openness to experience to predict perceptions of social presence.

Research Questions

Among the identified studies investigating effects of personality in online distance learning, some have failed to report sufficient descriptive information of their sample (e.g. Keller & Karau, 2013; Shih et al., 2013; Chen & Caropreso, 2004). However, this information is critical to assess the extent to which these students are representative of the broader population, an assumption that may be doubted (Harris & Gibson, 2006). Thus, our first research questions is concerned with the personality characteristics of online distance students:

Research question 1: *What are the personality characteristics of the online distance students?*

Our search efforts have yielded virtually no research that has considered personality as a predictor for social presence. The single exception being the study by Hingorani (2008), which, however was based on the Myers-Briggs Personality inventory. Thus, our second research questions was concerned with the relationship of personality dimensions with social presence:

Research question 2: *How are personality dimensions related to perceptions of social presence?*

As a final drawback of the literature of personality in online distance learning, investigations were limited to dimensional analyses. With these approaches, personality dimensions are analyzed separately with respect to the criterion variable. However, this fails to account how personality is actually represented in each person, as a unique configuration of different personality dimensions; a typology (Schnabel et al., 2002). It may well be the case that changing the unit-of-analysis from single personality dimensions to distinct personality will yield more valuable results. Thus our third research questions was concerned with the relationship of personality configurations with social presence.

Research question 3: *How are different personality configurations related to experiencing social presence?*

Method

Sample. Respondents were 201 students at the largest distance university in Germany, FernUniversität in Hagen. This convenience sample consists of students enrolled in either B.A. Educational Science (173 students) or M.A. eEducation (28 students), collected over two semesters: winter semester of 2017/2018 and summer semester of 2018. Of these students, 176 were female, 24 were male. Mean age was 37 years (min. 21, max. 75, SD=9).

Procedure. Students were recruited for the survey through the learning management system Moodle, in which the activities of these courses took place. Nearing the end of each semester, they were asked to participate in the survey with no course credit or reward attached to participation. For a duration of two weeks, a link in the learning environment directed them to the survey, which was created via LimeSurvey (<http://www.limesurvey.org>). Students were informed that the questions are concerned with their subjective experiences and preferences and were asked to answer accordingly. The survey took them a total of about 15 minutes to complete.

Measures. BFI-25 is a 25-item scale for assessing personality traits according to FFM. Each trait is assessed with 5 items. The German version of the BFI-25 has been validated in a randomly chosen and representative sample and shown adequate reliability and factor loadings (Gerlitz & Schupp, 2005). Out of the many choices of FFM inventories, this scale was chosen as a compromise between test duration and validity, as shorter inventories often display psychometric disadvantages (Rammstedt & John, 2007; Schupp & Gerlitz, 2008). Personality was measured on a 7-point Likert scale. Social Presence measurement was based on a recently developed measure (Weidlich, Rajagopal, Kreijns, & Bastiaens 2018; Kreijns, Weidlich, & Rajagopal, 2018). Crucially, this measure is based on a narrow and precise definition of social

presence that goes back to the original conception of Short et al., (1976), emphasizing the psychological sensation of the other being “real” and “there” in mediated communication. Social presence was measured on a 5-point Likert scale.

To assess the factorial structure of the FFM for the purposes of this study, an Exploratory Factor Analysis was conducted. Prior to this, one item (BF21) was removed due to low KMO measure of sampling adequacy (<.6, Hair et al., 2014). The remaining 24 items were entered. Bartlett’s test of sphericity confirmed factorability of the data. Because it is most robust to deviations from normality assumptions, principal axis was used as extraction method. As rotation method, oblimin was used due to expected correlations between some dimensions of FFM. The resulting analysis yielded a five-factor-solution with factor loadings ranging from .44-.95, with only one item showing substantial crossloading (BF17). Therefore, this items was eliminated for the subsequent analyses. The five-factor structure explained 54.2% of variance. All measures showed sufficient internal consistency with Cronbach’s alpha ranging from .69 to .91 (see Table 21).

Table 21: Overview of measures used in this study

Variable	Description	#Items	Cronbach’s Alpha
Openness	A general appreciation for art, emotion, adventure, unusual ideas, imagination, curiosity, and variety of experience.	5	0.87
Conscientiousness	A tendency to display self-discipline, act dutifully, and strive for achievement against measures or outside expectations.	4	0.72
Extraversion	A preference for breadth of activities, from external activity/situations, and energy creation from external means.	5	0.84
Agreeableness	A general concern for social harmony and valuation of getting along with others.	4	0.69
Neuroticism	A tendency to experience negative emotions, such as anger, anxiety, or depression.	5	0.78
Social Presence	The psychological sensation of the other being “there” and “real”.	10	0.91

Results

Descriptive results of personality characteristics were such that Agreeableness was the most endorsed personality dimension, followed by Openness and Conscientiousness, whereas Neuroticism was the least endorsed personality dimension, followed by Extraversion (see Table 22). As a result, Openness and Agreeableness were strongly left-skewed with medians of between 5.2 and 5.75 (on a 7-point scale), respectively, indicating that a large proportion of the sample are very open to experience and agreeable. Neuroticism and Extraversion were relatively normally distributed without skewness and yielded medians of 4 and 4.2, respectively. Factoring in gender of students showed that the only considerable difference was with respect to Neuroticism, with a mean difference of approximately ½ SD, yielding a statistically significant effect at the <.01 level.

Bivariate Pearson correlations yield five statistically significant relationships among personality dimensions, with effects between .21 and .33, one of which is negative (see Table 22).

Table 22. Descriptive data of personality characteristics

	Open	Consc	Extra	Agree	Neuro
Mean	5.14	5.12	4.21	5.69	4.03
Median	5.20	5.25	4.20	5.75	4.00
SD	1.16	1.06	1.27	.94	1.13
Skew	-.26	-.16	.06	-.97	-.02
Mean ♂ (N=24)	5.30	4.90	4.07	5.44	3.58
Mean ♀ (N=176)	5.12	5.20	4.22	5.73	4.11
Open	-	.31***	.28***	.21**	-.13
Consc		-	-.02	.33***	-.13
Extra			-	-.03	-.31***
Agree				-	-.06
Neuro					-

Note. **p < .01; ***p < .001

In order to detect distinct groups of personality configurations, a cluster analysis was performed on all five FFM variables. Cluster analysis is similar to factor analyses with the main difference in practice being, broadly speaking, that cases (i.e. respondents) instead of items are clustered into meaningful groups. A two-step clustering algorithm was chosen, which in a first step pre-clusters the data and then goes on to confirm the resulting structure in hierarchical clustering. Log-likelihood was chosen as distance measure and Schwarz Bayesian Criterion was chosen as clustering criterion. The algorithm proposed a three-cluster solution with a silhouette coefficient of .3, indicating “fair” cluster quality. The resulting clusters are roughly equal in size with a ratio of largest to smallest cluster of 1.31. Inspecting the relative importance of the BFF variables with respect to cluster structure suggests that Agreeableness (Predictor Importance, PI: 1) was the strongest predictor, followed by Neuroticism (PI: .82) and Conscientiousness (PI: .68). Extraversion and Openness were relatively weak predictors, with PI’s of .37 and .34., respectively. According to their primary characteristics as derived from the three strongest predictors, clusters will be labeled (1) “unagreeable & unconscientious”, (2) “agreeable & conscientious”, and (3) “agreeable & neurotic” (see Table 23). Chi-square test of contingency across cluster membership and gender revealed no statistically significant relationship, $X^2(2, 200) = 2.17$, $p = .34$. One-Way ANOVA suggested no age differences between clusters, $F(2,125) = 1.56$, $p = .21$.

Table 23. Personality clusters.

Cluster	1	2	3
Size	N=65 (32.3%)	N=59 (29.4%)	N=77 (38.3%)
Label	“Unagreeable & Unconscientious”	“Agreeable & Conscientious”	“Agreeable & Neurotic”
Prim. Characteristics	Low Agreeableness Med. Neuroticism Low Conscient.	High Agreeableness Low Neuroticism High Conscient.	High Agreeableness High Neuroticism Med. Conscient.
Second. Characteristics	Med. Extraversion Low Openness	High Extraversion High Openness	Low Extraversion Med. Openness

Note. “Low” $\approx$ 25th Quartile, „Medium“ $\approx$ Median, „High“ $\approx$ 75th Quartile

In order to examine relationships between personality dimensions and social presence, bivariate correlations were calculated. Result suggest no notable associations between BFF dimensions and social presence, with the strongest association between Conscientiousness and social presence with $r=.13$, yet failing to reach significance. Similarly, linear multiple regression with BFI dimensions as predictors and social presence as dependent variable shows a non-significant model $F(5,195) = 2.04$, $p = .075$, $R^2 = .05$.

Using the results of the cluster analysis, we can go beyond evaluating single personality dimensions in terms of social presence and instead investigate personality configurations. Thus, a one-way ANOVA with the three-level-factor “personality cluster number” and social presence as dependent variable was calculated. ANOVA assumptions were met for normality, $p = .131$ (Shapiro-Wilk), but not for equality of variances, $F(2, 198) = 4.01$, $p < .020$ (Levene’s Test). To account for unequal variances, Welch’s Test was used for omnibus analysis. Results indicate a significant difference between the three groups $F(2,198) = 3.71$, $p = .018$, while post-hoc tests (Games-Howell) indicate a significant difference between cluster 1 and cluster 3, $F(2, 125) = 2.61$, $p=.027$. The remaining comparisons yield no significant difference (see Table 24).

Using ANOVA and post-hoc tests from the family of robust methods that are known to yield better estimates under violated assumptions operating with trimmed means (trim level 0.2, Mair & Wilcox, 2019), we find stronger support for an effect, $F(2, 198) = 5.31$, $p = .007$. Post-hoc test yield a significant mean difference in social presence between cluster 1 and 3, $p = .003$ with an effect size of $\psi\text{-hat} = .35$, $[95\%CI:.07;.63]$, indicating a medium-sized effect (Mair & Wilcox, 2019).

Table 24. Personality clusters and social presence.

	Cluster	Label	N	Mean	SE	Post-hoch mean differences		
Social Presence	1	"-A & -C"	65	2.51	.09	} n.s.	} n.s.	} p=.027 (classic) p=.003 (robust)
	2	" +A & +C"	59	2.56	.11			
	3	" +A & +N"	77	2.82	.07			

Discussion

As a construct vernacular to social psychology, social presence has traditionally been conceptualized as a state variable, thereby also implying a certain degree of amenability to manipulation. The goal of this study was to investigate the potential value of adding a trait perspective that may place limits on the degree to which social presence is amenable to situation, context, and, thus, intervention. Instead, complementing our understanding of social presence, a trait-level view may introduce determinants that are outside of what is usually considered influencing factors. In this study, personality was used as a starting point to investigate social presence in relation to individual differences.

RQ1: What are the personality characteristics of the online distance education students?

Using the Big Five to characterize the personality profile of this online distance sample, we found Agreeableness, Openness, and Conscientiousness to be the most endorsed personality dimensions, whereas Extraversion and Neuroticism were the least. This is in line with previous research in online education settings (Randler et al., 2014; Cohen & Baruth, 2017) and may be explained by selection effects due to the unique characteristics of online distance learning (Caspi et al., 2006). Meanwhile, the correlations among personality dimensions were largely consistent with the broader personality literature (Digman, 1997; Anusic et al., 2009). Together, these findings indicate that the present population is both relatively typical with respect to the expected population of online distance learning students and as well as in terms of personality structure more generally.

Results of a cluster analysis suggested a three-cluster solution with clearly distinguishable personality profiles. Agreeableness being the strongest cluster predictor, we found one cluster with low Agreeableness and two clusters with high Agreeableness. The latter two can further be distinguished through one being high on Conscientiousness and the other being high on Neuroticism, with the remaining two personality dimensions adding less information. We can compare these clusters with the literature on personality prototypes that has consistently yielded three major prototypes, *resilient*, *overcontrolled*, and *undercontrolled* (Asendorpf & van Aken, 1999; Schnabel et al., 2002; Alessandri et al., 2014). The *resilient* prototype is mainly

characterized by low Neuroticism and high Conscientiousness, as well as medium to high scores on the remaining dimensions. Thus, it bears strong resemblance to the cluster “Agreeable & Conscientiousness (2) in our sample. The most indicative characteristic of the *overcontrolled* prototype is high Neuroticism and low Extraversion, thus, being similar to what we identified as “Agreeable & Neurotic” (3), the only slight deviation being the Agreeableness dimension. Finally, the *undercontrolled* prototype is characterized by low Conscientiousness and low Agreeableness, thus, mirroring our cluster “Unagreeable & Unconscientious” (1). These convergences lend support toward the validity of personality assessment in the present study, as it has yielded an analogue of what is a prevalent set of personality configurations with wide-ranging real-world relevance, for example with respect to understanding mental health (Bohane et al., 2017) or performance in simulator navigation training for candidate Navy Officers (Saus et al., 2012). Invariance of these configurations across gender and age further aligns these findings with the broader literature (Asendorpf & van Aken, 1999).

RQ2: How are personality dimensions related to perceptions of social presence?

Results of correlations and linear regression suggest no straightforward relationships between a given personality dimension and social presence. This goes against our hypotheses of Extraversion and Openness predicting social presence experiences in online distance learning, based on the larger literature of how personality dimensions interact with interpersonal aspects in online distance learning (e.g. Hingorani, 2008; Randler et al., 2014) and social networking sites (e.g. Seidmann, 2013, Hollenbaugh & Ferris, 2014). Thus, it appears that the dimensional perspective that has individual personality dimensions as unit-of-analysis provides little value in understanding the emergence of the social presence phenomenon.

RQ3: How are different personality configurations related to experiencing social presence?

Extending our analysis to account for personality configurations, however, paints a slightly different picture. Comparing mean values of social presence experience, we find a significant difference between cluster 1 (-A & -C) and cluster 3 (+A & +N), such that relatively agreeable and neurotic students appear more prone to experiencing social presence. The strongest cluster predictor, Agreeableness, being the most salient difference between these characteristics, one could intuit this to be the deciding factor. However, as cluster 2 (+A & +C) is similar in Agreeableness while failing to yield comparable social presence scores, this intuition should be discarded. In fact, it appears that no single cluster criteria can account for differences in social presence, but instead the whole configuration need be considered.

Drawing on literature of personality prototypes we can arrive at further interpretations. Our results suggest that overcontrolled students, that is, neurotic, introverted, and agreeable students are most prone to experiencing social presence. This finding is interesting, as a variety of interpersonal and social variables like trust in other people or participation in social events is usually weakly represented in overcontrollers compared to the other prototypes (Steca et al., 2010). With particularly low scores on Extraversion, overcontrollers are expected to be relatively socially reclusive and, thus, less prone to prosocial behavior. With these characteristics in mind, it is especially surprising that this personality type reports the most social presence. One explanation for this observation may come from the degree of sociability offered by an online environment compared to real-world interactions. Some research has found that high Extraversion not always translated to prosocial behavior in online environments, as extraverted users will not use the platform as an alternative to social activities (Amiel & Sargent, 2004; Ross et al., 2009). On the other hand, introverted users may be relatively more likely to satisfy their social needs in online environments (Amichai-Hamburger et al., 2002; Ebeling-Whitte et al., 2007; McIntyre et al., 2015), thereby, over time, exercising stronger adaption to the unique specifics of mediation communication. Experience of heightened social presence may be a result of this.

This study should be considered a proof-of-concept for the notion that social presence is not exclusively socially and contextually determined. It appears that, in addition, there are also meaningful individual difference factors at work, which are currently still underexplored. Future research may want to further investigate trait-level differences with respect to social presence experiences, for example, by delving deeper into what may account for the observed differences between personality prototypes. This may be done by including into analyses known correlates of social presence, as represented in the SIPS model. It is entirely possible that precursors of social presence, sociability and social interaction may account for social presence differences. Another fruitful approach may be a qualitative investigation of the personality prototypes, in order to arrive at a more in-depth understanding of their differential perceptions.

Limitations

Upon interpreting these findings, some limitations should be kept in mind. Firstly, all measures in this study were based on self-report, such that certain response biases like social desirability can't be ruled out. Additionally, although our sample size of 201 students should be considered adequate for typical analyses like linear regression, the most interesting findings of this study hinge on the results of our cluster analysis. A three-cluster solution essentially divided the total sample into three cells with sizes between 59 and 77, which were then used to assess our effect of interest, thus effectively limiting our sample size. However, a sensitivity analysis of

ANOVA, fixed effects, omnibus, one-ways, with $\alpha = .05$, $1-\beta = .8$ using G*Power (Faul et al., 2007) yields a minimum detectable effect size f of .22. As this is lower than our detected effect of 3.71, we can conclude that our study was sufficiently powered above the 80% level. Another limitation may be found in our assessment of the Big Five personality inventory for which we chose the BFI-25 as a compromise between test duration and validity. Still, this compromise may affect the validity of our findings as each sub-facet making up a personality dimensions could not be assessed with a sufficient amount of items. However, the adequate results in terms of factor structure and internal consistency of scales support the assumption that basic requirement for measurement have indeed been met. However, to ensure robustness, this effect should be replicated in future studies before being extended conceptually.

Conclusion

This study aimed to address a significant gap in our understanding of what determines social presence experiences in online distance learning environments. Toward this goal, the value of a trait-level view of social presence was investigated by assessing personality as a potential determinant. Results suggest no straightforward relationships between single personality dimensions and social presence. However, drawing on a cluster-based segmentation of our sample, we can identify three central personality configurations. Using these to compare social presence experiences, one personality configuration emerges as critical. Students characterized by high Agreeableness, high Conscientiousness, high Neuroticism, and low Extraversion (e.g. “overcontrolled” personality prototype) report significantly higher degrees of social presence than their peers with different personality profiles. These results point to the value of considering personality in our attempts to model the phenomenon of social presence and, possibly, limit the efficacy of our efforts of enhancing social presence in online distance learning through interventions.

SECTION C

DISCUSSION AND FUTURE DIRECTIONS

6 FINDINGS AND DISCUSSION

In order to present and discuss the main findings of this thesis in a way that demonstrates how the high-level challenges of social presence research have been addressed, the following sections will be structured around said challenges: identification, explanation, and application. After examining the contributions toward these challenges, we can derive central conclusions of this thesis and formulate preliminary answers for the questions of “what?”, “why?”, “when?”, “where?”, “who?” and “how?” (Whetten, 1989, see chapter 3.3), in order to satisfy fundamental requirements for a tentative theory of social presence. Finally, the last section discusses some limitations of this thesis and outlines promising avenues for improvement with respect to recent developments in educational technology research.

6.1 Identification

Many research endeavors concerned with social presence have not displayed sufficient effort towards a proper *identification* of social presence. Although some researchers have put a stake in the ground by restricting themselves to a consistent and well-defined notion of social presence, the greater literature hasn’t shown this restraint. As a consequence, the central variable of this line of research, social presence itself, has a history of being inconsistently defined (Lowenthal & Dunlap, 2017) and measured (Kreijns et al., 2011) causing the body of evidence to be highly eclectic and progress relatively slow. In order to improve on this undesirable state of the literature, the first overarching research question of this thesis was concerned with definition and measurement.

A thorough literature review suggested that definitions and measures of social presence abound, many of them convoluted. To provide researchers with a psychometrically sound alternative on which to ground future investigations, this first research question was answered by reporting on a scale development study (see chapter 5.1). The Rasch Measurement Model (Rasch, 1960), an item response theory approach overcoming many of the shortcomings of classical test theory. This yielded a psychometrically sound ten-item measure covering a large part of the construct. Due to these desirable properties and, crucially, a narrow and precise underlying definition that can be traced back to Short et al. (1976) that emphasizes the “realness” of the other in the mediated communication, the measure is shown to be valid and reliable. Although validation is an ongoing process and more validation work needs to be done, a scale meeting requirements of validity and reliability was presented. This way, it can confidently be concluded that all studies in this thesis investigate the same construct, thus, avoiding the jingle-fallacy (Thorndike, 1904) that has been widespread in social presence research (Kreijns et al., 2014).

This is important because, besides being a precondition for effective scholarly communication about the subject, (1) a clear understanding lays the foundation for valid measurement (Clark & Watson, 1995; 2019) and (2) a psychometrically sound and theoretically derived measure is the gateway for rigorous and cumulative research (Flake et al., 2017). The stressing of the issue of definition and measurement in this thesis is mirrored in recent calls to better measurement practices in the broader field of psychology (Fried & Flake, 2018). Thus, through answering the first research questions and establishing a valid measure, the first challenge of social presence research has been addressed and a foundation for rigorous investigation of social presence was provided.

6.2 Explanation

After the necessary groundwork of identifying what social presence is and what it isn't, research can build on this by attempting to provide an *explanation* of social presence. This was the second high-level challenge for social presence identified in this thesis. To be able to provide us with actionable knowledge, an explanation of social presence should encompass answers from two sides. First, there is a need for clearly delineating effects of social presence. That is, what are the merits of social presence and how meaningful are they for online distance learning? Second, we need a better understanding of the processes, conditions, and mechanisms governing this phenomenon. That is, when and under which conditions and for whom can we expect it to arise in online distance learning?

Explanation of effects. Although much research has been produced on the effects associated with social presence (see for example the meta-analysis of the relationship with satisfaction and perceived learning, Richardson et al., 2017), we should remain cautious. Inferences based on the literature may turn out to be flawed as much of research has used and continues to use convoluted definitions and measures of social presence (see chapter 5.1). Yet, having a solid base of evidence of the expected effects is critical to estimate how beneficial social presence really is and what we can expect from fostering it. For this reason, the third research question of this thesis was concerned with the investigation of these social presence effects under conditions of a more narrow definition and precise measurement. Empirical evidence from two studies in this thesis (chapter 5.2 and chapter 5.3) is mixed, in that some effects associated with social presence in the literature were found (i.e. social space) or replicated (i.e. perceived learning) whereas others remain questionable (i.e. satisfaction and learning achievement).

Social space. A theoretically predicted relationship was between social presence and social space. Theory suggests a causal flow such that feelings of social presence contribute to the development of a sound social space. An intimate connection of social presence with perceptions of group-level cohesion and climate is a common assumption of the social presence literature.

This is demonstrated, for example, by stating this connection as a rationale for the importance of social presence (Rovai, 2002; Ackaoglu & Lee, 2018) or, more implicitly, by collapsing the two into social presence (see Lowenthal & Snelson, 2017 for review). This assumption can be traced to the second part of the original definition by Short et al., (1976) "...and the subsequent salience of the interpersonal relationships" (p.65). Until now however, little actual research has come since to support this intuitive notion.

Critically, supporting what is a crucial hypotheses of the SIPS model, the evidence in this thesis has shown social presence and social space to be connected yet distinct constructs. As reported in chapter 5.2, these construct covary sufficiently to attribute a meaningful association. Thus, group cohesion, learning climate and feelings of community, as captured by the construct of social space, are facilitated by social presence. With a definition of social presence restricted to the "realness of the communication partner" these findings are far from being trivial and can be considered an advancement of our theorizing on social presence. The value of social connections is a staple of the distance education literature (Yu et al., 2010; Stürmer et al., 2018; Stone & O'Shea, 2019; Kara et al., 2019) and is, thus, considered an imperative for providing good learning experiences. Toward this goal, identifying conditions and antecedents of a sound social space is an important contribution toward generating design recommendations. With the findings of this thesis, we can have more confidence in the central role of social presence in fostering such socio-emotional processes that are associated with a sound social space.

Subjective outcome variables. Regarding outcome variables associated with these socio-emotional processes, chapter 5.2 also investigated perceived learning, confidence, and satisfaction with the learning experience. Although these subjective valuations are not synonymous with actual cognitive learning (Sitzman et al., 2010), they do indeed play an important role in the realm of online distance learning by providing the affective and emotional underpinnings of learning experiences (Richardson et al., 2010). Here, both social presence and social space was associated with perceived learning and perceived learning was in turn associated with satisfaction and confidence. Satisfaction was also predicted by socio-emotional variables but, in contrast to the general literature, not by social presence but instead by social space (see chapter 5.2). This, again, may be explained by use of convoluted definitions and measures of social presence. When we consider how substantially different social presence has been defined, ranging from a mere perceptual phenomenon to all-encompassing affective experiences, it must be conceded that we know little about what is truly the active ingredient when effects are found. Thus, well documented patterns of associations between social presence, satisfaction, and perceived learning in the literature (e.g. Richardson et al., 2017) may have been, to some extent, not robust. In a quasi-experimental design (chapter 5.3), there was further support for the general

notion that these socio-emotional aspects matter. Here, satisfaction was found to be influenced by relatively subtle changes in the design of the learning environment via increased social interaction, social presence, and social space. However, a clear cut picture of which variables specifically may have caused this was not inferable from this design.

Combining the results of these two studies, we find that, in line with expectations, subjective outcomes of the learning experience are, at least in part, determined by the degree to which these socio-emotional aspects are present. Constructing the causal chain from social presence to a sound social space and both together affecting subjective outcomes, there is increasing evidence that these socio-emotional aspects matter in online distance learning. This is in line with the broader literature according to which these subjective outcomes can be central towards combating, for example, student attrition (Levy, 2007; Schreiner, 2009; Park & Choi, 2009; Joo et al., 2011; Joo et al., 2013; Lee et al., 2013). However, we find that effects associated with social presence might be more nuanced than expected. This is in line with evidence demonstrating that different conceptions of social presence are associated with different effects (e.g. Caspi & Blau, 2008). Some might be subject to hidden moderators and others may be simply attributed to related socio-emotional variables (e.g. a sound social space) that in the past have been confounded with social presence. Although some new evidence was presented in this thesis, these important relationships should be reevaluated in future research while also considering a broader array of contextual moderators that may be at work here.

Learning achievement. In terms of cognitive learning outcomes, no evidence of a connection was found in this thesis. Learning achievement was not significantly affected by the intervention in chapter 5.3, adding to the inconclusive literature on this. Although many scholars seem convinced that social presence should affect cognitive learning and use this assumption to argue for the importance of social presence considerations (e.g. Wei et al., 2012; Molinillo et al., 2018), studies demonstrating this with measures that go beyond self-reported learning are mixed (Wise et al., 2004; Mackey & Freiberg, 2010; Zhan & Mei, 2013; Joksimovic et al., 2015; Maddrell et al., 2017; Dahlstrom-Hakki et al., 2020).

The mixed evidence is less surprising when we consider how these cognitive learning improvements are expected to arise. Whereas the theoretical bridge from socio-emotional aspects to affective outcome measures like satisfaction can easily be intuited, the theoretical chasm between socio-emotional variables and cognitive learning processes is deeper and, thus, more challenging to fill theoretically. In fact, the current literature on social presence provides little in terms of plausible psychological mechanisms that account for how social presence may lead to gains in learning achievement. However, to establish social presence as an important design

characteristic for online distance learning environments, this missing link to learning achievement needs to be addressed by producing a more conclusive confirming or disconfirming body of evidence, with a theoretical account modeling this connection as well as suggesting moderators that may explain differential effects.

Summary of effects. In conclusion, we find that although the importance of socio-emotional aspects for online distance learning is evident in the findings of the empirical studies, the previously (perhaps naively) assumed simple and linear relationships cannot be wholly supported. Instead, the data points to a more nuanced picture of effects associated with social presence, as (1) social presence has been shown to contribute to a sound social space, (2) subjective outcome variables are differentially related to both social presence and social space, and (3) there was no evidence for a connection between socio-emotional aspects and cognitive learning outcomes. Upon further deliberation, we can conclude that all three of these findings follow logically from inconsistencies in the literature. These inconsistencies need to be continually addressed in social presence research. As an interim result, our current state of knowledge limits the merits of social presence to its contributions to socio-emotional and affective aspects of the learning experience. Yet, more research is needed to tease out these effects more rigorously and context-specific while also assessing the elusive connection to cognitive learning processes.

Explanation of processes. After covering the important steps of identifying what social presence is, what it isn't, and what we can hope to expect from it, a better understanding of how the construct arises should follow. This implies modeling and investigating variables that may help us understand the construct, specifically by looking at means and conditions by which it emerges. As research in the past has often focused on outcome variables associated with social presence, much less emphasis has been put on understanding how and when the sensation of social presence arises, although this understanding is the basis for generating much needed design recommendations (Cui et al., 2013) and deriving empirically-tested and practically relevant recommendations is an important goal for an applied science like educational technology (Luppici, 2005; Rushby & Surry, 2016). For this reason, the second research question of this thesis was concerned with identifying precursors and mechanisms associated with social presence. Results suggest two basic mechanisms that account for social presence in online distance learning environments, both of which are underpinned by a well-known psychological process, impression formation.

Impression formation. Combined evidence from two empirical studies (chapter 5.2 & chapter 5.4) suggest two basic mechanisms that may account for the emergence of social presence, social interaction and ambient awareness. Underlying both these mechanisms is a process called

impression formation, a concept with a rich history in social psychology (see Uleman & Kressel, 2013 for a review) but that has rarely been explicitly investigated with regards to social presence, despite its apparent fit as an explanatory variable. Impression formation happens when, by accumulating information about others, e.g. their appearance, personality, preferences, and demeanors, we start constructing a mental picture of who this person is (Hancock & Dunham, 2001). Having a mental picture of another person at hand is what makes possible that we perceive this person as “real” and “there” in mediated communication, as suggested by Rettie (2005) and Kreijns et al., (2013). These impressions become more salient and richer over time, as we accumulate information through continued exposure. This thesis proposed two such exposure mechanisms, direct social interaction and, more indirect, ambient awareness.

Social interaction. As shown in chapter 5.2, impression formation can occur through mediated social interaction in computer-mediated communication, as is the case when using message boards and chats in online environments. Similar to face-to-face interaction, we learn about the other person(s) via content and form of these interpersonal communications; figures of speech, choice of words, length of messages, or volunteered personal information together form a picture in the minds’ eye. Although these effects have long been established in the literature on computer-mediated communication (Walther, 1993; 1996), there has been little explicit research empirically linking mediated social interaction with social presence through impression formation. The evidence reported in chapter 5.2 suggests that social interaction in the online learning environment indeed was associated with salient impressions of others, e.g. their appearance, personality, and competence. That is, students reporting higher degrees of social interaction felt that they knew more about their fellow students than those who interacted less. As additional evidence, students reporting more specific interpersonal impressions also indicated a general experience of having rich and salient impressions of their peers. Crucially, impression formation itself was related to social presence. These associations close a critical explanatory gap of social presence research, that is, how this construct emerges psychologically. Thus, a central finding is simply that social interaction leads to social presence via the mechanism of impression formation.

Ambient Awareness. However, as can be intuited, direct social interaction is not the only way for us to form impressions of others. From everyday experience, we know that we can just as well learn about others without direct communication. As an explanation for this phenomenon in online environments, the notion of ambient awareness is put forward in chapter 5.4. With ambient awareness, impressions are not gathered through the content and form of communication but through information available in the learning environment that a student may come across while browsing, searching for information, or looking to collaborate with peers. Building on the

work of Levordashka & Utz (2016), who conceptualized this phenomenon for Twitter, ambient awareness in online learning environments was investigated with a similar paradigm.

Yielding what is convincing evidence to support the notion of ambient awareness in online learning environments, many students reported individuating impressions without engaging in social interaction. Although these impressions were fewer, more students reported having learned about their peers through this mechanism than through social interaction. This suggests that these environments may share some important properties with social networking sites, in terms of their interpersonal effects. Even though there is a growing body of literature on the apparent socio-emotional differences between SNS's and learning environments (Thomas & Eryilmaz, 2014; Akcaoglu & Lee, 2018; Quong et al., 2018; Thai et al., 2019), there also seem to be some important commonalities, as seen in their propensity to foster ambient awareness. Crucial for enhancing our knowledge of the processes governing social presence, we found that ambient awareness plays a role as a mediator between sociability and social presence. Thus, besides direct social interaction, a second important precursor for social presence is the more indirect, yet possibly more pervasive ambient awareness. This contributes to our state of knowledge by providing two different mechanism through which we can learn about fellow students and, thus, facilitate social presence in online distance learning environments. Moreover, as these mechanism have different impression formation properties, that is yielding different impression configurations, they may be considered complementary in fostering a rich degree of social presence.

Personality. As a construct rooted in social psychology, social presence has historically been conceptualized as a psychological state, implying that it is highly context-dependent and, thus, amenable to contextual factors. This is also implicit when researchers investigate course-level social presence with relation to factors of course design. However, this may only be one side of the coin, as there is a large literature demonstrating how trait-level personality differences can interact with perceptions and behaviors in online environments like social networking sites (Stopfer et al., 2013; Seidmann, 2013; Lin et al., 2017). Thus, in order to consider this possibility, one research question of this thesis was concerned with the relationship of personality factors with social presence. In chapter 5.5, the most prominent and well-known personality model, the Big Five personality inventory (Costa & McCrae, 1995; John & Srivastava, 1999) was used as a starting point into the realm of individual differences that may predict social presence.

Contrary to expectations, there were hardly any notable associations between single personality dimensions and perceptions of social presence. This was especially surprising with regards to the personality dimension of extraversion, a trait that has consistently emerged as a

good predictor for social behaviors in online environments (Gosling et al., 2011; Seidmann, 2013; Hollenbaugh & Ferris, 2014). A closer look, however, revealed that different personality configurations, or prototypes, may be associated with social presence experiences. Three clusters emerged in the sample, with close resemblance to major prototypes of the personality literature (Asendorpf & Aken, 1999; Schnabel et al., 2002, Alessandri et al., 2014), one of which indicated a significantly larger degree of social presence. Interpreting these findings, we can see that the trait-level view of social presence may have merits in explaining social presence differences that cannot be explained by contextual factors. Thus, not all students may be similarly prone to experiencing social presence, possibly placing limits on our efforts to facilitate these experiences or asymmetrically benefitting certain students. The investigation of personality was intended as a mere starting point toward a more comprehensive examination of how individual differences may relate to social presence. There are many possible factors that influence how we perceive and engage socially with others, some of them associated with culture (Jung & Gunawardena, 2014), others determined more biographically, e.g. individualism vs. collectivism, personal vs. relational identity, or independent vs. interdependent self-construals (Varnum et al., 2010); all of them largely untested with regards to mediated communication. By default, there is reason to believe that these factors matter, as they have shown to interact with perception (Qi et al., 2020), intention (Gärling et al., 2003), cognition (Varnum et al., 2010), and behavior (Thielmann et al., 2020) in non-mediated settings. The added layer of complexity afforded through technological mediation makes this an area worthy of further investigation, with some preliminary evidence of its merits provided in this thesis. Thus, as a central finding of this thesis, we can conclude that single personality dimensions hold little explanatory power as to how social presence emerges but certain personality prototypes may be more or less prone to social presence experiences.

Summary of processes. To conclude this discussion of the processes governing social presence, we can summarize the updated state of knowledge by referring to two central mechanisms that give rise to social presence, social interaction and ambient awareness. Although they entail different perceptions, both are driven by accumulating individuating information about other students, i.e. impression formation. These impressions are what helps us form a picture of who the other person is, conjuring a mental image when engaging in mediated interaction. It is this relative saliency of the mental image that allows us to perceive the other as “real” and “there”, even when he or she is not, thus constituting social presence. Lastly, likely due to their undisputed personal and subjective nature, social presence perceptions appear to be, at least to some extent, susceptible to individual differences encoded in personality. Taken together, these findings form an increasingly complex picture of how social presence “works”, one that is multivariate in its emergence, sufficiently amenable to environmental interventions, as well partly susceptible to trait-differences.

6.3 Application

By making progress on the challenges of identification and explanation, systematically extending this knowledge in order to delineate strategies for enhancing social presence and socio-emotional aspects becomes possible. With a theoretical foundation at hand, hypotheses with practical implications can be derived and empirically tested. Similar to the dearth of practical guidance in learning design (Mor et al., 2015) and technology-enhanced learning design (Bower & Vlachopoulos, 2018), socio-emotional learning design recommendations are, too, largely absent (Cui et al., 2013). Thus, the fourth research question is concerned with the application of social presence, by means of testing strategies of fostering social presence in online learning environments.

To this end, an ecological framing of socio-emotional aspects was adopted, following Kreijns (2004) and Kreijns et al. (2002). By drawing from ecological psychology, which is concerned with the subtle but pervasive influence of environmental cues, i.e. affordances, on our perceptions and behaviors, we can analogously propose that certain characteristics of the online learning environment, i.e. social affordances, may influence socio-emotional perceptions and behaviors. The extent to which these design features enhance the environment in this regard is called sociability and this characteristic is deemed crucial as facilitative condition for socio-emotional processes to develop naturally by instigating the processes governing social presence, as presented above. As the literature suggests that many students in online distance learning have experienced environments that are not sociable (Deng & Tavares, 2013; Kurtz, 2014; Sadowski et al., 2017), this circumstance was identified as an important lever for enhancing social presence and improving online distance learning experiences in the socio-emotional realm.

As a characteristic of the learning environment, sociability is subject to deliberate design decisions, the hypothesis being that there will be mechanisms or features that can be introduced to the environment such that sociability is enhanced. In a correlational design, chapter 5.2 showed that these variables are related in the way that was theoretically predicted by the SIPS model with sociability as an exogenous variable predicting social interaction. Chapter 5.3 then put the central assumption of sociability as a lever for socio-emotional processes to the test. A quasi-experimental study with high ecological validity provided evidence for the efficacy of enriching an extant online environment with carefully selected affordances to account for the socio-emotional dimension of the learning experience, with promising effects on downstream variables of the SIPS model. Notably, there was no direct intervention, in the sense of facilitative strategies that specifically encouraged students toward these socio-emotional aspects. Instead, in line with the ecological notion, the environment influenced student perceptions and behaviors through the

social affordances provided, thus enhancing social presence and socio-emotional processes in a subtle yet pervasive way.

Making online learning environments more sociable in order to foster social presence is a notion that has been proposed as far back as 2002 (Kreijns et al., 2002). Yet so far, there has been little research translating this notion into a rigorous research design by, for example, providing experimental evidence for efficacy of this approach. As a central finding of this thesis, results of chapter 5.3 suggests that this affordance enrichment approach is both an easy-to-implement and effective strategy, thus highlighting the key role of sociability for setting the stage for other socio-emotional variables to develop. As an effective facilitator, it should, thus, be seen as an important design variables for future research and practice, an instrument to be added to the repertoire when designing for socio-emotional aspects. Although, the concept of sociability has been sporadically investigated in conjunction with socio-emotional learning design (Preece, 2001; Barab et al., 2001; Abedin et al., 2012), this design consideration has not been researched extensively. Putting a stake in the ground for fostering social presence through sociability, this thesis provides efficacy evidence for one specific instantiation of this approach. However, more ways of operationalizing the construct and testing these empirically may lead to a more comprehensive and context-sensitive understanding of sociability. As an instance, it is entirely possible that sociability may be better understood as multidimensional, for example with regard to the types of social interaction it facilitates (i.e. quadrants of social interaction, see Kreijns et al., 2013).

Concluding this discussion of how to apply social presence, we can highlight the central role of sociability in setting the stage and allowing for socio-emotional processes in general, but most notably, social presence to develop. As potentially both an objective characteristic that can be designed as well as a subjectively attribute that needs to be perceived by students in the environment, sociability is a complex and still under researched construct. Nonetheless, this thesis provides some early support for an sociability-based approach to fostering social presence and socio-emotional aspects, and thus, constitutes a welcome addition to the currently meager assortment of socio-emotional design recommendations.

6.4 A tentative theory of social presence

Taken together, we can conclude that central challenges of social presence have been addressed in this thesis, albeit not conclusively so. Putting these pieces together forms a complex and nuanced picture of social presence and socio-emotional processes in online distance learning. Figure 15 is a visual representation of this, an extended view of the SIPS model first presented in chapter 3.3. The extended model depicts all constructs, associations, and mechanisms investigated in this thesis. Descriptively, it provides tentative explanations for how these socio-emotional

variables work together by displaying central causalities and contingencies. As a design model, it can also prompt researchers and practitioners towards systematically exploring ways of enhancing learning experiences in the socio-emotional dimension by guiding their attention to relevant variables. Over time, the SIPS model should be further tested and extended. For example, different context-specific variants of the SIPS model could be developed to account for the diversity of possible learning contexts, student populations, or learning technologies. Lastly, socio-emotional aspects of the SIPS model could be embedded within a broader framework of variables relevant for online distance learning, as has been recently proposed for CSCL (Kreijns et al., 2018).

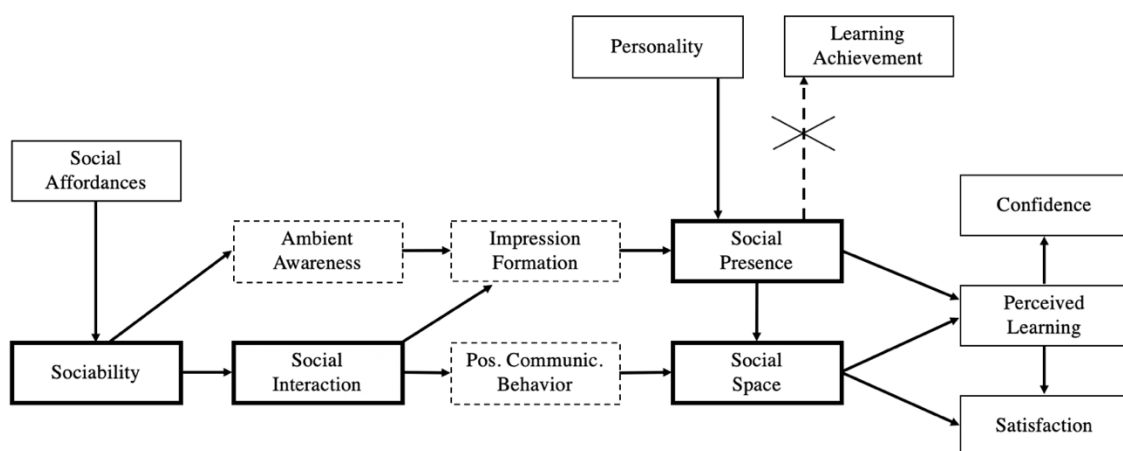


Figure 15. A visual summary of the results of these studies, framed within the SIPS model. Bold boxes signify central SIPS variables. Dotted boxes represent process variables or mechanisms that help explain how SIPS variables emerge. Dotted lines depict relationships that were not confirmed.

As identified in the literature review of this thesis and despite social presence *theory* being a popular term in the literature, social presence is a phenomenon still not well understood as our knowledge leaves many questions unanswered. Because many of these questions are fundamental to theory building, one goal of this thesis was addressing the *what, why, when, where, who, and how* (Whetten, 1989) of social presence. Drawing together the findings of this thesis, we can provide tentative answers:

“*What is social presence?*”

Social presence is a relatively specific psychological phenomenon observed in mediated communication, in which the subject has the experience of the communication partner being “real” and “there”. In other words, social presence is a failure to perceptually integrate or maintain the mediated nature of a communication episode.

“*Why is social presence important or desirable?*”

Social presence may be desirable in online distance learning because it contributes to the socio-emotional dimension that has been shown to be traditionally lacking in these

technologically-mediated learning contexts. Improving learning experiences in this respect is associated with positive affective outcomes so that students report more satisfaction, confidence, and perceived learning. Although still an open question, social presence may not be similarly helpful to actually improve learning achievement of students as it is currently not associated with cognitive benefits but with social and affective benefits.

“When can we expect social presence to arise?”

Feelings of social presence are more likely when students have opportunities to form salient impressions of their fellow students. Together, these impressions form a mental picture that may compensate for the relative lack of social cues, thus, allowing for the illusion of non-mediation.

“Where can we expect social presence to arise?”

Students will report higher degrees of social presence in learning environments that facilitate social interaction and allow for experience of ambient awareness, because this is how interpersonal impressions are formed. As these processes are partly determined by features of the learning environment, social presence is more likely to arise in more sociable learning environments.

“Who is expected to perceive social presence?”

It appears that not all students are similarly prone to experiencing social presence under conditions of sociability. Instead, different personality configurations appear to be associated with the degree to which social presence is experienced.

“How can social presence be fostered in online distance learning?”

Social presence can be fostered by increasing the sociability of the learning environment. Mediated environments may be made more sociable by allowing for more easy and spontaneous communication with fellow students and by providing more individuating information about them. The notion of social affordances and design philosophies of social networking sites may provide guidelines for designing more sociable online learning environments. Enriching extant environment with these affordances has been shown to be effective.

Recent observations have concluded that much of the literature in educational technology is either wholly bereft of theory or makes only vague reference to theory as a framing tool (Hew et al., 2019). In fact, it appears that despite our discipline producing an ever-increasing body of literature, many empirical observations do not feed back into theories or frameworks but instead remain on-off and, thus, fail to contribute to systematic cumulation of knowledge (Bennett &

Oliver, 2011). These observations apply to social presence too, where decades of research have not yielded a robust body of work leading to a theory with falsifiable predictions (Biocca et al., 2003, Öztok & Kehrwald, 2017). Building on the work that has identified some of these challenges and taken steps to address them (Kreijns et al., 2004; Kreijns et al., 2011; Kreijns et al., 2014), this thesis attempted to substantially extend this line of research and contribute to answering these central questions. It is hoped that the above tentative answers lay the foundation for a more robust research into the phenomenon of social presence, and consequently, a more comprehensive modeling, a deeper understanding, and an evidence-based enhancing of social presence in online distance learning environments.

6.5 Avenues for improvement

Although steps were taken to address central challenges of social presence, not all weaknesses of social presence research could be considered in this thesis. Thus, some important general limitations in terms of methodology, learning contexts, and learning technologies, as well as accompanying avenues for improvement will be discussed in the following paragraphs.

Methodological innovations. The studies in this thesis were planned and conducted in the spirit of the largely quantitative and questionnaire-based approaches that have characterized social presence research in the past. Of course, relying on self-report of students to measure central constructs comes with limitations. Besides the problem of establishing concurrent validity without another data source to correlate it with, self-reporting social presence is also prone to validity threats, e.g. biases like social desirability (Nederhof, 1985) or simply general lack of introspection (Silvia & Gendolla, 2001). Besides this, many studies investigating social presence fall prey to the possibility of method bias when their focal variables are assessed via self-report, especially when scales share common properties (Podsakoff et al., 2012). Without alternative data sources, it is difficult to gauge to what extent associations and effects may be due to biases that stem from methodological choices or actually from true underlying psychological mechanisms. The prevalence of self-report in social presence research is mirrored in an increase of this data source in the broader field of psychology (Sassenberg & Ditrich, 2019), a trend that has been criticized (Baumeister et al., 2007).

Therefore, future research on social presence should consider expanding its array of instruments to behavioral and/or qualitative approaches. As of now, scholars have yet to come up with valid alternatives to self-reporting social presence. A great step forward would be a behavioral data source. Although there are behavioral indicators within the Community of Inquiry framework, they are derived from a different conception of social presence that is more expansive, with the course as unit-of-analysis. Thus, they are not applicable to a narrow definition of social presence that focuses on the individual experience of students. Instead, inspiration toward the

goal of a behavioral instrument may be taken from Zhang et al., (2015), who inferred presence by asking respondents to indicate the location of a communication partner (via telephone and video call) on a map. Consistent with an illusion of non-mediation, louder voices were judged to be closer. Thus, a continuous measure of social presence could then be obtained through the strength of the association between voice intensity and closeness ratings. More generally, a fruitful way to conceive of a behavioral social presence measure seems to be a *behavioral realism* approach (IJsselsteijn et al., 2000), the underlying idea being that the more social presence is experienced, the more a person will behave as if the interaction was in fact not mediated. Future research may try to incorporate a behavioral realism approach to measuring social presence in online distance learning. This could be done, for example, by using a within-person design to compare mediated to non-mediated communication episodes in collaborative learning and using congruence of specific behavioral markers as a measure of social presence in the mediated episode. Recently, a measure consisting of a combination of such behavioral markers and self-report items has been proposed in the more general (non-social) presence literature (Haans & IJsselsteijn, 2018). As general presence differs from social presence by largely neglecting the social aspects and instead being more concerned with spatial illusions of non-mediation, a direct adaption may not be straightforward. Still, social presence research may take inspiration from such fidelity-based approaches for detecting the phenomenon.

On the other hand, a phenomenon as innately subjective as social presence may also benefit from more qualitative approaches. A remaining gap in our understanding is the question of the phenomenological essence of social presence, that is, how mediation *feels*. While strictly quantitative approaches may turn out to be limited toward this goal, qualitative approaches could sample rich and nuanced experience, for example by asking students to introspect “live” while interacting in mediated spaces. In the past, research has mainly been deductive, in that from the conceptual idea of social presence that has been handed down from the original Short et al. (1976) work, questionnaires were designed and data was interpreted within this framing. This approach may be prone to severe threats of validity (Slater, 2004). A notable exception to this tradition is the more inductive and qualitative work of Benjamin Kehrwald (2008; 2010). As has been proposed by other scholars (Lowenthal, 2010; Oztok & Brett, 2011), getting at the more experiential side of social presence through qualitative or mixed-methods studies might inform future research about the intricacies and dependencies of experiencing social presence, allowing for an increasingly fine-grained and context-sensitive modeling and, thus, theorizing of the phenomenon. One of many unexplored avenues for investigating intricacies of social presence is experience sampling (Larson & Csikszentmihalyi, 2014), through which students may periodically be asked to introspect and report their feelings and perceptions in mediated communication episodes.

As new data-driven approaches like Learning Analytics are gaining traction in education (Tsai et al., 2018), here too, methodological advances in social presence research may be possible. Although the rich availability of student data from online learning platforms allows for better prediction of future learning-related behaviors, like (dis-)engagement or drop out (Kim et al., 2016), it remains challenging to model latent variables like goal orientation or cognitive load (Gasevic et al., 2015). The same likely holds true for social presence. Recently, a step toward this has been made where social presence was inferred from online discussions via text and network analytics (Rolim et al., 2019). However, this study also operated with a Community of Inquiry definition of social presence, making it hard to apply these findings to social presence as defined in this thesis. Especially recent developments of temporal analysis of learning data (Knight et al., 2017; Chen et al., 2018) may provide a high fidelity approach for modeling emergence of social presence and/or socio-emotional processes as they develop over time. Through this, critical communication episodes may be identified and investigated, further enhancing our knowledge of what makes social presence work, going far beyond mere summary statistics of what may turn out to be a rather dynamic phenomenon. If we were to model individual students current level of social presence, adapting activities and interfaces to achieve a desired degree of socio-emotional experience may also be possible. However, for this to work the social presence construct still needs to be better understood on a theoretical basis and intermediate methodological advances for measuring the construct are needed.

Contextual diversity. The studies in this thesis were mostly conducted in an online distance education setting with undergraduate social science students. Of course, there may be limitations when drawing conclusion onto other learning contexts, educational sectors, and student populations. This, too, mirrors the state of social presence research as the overwhelming majority of research has been conducted in formal higher education and within strictly online distance settings, as this is where social presence is vernacular (Richardson, et al., 2017; Mykota, 2018). Extending the scope of research to other arenas of technology-enhanced learning may help us better understand generalizability of the social presence literature and add important nuance to our knowledge base.

As an example that is relatively close yet sufficiently different to serve as a starting point with respect to contextual diversity, MOOC's may be considered. Many trademarks of formal higher education are also shared by MOOC's, although these relatively new non-formal offerings see some of the classical problems exacerbated, like low levels of engagement (Sunar et al., 2016) and dropout (Jordan, 2015). Very recently, research has started to investigate MOOC's from a social presence viewpoint, for example as a way to increase engagement in a gamified MOOC

(Antonaci et al., 2019) or assessing its relationship with discussion forum participation (Poquet et al., 2018). Social presence and socio-emotional aspects may be especially challenging in MOOC's due to their frequently short duration, large group size, and likely student heterogeneity, all of which may be inhibiting factors for social presence (Richardson et al., 2017; Lowry et al., 2006; Gunawardena, 2017). So far, we know very little about the potential relevance of social presence in MOOC's, although social context, i.e. a lack thereof, has been shown to be a barrier to individual learning success in MOOC's (Henderikx et al., 2018). Paralleling recent shifts in perspective as to incorporate individual intentions and preconditions in order to conceptualize student success or failure in MOOC's (Henderikx et al., 2017), the potential role of social presence may also be determined by individual student intentions and should not be considered universal.

Outside higher education offerings, there are many other learning contexts that increasingly make use of technology to mediate interaction. Accordingly, social presence may also play an important role here. For example, within formal contexts, K-12 education is using methods of online and blended learning, albeit still much less prevalent than in higher education (Borup et al., 2016). So far, the literature is mute with regards to the relevance of social presence in this learning arena. It is not self-evident that our knowledge of adult learning in online distance education similarly applies to the unique characteristics of adolescent populations (Borup et al., 2014). Another example to consider may be human resource development, an area that is also increasingly becoming mediated through technology, to what is now known as virtual human resource development (VHRD) (Bennett, 2014). To the extent that these learning experiences also rely on communication, interaction, and collaboration while being mediated, here too, social presence is worthy of investigation. Accordingly, social presence has been very recently introduced to VHRD (Bickle et al., 2019). These are just two examples of educational contexts outside of higher education in stages of relative ignorance about social presence. There are many more, all with their own unique specifics and learning-related dependencies. It is likely that most of them, to the extent that they rely on mediated human interaction, might profit from a better understanding of the psychology of this mediation, as is provided with social presence. In connecting these literatures, we need to think hard about what exactly makes these contexts unique and to what extent we can identify commonalities. Through this, we can clearly identify research needs that are specific for a give educational context. What is learned through these investigations may then extend our knowledge by flowing back to a more general theory of social presence

Technological applications. Another limitation of the research in this thesis stems from the relatively technologically constrained context in which most research of this thesis was conducted. Students sampled in these studies engaged in their learning activities primarily through the learning management system Moodle. Within this, there were different ways of communicating both synchronously and asynchronously as well as text-based and audio- or video-based. However, from experience it can be concluded that asynchronous and text-based mode of communication is by far the most prevalent mode of communication in these learning environments, mirroring both the state of the online distance learning landscape (Legon & Garrett, 2018) as well that of social presence research (Mykota, 2018), for both of which this mode of communication is still typical. Intuitively, demonstrating and investigating social presence in these supposedly low-fidelity interaction modes (compared to, for example, synchronous video chat) provides evidence for the apparent universality of the phenomenon. Yet, it is entirely plausible that different communication modes with different degrees of fidelity will also impact the emergence of social presence and possibly, the way it interacts with other variables. As this was not explicitly modeled in this thesis, future research may account for such differences as potential moderators.

Going beyond relatively restricted modes of interaction, an extension of social presence research to a more diverse set of technologies, especially ones that allow for high-fidelity or even hyper-realistic representations has been shown to be a promising way forward in the recent years (Oh et al., 2018). Investigations of these technologies for learning purposes may benefit from a social presence perspective and, in turn, these applications may also inform our understanding of social presence itself. Two obvious contenders for this are Augmented Reality (AR) and Virtual Reality (VR), as they essentially blur the line between the virtual and the “real”, albeit in rather distinctive ways. Where VR simulates a full environment with varying degrees of sensory experience, thus, blocking out the physical world, AR uses graphical information to enhance reality. Although VR is not new, with early applications in flight simulation and military training, the recent development of consumer headset products like Oculus Rift or HTC Vive make VR experiences both accessible and affordable (Cipresso et al., 2018). AR on the other hand is less technology-intensive and can be experienced with a smartphone. The educational promise for certain student populations and the recent availability has made both technologies increasingly interesting for applications within education (Freina & Ott, 2015; Garzon & Acevedo, 2019; Howard & Gutworth, 2020).

In terms of their potential for social presence, however, these technologies may be rather distinct. In VR, the experience of immersion and, thus, presence within a virtual world is a key characteristic of the technology (Oh & Bailenson, 2017). For this reason, it should come as no

surprise that presence is a lively topic in VR research (e.g. Oh et al., 2018; Haans & Ijsselstein, 2018; Herrera et al., 2020). However, presence in VR research does not always mean social presence but can also mean place presence or what is called *place illusion* (Slater, 2009). Thus, VR research has focused on either place presence in the environment and/or social presence with virtual humans, agents or avatars (e.g., Zibrek & McDonnell, 2019). It should be noted that, mirroring the issues of the concept in online distance learning, these distinctions are not always entirely clear and adhered to (Sadowski & Stanney, 2002). Interestingly, although both are concerned with variants of the presence phenomenon and both appear to grapple with definitions and measurements (Slater, 2004; Hein et al., 2018), these two lines of research have developed largely independent from each other. In an extensive review, Oh et al. (2018) show that experiencing social presence in VR is governed by an array of factors, like visual representation and interactivity of the virtual communication partner, depth cues and audio quality in the environment, and haptic feedback while interacting.

In AR however, the persisting link to physical reality prevents the place illusion and, thus, affords different psychological effects. Correspondingly, creating a sense of social presence with other humans may also be harder, as the persistent comparison exacerbates differences between reality and virtuality. Nevertheless, different configurations in augmented reality have shown to affect social interaction and social connectedness both with virtual and real human interactants (Miller et al., 2019). To harness the unique educational potential in AR and VR learning scenarios, it is important that students temporarily suspend their disbelief, the virtual artifice deemed briefly plausible (i.e., plausibility illusion, Slater, 2009). Toward this end, social presence is invaluable. For example, a VR training to alleviate anxiety in triggering situation like public speaking cannot bear fruit if it does not truly feels like public speaking, i.e. the audience is perceived to be socially present. Future research should consider the unique properties of these rather different technologies and evaluate how they social presence may be beneficial in attaining specific learning goals. Additionally, it may be promising to attempt a better communication between these largely separate lines of research.

7 TWO RESEARCH FRAMEWORKS FOR A NEW PHASE OF SOCIAL PRESENCE RESEARCH

As a research area that is still underdeveloped while at the same time being highly relevant for human relations in an increasingly technologically mediated world, but also for learning related social interaction in online distance contexts, rigorous and innovative research on social presence is in dire need. Of course, the studies presented in this thesis are only small steps towards a deeper understanding of social presence. Besides the limitations of this thesis and avenues forward presented in chapter 6.5, there are still more fundamental limitations that plague social presence research and may turn out to be obstacles in the way of maturity for this line of research. Both relate to the challenge of explaining social presence and both can only be overcome by widening the scope of theorizing beyond disciplinary lines of educational technology or online distance learning.

A common theme of this thesis is the notion of turning to neighboring research disciplines for answers. To avoid reinventing the wheel, we need to connect literatures that explicitly or implicitly may contribute to our understanding of social presence and socio-emotional aspects more generally. More likely than not, we can find valuable information to inform our thinking in other literatures. With relevant theories or data being scattered, it is likely that we, in the broader social science discourse, already have many of the pieces to the puzzle that is social presence. For example, to model the elusive connection from socio-emotional aspects of online distance learning to cognitive learning achievement we cannot simply rely on the literature of educational technology. This is because our knowledge about how learning works stems from cognitive and educational psychology and just as educational psychology is not at the frontier of using technology to support learning, educational technology cannot be expected to be at the frontier of learning theory. Thus, developments in theory that may extend above and beyond the cognitive aspects of learning to incorporate socio-emotional aspects will likely emerge first in these disciplines. Similarly, our understanding of social presence as a psychological phenomenon will remain limited if we do not turn to the broader psychological literature for answers, for example social and personality psychology. Building on the knowledge generated in this thesis, the following will provide examples of the power of this approach, by outlining two frameworks that address central issues of the social presence literature by way of integrating knowledge across disciplinary lines.

7.1 The elusive connection to learning achievement

Summarizing results of her large-scale review of social presence, Xin Chen concludes: “The most concerning question relates to the effects of social presence in improving distance learning. [...] Unfortunately, the outcomes are inconclusive” (2014, p. 3). The inconclusive body

of evidence is considered concerning because, while outcomes like a sound social space and affectively satisfying learning experiences are desirable, they still may be considered relatively distal factors. Educational research typically aims to identify proximal factors that are in close relation to actual learning achievement. The host of inconclusive findings regarding this connection makes necessary a critical evaluation of what many of these studies actually intended to investigate. By calculating correlations between social presence and measures of learning achievement while omitting plausible mediating mechanisms, these studies seem to assume a straightforward association between a socio-emotional interpersonal phenomenon which in the first instance is conceived entirely independent of learning-related cognitive processes, to be nevertheless mono-causally linked to the consequence of these processes, learning achievement. It should hardly come as a surprise that such a simplistic view of what is very likely a complex interplay of different psychological mechanisms with many conceivable moderators yields an inconclusive literature.

However, if we want to determine the relative importance of social presence compared to other design considerations, conclusively establishing if and how social presence may affect learning achievement should remain a critical goal of future research. Ultimately, the extent to which this connection is empirically supported may determine whether social presence considerations will advance from a scholarly special interest to a firmly established online distance learning design recommendation. In order to provide a starting point for a more nuanced and theoretically sound investigations of how social presence may affect learning achievement, the following paragraphs will outline a possible research framework by first drawing on insights from this thesis and the extant social presence literature and then proceeding to integrate these aspects with a firmly established theory of educational psychology. Together, the proposition in this framework may help explain the inconclusive literature on social presence effects by providing a plausible account of the context-dependent nature of consequences of social presence. This chapter concludes with a set of testable hypotheses generated from this framework.

Relative social presence fit. Alongside the majority of simplistic accounts, there have also been studies showcasing nuanced and context-sensitive discussions of how social presence may exert purported effects. For example, Zhan & Mei (2013) concede that not all learning tasks or courses may necessitate a high degree of social presence. Depending on context, other scholars state that “[s]ometimes a low level of social presence will do the work” (Cui et al., p.676). Similarly, there have been mentions of possible threshold effects, where once a certain degree of socio-emotional design is reached, there may be diminishing returns (Wise et al., 2004; Weinel et al., 2011). Moreover, we could also conceive of too much social presence (Biocca et al., 2003), in which case it may be associated with negative effects (Chen, 2014). This could

occur, for example, in relatively short courses in professional settings where a rich interpersonal experience serves little purpose and is considered a distraction or in learning contexts in which a certain degree of anonymity is preferred. Thus, moving away from a simple more-is-better view of social presence to one that accounts for situative factors is a promising, yet largely unexplored perspective on social presence (Mykota, 2018). Further supporting this perspective, evidence for the potentially far-reaching effects of social surroundings on cognition can be found in the literature on social facilitation and social impairment effects (e.g. Belletier et al., 2019)

Given this perspective, we need not only account for the degree of social presence that is achieved in the course, as determined by variables described and investigated in this thesis or in the literature, e.g. sociability and social affordances (chapter 5.2 and 5.3), social interaction and ambient awareness (chapter 5.4), group size (Ackaoglu & Lee, 2016), communication technology (Zhan & Mei, 2013), instructor strategies (Lowenthal & Dunlap, 2018); we may also need to consider the degree of social presence that is desirable for a given learning experience. This may be determined by an entirely different set of factors, e.g. learning context (academic/professional), course duration (long/short), students motivation (intrinsic/extrinsic), instructional design (inquiry-based/guided), and others. Failing to match actual and desired levels of social presence would then lead to two possible types of mismatch. The first type of mismatch is what is usually associated with the more-is-better view of social presence. Here, achieved levels of social presence do not extend to level that is desired, for example when intrinsically motivated academic students take part in semester-long course and after a few weeks feel socially isolated. Here, an increase in social presence would likely be associated with positive effects. A second type of mismatch would be the opposite, the perceived levels of social presence are too high for what is desired. This may be the case in a relatively outcome-oriented courses, for example when business professionals in a hierarchical management structure are expected to complete an online-based training to acquire new workplace-related skills. Here, relative anonymity may be preferred to facilitate question-asking and to allow for mistakes and misunderstandings. In this case, efforts toward higher levels of social presence may be less appreciated or even considered distracting (Ke, 2010). In severe cases, social presence would then yield negative effects. Although, this type of mismatch is expected to be less frequent, the possibility should be considered in a comprehensive framework. Finally, a third possible configuration is a social presence match. Here, the actual degree of social presence is in line with what is perceived to be desirable for the learning experience. In this case, there may be weak or no associations between social presence and outcome variables, as increasing levels of social presence may yield diminishing returns.

This view of social presence being relatively fitting or mismatched may help explain what has been a mixed bag of empirical evidence regarding effects of social presence (Chen, 2014).

For example, chapter 5.2 and chapter 5.3 may have not found an association of social presence with satisfaction and learning achievement, respectively, because for these specific learning contexts, the amount of achieved social presence was perceived as adequate. Future investigations may account for this added layer of complexity by assessing not only the levels of social presence but also the degree to which these levels were perceived as adequate. As an important benefit of this perspective change, it allows for systematic comparison. If we refrain from considering absolute values and instead consider relative social presence (mis)match, we can plot two or more courses on a continuum according to how close or far they are from a perceived ideal. On this scale, rather different courses suddenly become comparable and room for improvement is immediately observable. This may not necessitate the development of another measure. Rather, students may respond to items measuring achieved level of social presence and then indicate their subjective ideal value for this item. (Mis)match would then be calculating the gap between achieved and desired levels and measures to close this gap may be taken, an approach similar to what has been called relative proximity theory using transactional distance (Swart et al. 2014; Mcleod et al., 2019).

Interaction integration. Using the examples of chapter 5.2 and 5.3 again, let's consider an alternative explanation, one in which we do not assume optimal or even adequate levels of social presence. What could then explain the missing association? As a second proposition in this framework, how and if a social presence mismatch actually affects outcome variables may itself be determined by another contextual factor, one that is associated with the instructional design of the learning experience. Because social presence is a phenomenon unique to mediated social interaction, the degree to which a social presence mismatch affects outcome variables should then be moderated by interaction integration. Interaction integration captures the notion that learning tasks can differ in their reliance on interaction toward reaching a learning goal. For example, computer-supported collaborative learning (CSCL) is characterized by high interaction integration, because reaching learning goals is highly dependent on the social interaction of students. Essentially, social interaction is a necessary but not sufficient condition for collaborative learning because learning fails without successful interaction (Johnson et al., 1985). On the other hand, while non-collaborative courses in online distance education may include opportunities for social interaction, they are not integral to attaining the learning goals. Students may choose to abstain from participating in chats or message board discussions without this hampering their ability to successfully complete the learning task or course. Thus, interaction integration is low (see Fig. 16).

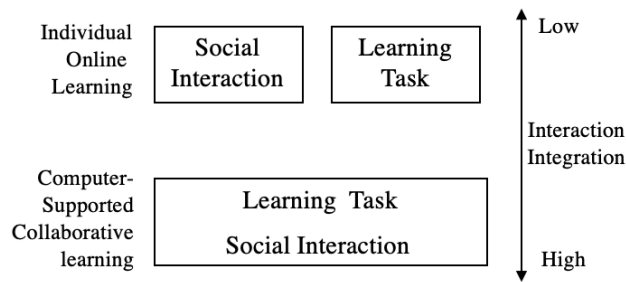


Fig 16. Two examples of learning scenarios along the axis of interaction integration.

The degree to which a social presence mismatch affects outcome variables may, thus, be affected by interaction integration. That is, interaction integration is a moderator of this relationship because in learning scenarios in which social interaction is not integral to reaching learning goals, a social presence mismatch is expected to be relatively inconsequential. While students may find a mismatch irritating, they may simply choose to abstain from social interaction. However, if interaction integration is high the situation changes, as students are more saliently confronted with this mismatch. At worst, it may be an obstacle in the way of attaining desired learning goals. For example, if achieved levels of social presence are too low in a semester-long course that relies on functioning group dynamics to complete a complex project, students may feel that this mismatch actively impedes the groups ability to complete the project. Thus, even a relatively slight mismatch may be exacerbated by very high interaction integration.

The possible role of interaction integration has rarely been explicitly addressed in social presence research thus far, with the exception of Oyarzun et al. with their notion of contextual versus designed interaction (2018). Yet we can find traces of this notion in the literature. Generally, pedagogical characteristics of the investigated scenario are noted in research papers, but usually to illustrate the problems that social presence is hoped to address. For example, Poquet et al. (2018) mention that in MOOC's "learners are not obliged to engage in social activities or complete the assigned assessment in the course" (p. 44), thus, indicating low interaction integration of this learning scenario. But because they do not go on to assess outcome variables associated with social presence, this observation remains inconsequential aside from providing a rationale to study social presence in this context. As an example of recognizing the role of interaction integration, we can refer to the sustained research efforts in social presence and social aspects within CSCL by Karel Kreijns and colleagues (Kreijns, 2004; Kreijns et al., 2013). The importance of social presence in CSCL is usually conveyed by referring to the integral role of social interaction for learning success in these scenarios. In other words, this line of research recognizes the importance of an adequate level of social presence (i.e. the avoidance of a mismatch) due to the fact that interaction integration is typically high in in collaborative learning.

In order to better understand differential effects of relative social presence (mis)match, interaction integration should be explicitly considered. This could be done by quantifying interaction integration in a learning scenario and testing this as a moderator between social presence (mis)match and outcomes variables that are expected to be affected. In contrast to other variables in this framework, interaction integration should be conceived as an objective characteristic of the learning design to be judged by the designer or researcher conducting the research. Using the relative social presence (mis)match perspective to assess different learning scenarios on one continuous scale while also providing discrete judgements of interaction integration would allow for empirical testing of these propositions.

Control-value of mediated social interaction. After introducing these possibly confounding factors when assessing effects of social presence, we can now propose a theoretical link to learning achievement. In order to model how a decidedly subjective interpersonal phenomenon like social presence may account for gains in learning achievement, we need a theoretical framework that can bridge the theoretical chasm between these affective components of the experience and cognitive processes that relate to learning. What has been missing in past research is such a theoretical causal chain, one that is rooted in established findings of the psychology of learning while being applicable to the mediated nature of online distance learning. Indeed, as experiencing social presence and engaging in cognitive processes responsible for learning appear on entirely different psychological planes, it appears difficult to argue that the former has intrinsic properties directly contributing to the latter. Instead, we are looking for mechanisms that explain how affective, socio-emotional experiences in mediated social interaction indirectly influence cognition. As an integrated account of emotion, motivation, and cognition, the well-established control-value theory of achievement emotions is a prime contender for this theoretical bridge (Pekrun et al., 2002; Pekrun, 2006).

Control-value theory (CTV) provides a integrative framework of how achievement emotions, the affective states tied to a given learning activity, are influenced by two central variables: (1) subjective control over learning processes and goal attainment and (2) subjective value of these learning processes and goals (Pekrun, 2006). Subjective control refers to the perceived causal influence of the student over his or her goal attainment. Subjective control is high when a student feels that he or she can produce a desired outcome, in this case reaching task- or course-level learning goals. Subjective value, on the other hand, refers to the perceived intrinsic or extrinsic benefit associated with an activity. Subjective value is high when a student feels that he or she is engaged in a process that is enjoyable (intrinsic) or is linked to a desired future state (extrinsic). When students more or less consciously assign values to these variables of control and value it is called an appraisal (Pekrun, 2006). Such appraisals are dynamic and can be shaped

by experience. Thus, they are subject to the design of the learning activity. Appraisals can influence the achievement emotions that are associated with the task- or course-level learning processes while these achievement emotions themselves are believed to be tied to motivation for future learning and subsequent expenditure of cognitive effort toward learning. That is, achievement emotions are considered beneficial for learning achievement by virtue of affecting motivations toward learning.

These core tenets of CTV have received wide empirical support (Goetz et al., 2006; Pekrun et al., 2011; Perry et al., 2001). Likely due to these successes, other scholars increasingly aim toward including learning related affect like emotions and motivation into what have previously been strictly cognitive accounts of human learning processes (Park et al., 2014; Plass & Kalyuga, 2019). Robust evidence for the viability of integrating emotion and affect with cognitive processes, with CVT as a guiding framework (Stark et al., 2018), as well more generally (Endres et al., 2020), is slowly accumulating. Thus, with the variables of CTV and our understanding of social presence at hand, a theoretical framework can be constructed.

Interpreting interaction integration and social presence (mis)match in terms of control and value appraisals, we can propose the following: (1) Interaction integration affects value appraisals of mediated social interaction with respect to the learning task. A student studying in a high interaction integration scenario will place high value on social interaction as it is integral toward the overall goal of completing the learning task or course. Conversely, a student will assign less value to social interaction if it is not integral to the learning task. (2) Social presence (mis)match affects control appraisals because a student experiencing severe mismatch will lower expectations of control over success within this learning task. In line with the interaction integration moderation proposition, students will feel less (more) in control of their success if a mismatch occurs in scenarios with high (low) interaction integration. This yields a total of four appraisal configurations, leading to different context-specific emotions that may be exemplified as: satisfaction, frustration, indifference, and irritation. For example, a student may be irritated by the fact that social interaction is mismatched in terms of social presence, making fruitful social interaction less probable. Yet, this mismatch will only yield the stronger valenced emotion of frustration, if mismatched interaction is in the way of achieving learning goals, that is, when interaction integration is high (Fig. 17).

		Social presence	
		match	mismatch
Interaction Integration	high	+ control + value	- control + value
	low	+ control - value	- control - value

Consequent emotions:
+ C + V = satisfaction
- C + V = frustration
+ C - V = indifference
- C - V = irritation

Fig 17. Four appraisal configurations and consequent emotions based on social presence (mis)match and interaction integration.

Emotions resulting from the appraisal configurations are expected to be context-specific, that is, they relate to the experience in which these appraisals take place. This is called activity focus of achievement emotions in CVT and is distinguished from outcome focus (Pekrun et al., 2007). A student experiencing frustration due to a social presence (mis)match in a high interaction integration scenario will likely not experience this negative emotion with respect to the general content of the learning task but instead toward the nature of the learning experience, its technical or instructional design.

Empirical CVT studies have provided evidence that the valence and strength of achievement emotions (positive/negative, strong/weak) is associated with expenditure of behavioral (Garn et al., 2017) and cognitive engagement or effort (Stark et al., 2018). This activating effect of emotional states on engagement levels is what is often subsumed under the concept of motivation (Linnebrink-Garcia & Patal, 2016). Recognizing motivation as situated, that is, having the student-in-context as unit of analysis (Nolen et al., 2015) then allows us to trace motives for future engagement back to the appraisals of previous mediated interaction. A frustrating episode of mediated interaction thwarts future cognitive engagement in comparable interaction episodes. While there are likely many other contextual factors that affect nature, valence, and activation of resulting achievement emotions, social presence (mis)match may rise above other factors in situations where social interaction lies at the heart of the learning task (i.e. high interaction integration).

Depending on resulting emotions associated with mediated interaction, willingness to expend mental effort toward the learning task either increases or decreases (Stark et al., 2018; Endres et al., 2020). Expenditure of such effort is known to be associated with learning, as is conceptualized in many different theories and frameworks of human learning via concepts like germane cognitive load, mental effort, or cognitive engagement (Sweller, 2010; Paas et al., 2005; Pintrich, 2003, Kahu, 2013). With respect to the above propositions, we can assume a relationship such that cognitive effort is a mediator between context-specific achievement emotions and

learning achievement. With this final step, we can conclude the theoretical account of how affective and motivational factors may bridge the gap between socio-emotional processes in learning environments and cognitive processes related to learning (Fig. 18).

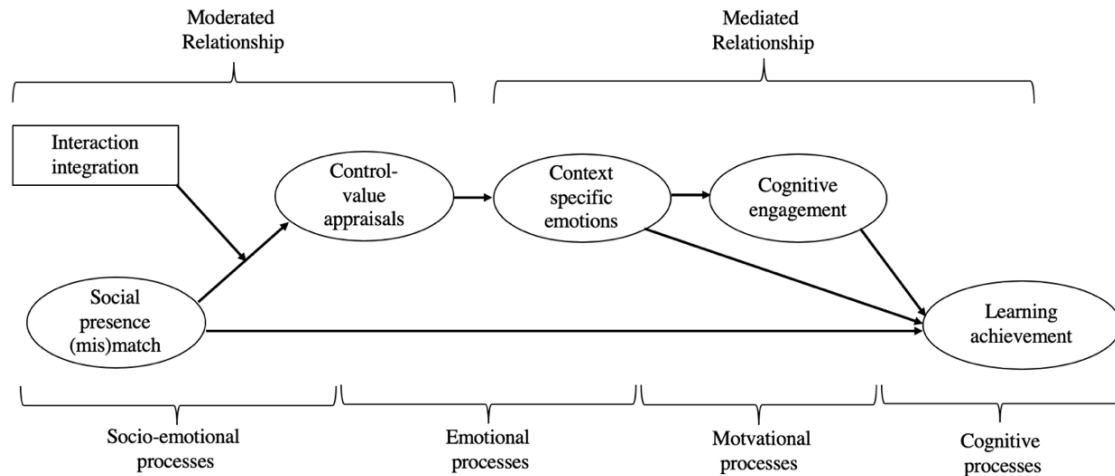


Fig 18. Hypothetical causal chain connecting social presence to learning achievement via control-value theory. Top parentheses indicate whether primary mechanism is moderation or mediation, whereas bottom parentheses indicate realm of literature in which variables and causalities usually are investigated.

To assess the validity of these propositions, we can generate falsifiable research hypotheses from this framework that allow for empirical testing.

- (1) Depending on the extent and type of social presence (mis)match, associations between social presence and affective appraisals will differ. For example, social presence will positively contribute to satisfaction if the mismatch was such that achieved levels were lower than desired levels. Conversely, social presence will negatively associated with satisfaction if the mismatch was such the achieved levels were higher than desired.
- (2) These effects will be exacerbated or thwarted depending on degree of interaction integration. For example, in scenarios will high interaction integration, social presence mismatch should yield more severe appraisal consequences than in scenarios with low interaction integration.
- (3) Interaction integration and social presence (mis)match will yield specific control-value appraisals. For example, high interaction integration will yield higher value-appraisals toward social interaction in a given task than low interaction integration. Social presence match will yield higher control-appraisals than social presence (mis)match.
- (4) These task-value appraisals will yield contextual achievement emotions like satisfaction or frustration with different valences and strengths which, in turn, are associated with learning achievement.

- (5) The effect of contextual achievement emotions on learning achievement is due to increased cognitive engagement toward the learning task. Cognitive engagement, thus, a mediator for this effect on learning achievement.

With this framework we can hypothesize a connection from the socio-emotional phenomenon that is social presence to what is essentially the goal of all educational research, learning achievement. In addition, this framework provides important moderators and mediators that may be kept in mind for future investigations that aim to test the validity of its propositions. Future research should carefully delineate relative social presence (mis)match and interaction integration when estimating effects of social presence in general and with respect to learning achievement specifically.

7.2 Construal level theory of social presence

Another central issue of social presence research is that its theorizing is “exotic”. Although the concept originated in social psychology, it has since strayed from this literature and developed a life of its own, in research lines on computer-mediated communication, online distance learning, and CSCL. These frequently do not provide explanations of psychological mechanisms governing the phenomenon of social presence, although our relatively nebulous understanding may benefit from the provision and investigation of such deep accounts. Undoubtedly, the responsible psychological mechanisms developed in times long before the possibility of mediated communication and, thus, are relatively generalized across situations (Premack & Premack, 1995). Thus, a deep understanding of social presence may necessitate a relatively general theoretical account, one that is not specific to mediated interaction (Biocca et al., 2003). Although this thesis has established and investigated processes and mechanisms governing social presence, a deeper psychological explanation may need to provide answers to the general question of how one forms mental representations with differing degree of abstractness. As these differing degrees of abstractness are likely what causes us to perceive the other person as “real”, we can formulate our questions regarding social presence much more broadly.

Opting against “killing” social presence might not be enough (Oztok & Kehrwald, 2017). In order for it to thrive, we should consider taking social presence sufficiently serious to embed it into a larger theoretical framework of how we form mental representations. Equipped with the insights from this thesis, the following will attempt to provide such an integration, into one of the most comprehensive and integrative theoretical frameworks in social psychology: construal level theory (CLT).

Construal level theory. Proposed by Yaacov Trope and Nira Liberman, CLT is concerned with how psychological distance affects individuals' thoughts and behavior (Trope et al., 2007). According to CLT, there are different dimensions of psychological distance, the most important ones being time, space, social, and hypothetical. These dimensions, thus, affect how we think and act. Supported by many lines of converging evidence (Fiedler, 2007; Soderberg et al., 2015), CLT has produced relatively general-level principles about human psychology that may be applied to what we have learned in this thesis. Two central tenets of CLT are that (1) dimensions of psychological distance are intrinsically connected and (2) this psychological distance affects the level of abstractness of mental representations (i.e. level of construal) and vice versa. These basic propositions are depicted in Figure 19. At first glance, we can intuit an affiliation to the topic of this thesis, as there seems to be a natural fit between psychological distance and social presence (as well as related socio-emotional variables). However, to investigate potential connections, let's consider the central propositions of CLT more thoroughly.

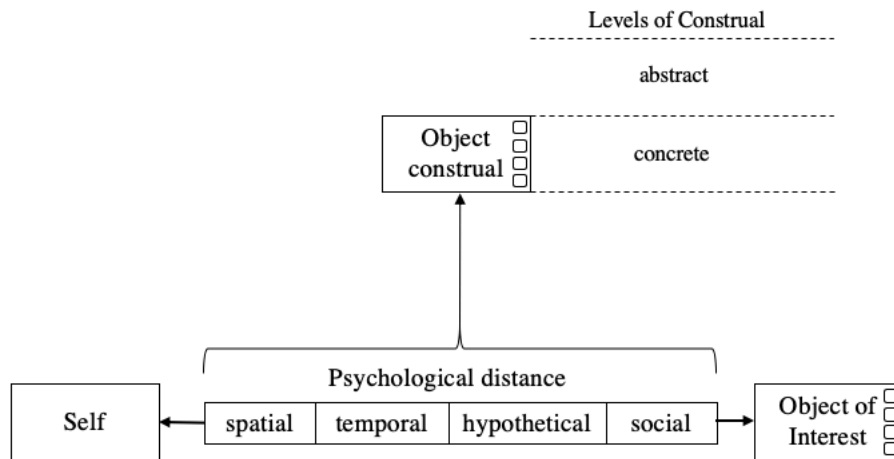


Figure 19: Basic propositions underlying CLT. The first proposition is visually displayed on the horizontal plane of this figure, whereas the second proposition of CLT can be seen along the vertical plane. Level of construal is determined by psychological distance, which itself is determined by distance dimensions. In this situation of small psychological distance, trait details of object of interest are fully carried over to low-level construal.

Regarding (1), CLT proposes that psychological distance to objects of interest (e.g. other people, items, events) arises when the object is removed from direct experience. This distancing can occur along the dimensions of time, space, social, and hypothetical (Trope et al., 2007). For example, the prospect of bad weather is psychologically distant if it is expected to happen in the far future (temporal), somewhere far away (spatial), if it is deemed unlikely to occur at all (hypothetical), or if it will affect a stranger rather than a friend or myself (social). The “or” is important here because, crucially, one dimension can by itself trigger psychological distance. However, as in this example, these dimensions often covary so that distance on one dimension may prompt distance on another (Fiedler et al., 2012). Connections to the realities of online distance education are straightforward here. Students are geographically distant from each other, that is why communication is mediated. Moreover, this communication is commonly temporally

removed, as is the case in asynchronous message board communication. In terms of the social dimension, fellow students are often strangers, especially at the beginning of a course or program. Lastly, relating to the probability dimension, successful and satisfying social interaction can be an uncertain prospect, as students at the beginning of a course may not know what to expect and learning environments may not be sociable. Given these circumstances, great psychological distance between students in online distance learning is all but guaranteed. Moreover, according to CLT, mechanically reducing just one of these distance dimensions, for example by switching to synchronous communication media may not always be enough, because interrelationships of these dimensions are complex and may need to be considered more holistically (Trobe & Liberman, 2010). But why is it important to traverse this psychological distance in online distance learning in the first place? The answer comes from (2), another central tenet of CLT, the relationship between psychological distance and level of construal.

Toward a construal level theory of social presence. Levels of construal refer to the abstractness of inner representation, that is, how concretely an object of interest is represented. Just as we begin to see the forest for the trees when we physically move backwards, conceptual abstraction is heightened with increased psychological distance. For example, an upcoming vacation in the Bahamas will be imagined more detailed and concretely (e.g. smell of the ocean) when it is set to happen next week and more abstractly and high-level (e.g. relaxation) when it is set to happen in six months. This is an example of psychological distance induced via temporal dimension (Liberman et al., 2002). This relationship is also understood to be reciprocal, so that, for example, specific memory retrieval cues (low-level construals) also prompt more recent memories than abstract retrieval cues (high-level construals) (Semin & Smith, 1999). Most pertinent to social presence, these effects also apply to interpersonal perception. In CLT-language, a high degree of social presence is actually a low-level, i.e. detailed and concrete construal of our communication partner. Being temporally removed from a communication partner through asynchronous technology would, thus, to some degree contribute to a high-level construal; we then think of the other in terms of general features. Such a relatively abstract impression then prevents us from experiencing social presence. And vice versa, if we are not able to gather concrete impressions of the other, we will feel psychologically further removed from that person.

By making connection between social presence and CLT we can formulate central propositions that go above and beyond of what these concepts bring forth by themselves. See Figure 20 for a visual summary.

- (1) High-level construals of others do not allow for experience of social presence, as the abstractness of the communication partner highlights the mediated nature of

communication. Thus, we need to aim for lower-level construals of fellow students. This can be facilitated by decreasing psychological distance.

- (2) The externalities of psychological distance are, to some extent, malleable. In online distance learning, some dimensions of psychological distance are fixed (spatial distance), others can be reduced (social distance, hypothetical), others are potentially flexible, depending on the case-by-case situation (temporal).
- (3) Increasing sociability of the online learning environment is one way of decreasing psychological distance by closing the temporal distance of expected social interaction (temporal) and generally increasing the prospect of communication and interaction in the future (hypothetical).
- (4) Through mechanisms of social interaction and ambient awareness do we learn about fellow students who were previously strangers to us. Through this, social distance is increasingly reduced, contributing to more concrete rather than abstract construals of peers.
- (5) Experiencing social presence and a sound social space, in turn, reduces social distance, leading to a self-reinforcing effect, such that progressively lower-level construals reduce social distance over time, thus, highlighting the importance of course duration.

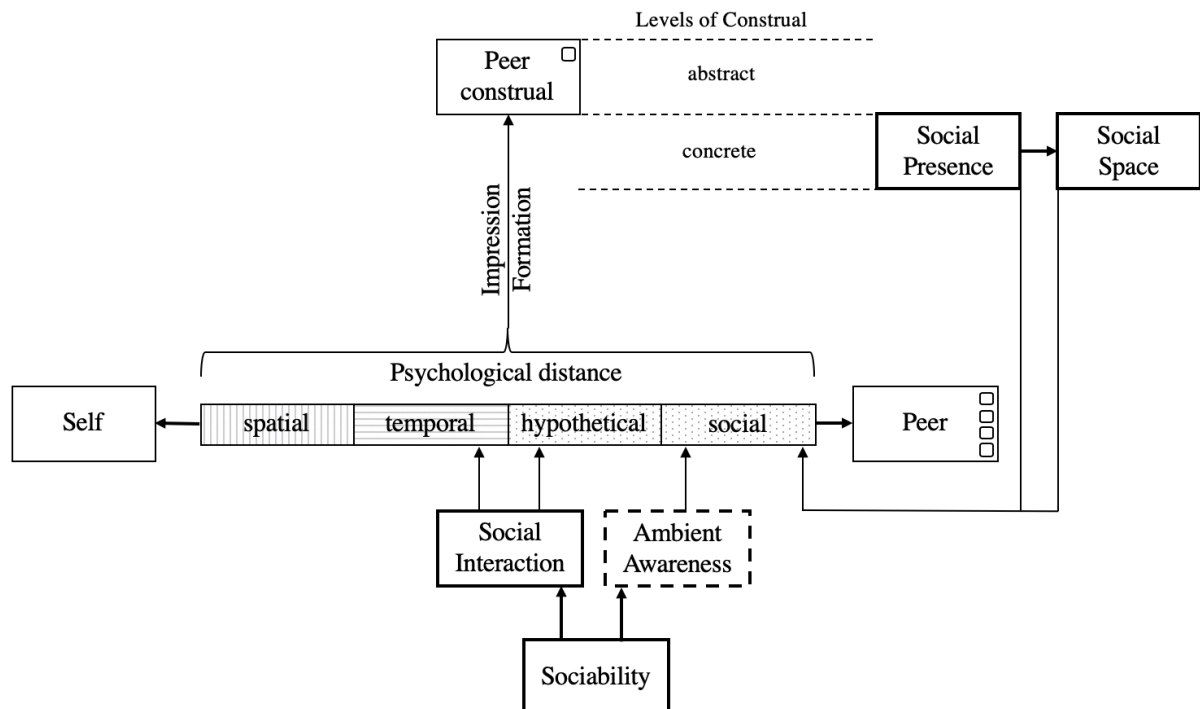


Figure 20. Extended propositions that flow from connecting CLT with SIPS model. SIPS variables are bold, fixed distance dimension (spatial) is vertically lined, potentially flexible dimension (temporal) is horizontally lined, malleable dimensions (hypothetical, social) are dotted. Due to relatively large psychological distance, peer construal is high-level and abstract, trait details of peers not fully being carried over.

To guide and inspire future research, the following outlines some potential research questions that may be generated from this. Surely, more can follow by more elaborately outlining a construal level theory of social presence.

- (1) How does construal level relate to social presence? Although theory would suggest a relationship between low-level construal and social presence, this hypothesis still needs to be empirically tested. A well-documented connection between these concepts would be a major asset for our theorizing of social presence. In extending this line of research, one could also investigate if level of construal merely influences the likelihood of social presence occurring or if level of construal actually changes the presence experience in a meaningful way.
- (2) How do psychological distance dimensions relate to each other in online distance learning? The CLT literature is still inconclusive as to if some dimensions may be more basic or influential (Trope & Liberman, 2010). Although they covary substantially (Fiedler et al., 2012), they clearly are not the same and may have unique properties. How exactly these dimensions are related to each other in online distance learning, where they are differentially malleable, would provide important answers toward decreasing psychological distance while learning from a distance.
- (3) How effective is it to induce a low-level construal of other students in order to affect psychological distance? The CLT literature emphasizes the reciprocal relationship of distance to construal (Semin & Smith, 1999). If student were to be prompted with low-level construals upon communicating in mediated spaces (e.g. presenting rich impression formation details), will this induce social presence and in turn decrease psychological distance?

7.3 Added value for social presence research

The above sections have provided theoretical integrations of social presence into broader theories of educational psychology and social psychology. The first integrative framework aims to establish grounds on which we can judge the importance of social presence with respect to the main dependent variable of education research, learning achievement. However, through considering the variables of interaction integration and relative social presence (mis)match, associations with other desirable outcome variables may also yield a more nuanced picture. Confirming or disconfirming these relationships will provide us with a more robust foundation on which to judge the merits of social presence for online distance learning. The second framework aimed to widen our view of social presence in order to conceptualize it as an instantiation of more general tenets of human psychology. As we do with other objects that are removed from direct experience, we also represent fellow students in mediated communication on different levels of abstractness. With respect to the unique characteristics of online distance

learning, we can use construal level theory to explain how these characteristics shape our mental representation of fellow students and thereby determine levels of social presence. Taken together, the frameworks may address two fundamental issues that have been obstacles toward the maturity of social presence research, determining the practical relevance of social presence and understanding social presence in a deep sense that is consistent with what we know about human psychology. As an immediate result of the above theoretical integrations, we can generate several novel and testable hypotheses to assess the validity of these proposition and thereby increase our understanding of social presence. On a meta-level, these integrations demonstrate that social presence need not be a muddled and exotic concept, but instead can be interfaced with more mainstream educational and social psychological theories

8 CLOSING THOUGHTS

Although we are still a far cry from what Isaac Asimov so vividly recounts of the Solarians in the *Naked Sun*, our own technological trajectory forecasts a shift from seeing to viewing that may not be unlike this fictitious world. We learn what the novel's protagonist lacks when he first visits Solaria, is in essence a key literacy of its people. This lack of literacy is the reason why Solarians are either shocked or amused when they discover his ignorance of what, to them, is a most self-evident psychological fact: that mediation changes everything.

This thesis is the result of opting against “killing” social presence. Rather, it outlines an attempt at a new phase of social presence research, one that takes the idea seriously by honoring its theoretical roots, constraining its meaning, systematically building an evidence base from the ground up, and outlining promising ways forward. This is important because we, too, still seem to lack this critical literacy. Countless studies of social presence have not produced a deep understanding of the psychological dependencies that govern mediated communication. We are as ignorant as the detective when he first visits Solaria, yet we are not visitors. This research was pressing and timely decades ago; today it is overdue. With increasing bandwidth allowing for more sophisticated and natural-seeming ways of communicating through technology, the illusion of non-mediation will likely be even more pervasive in the future; the difference between seeing and viewing more subtle and nuanced than ever. A thorough understanding how our social fabric is affected by the steady erosion of the distinction between the digital and the “real” will be both crucial for learning experiences as well as human interaction in general. Toward this goal, social presence may turn out to be critical.

The body of work presented in this thesis paints both a bleak and a hopeful picture. On the one hand, after decades of busy research, we are still surprisingly naïve as to the inner working of this phenomenon and what it actually entails, especially in relation to online distance learning. With this sluggish progress in mind, one can hardly be blamed for holding a pessimistic view of the future of social presence research. On the other hand, social presence apparently remains an evocative and stimulating idea, inviting countless scholars to explore it in a myriad of ways. Not expecting this momentum to slow down anytime soon, one could remain hopeful that through more and better research, at some point in the future, we will have learned enough about this phenomenon to put it to good use, both toward improving online distance learning as well as human relations in mediated settings more generally. And indeed, it appears that there are more and more instances of rigorous social presence research to build upon. As a central aspiration of this thesis, it is hoped that the work presented here will serve as a modest but constructive contribution toward a deeper understanding of social presence in online distance learning environments.

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Eigenständigkeitserklärung

Erklärung nach §7 der Promotionsordnung der Fakultät für Kultur- und Sozialwissenschaften der FernUniversität in Hagen vom 06. August 2018:

Hiermit versichere ich an Eides statt, dass ich die vorliegende kumulative Dissertation selbstständig und ohne unzulässige Inanspruchnahme Dritter verfasst habe. Ich habe dabei nur die angegebenen Quellen und Hilfsmittel verwendet und die aus diesen wörtlich, inhaltlich oder sinngemäß entnommenen Stellen als solche den wissenschaftlichen Anforderungen entsprechend kenntlich gemacht. Die Versicherung selbstständiger Arbeit gilt auch für Zeichnungen, Skizzen oder graphische Darstellungen. Die Arbeit wurde in gleicher oder ähnlicher Form weder derselben noch einer anderen Prüfungsbehörde vorgelegt und auch noch nicht veröffentlicht. Mit der Abgabe der elektronischen Fassung der endgültigen Version der Arbeit nehme ich zur Kenntnis, dass diese mit Hilfe eines Plagiatserkennungsdienstes auf enthaltene Plagiate überprüft und ausschließlich für Prüfungszwecke gespeichert wird. Es ist mir bekannt, dass wegen einer falschen Versicherung bereits erfolgte Promotionsleistungen für ungültig erklärt werden und eine bereits verliehene Doktorwürde entzogen wird.

Ebenfalls versichere ich, dass die veröffentlichten Studien in Ko-Autorenschaft mit mindestens 50% Beteiligung von mir verfasst worden sind.

Zuletzt versichere ich, dass die ausführliche Darstellung des wissenschaftlichen Zusammenhangs der Publikationen noch nicht veröffentlicht wurde und nirgendwo anders zur Veröffentlichung vorliegt.

Heidelberg, den 01. Juni 2020

Joshua Weidlich